

DETECTING ECCENTRICITY: WHAT IS THE CURRENT EVIDENCE?

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GW Eccentricity Corfu, Greece September 2026



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WHAT IS DISCOVERY: WHAT IS BAYES?

Bayes/Laplace told us that

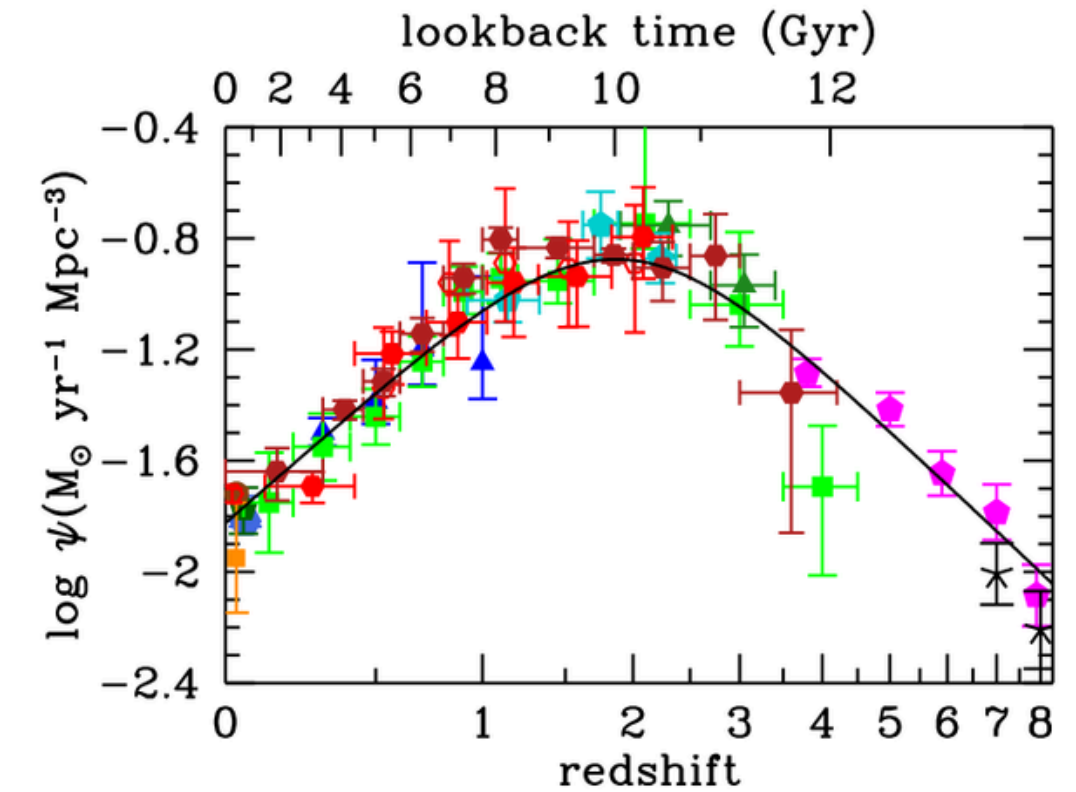
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The **prior** is **model dependent** and encodes our belief about the parameters **before seeing the data**



Mandau & Dickinson 2014

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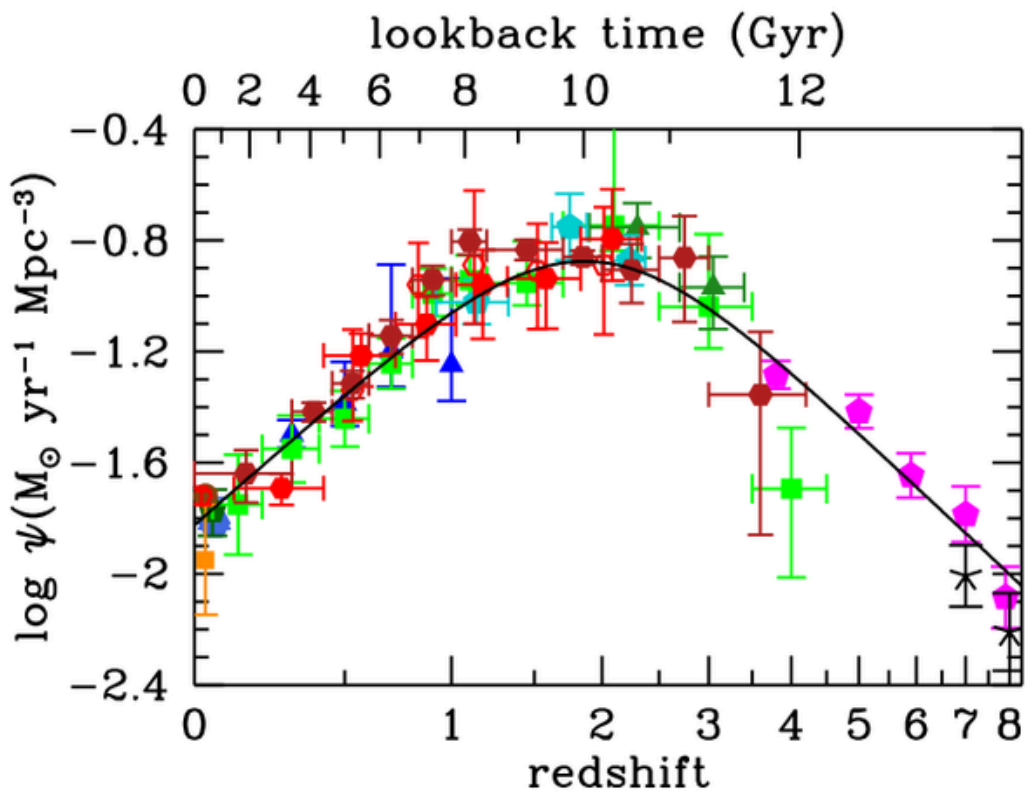
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Data Model

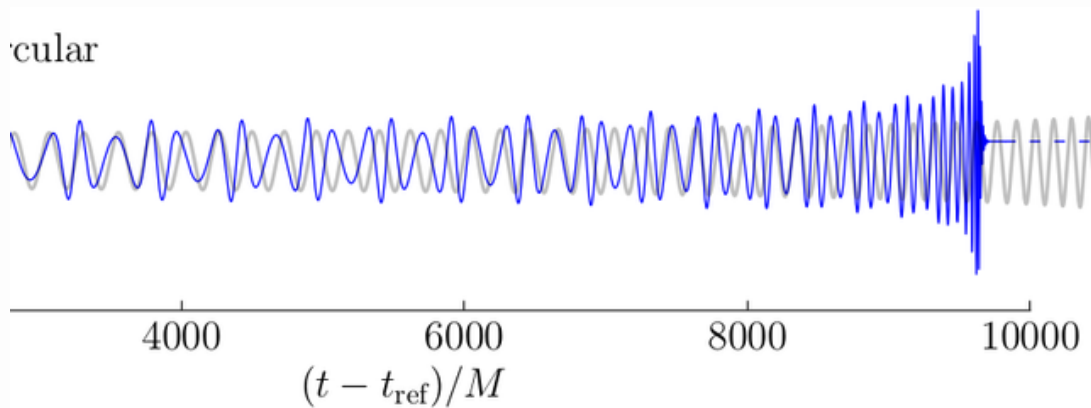
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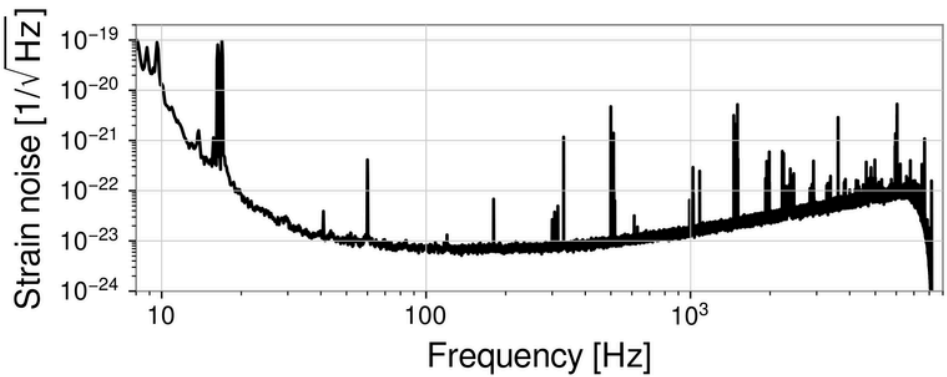
The **likelihood** is **model dependent** and encodes our belief about the **data generating process**

Signal model



SXS Collaboration 2025

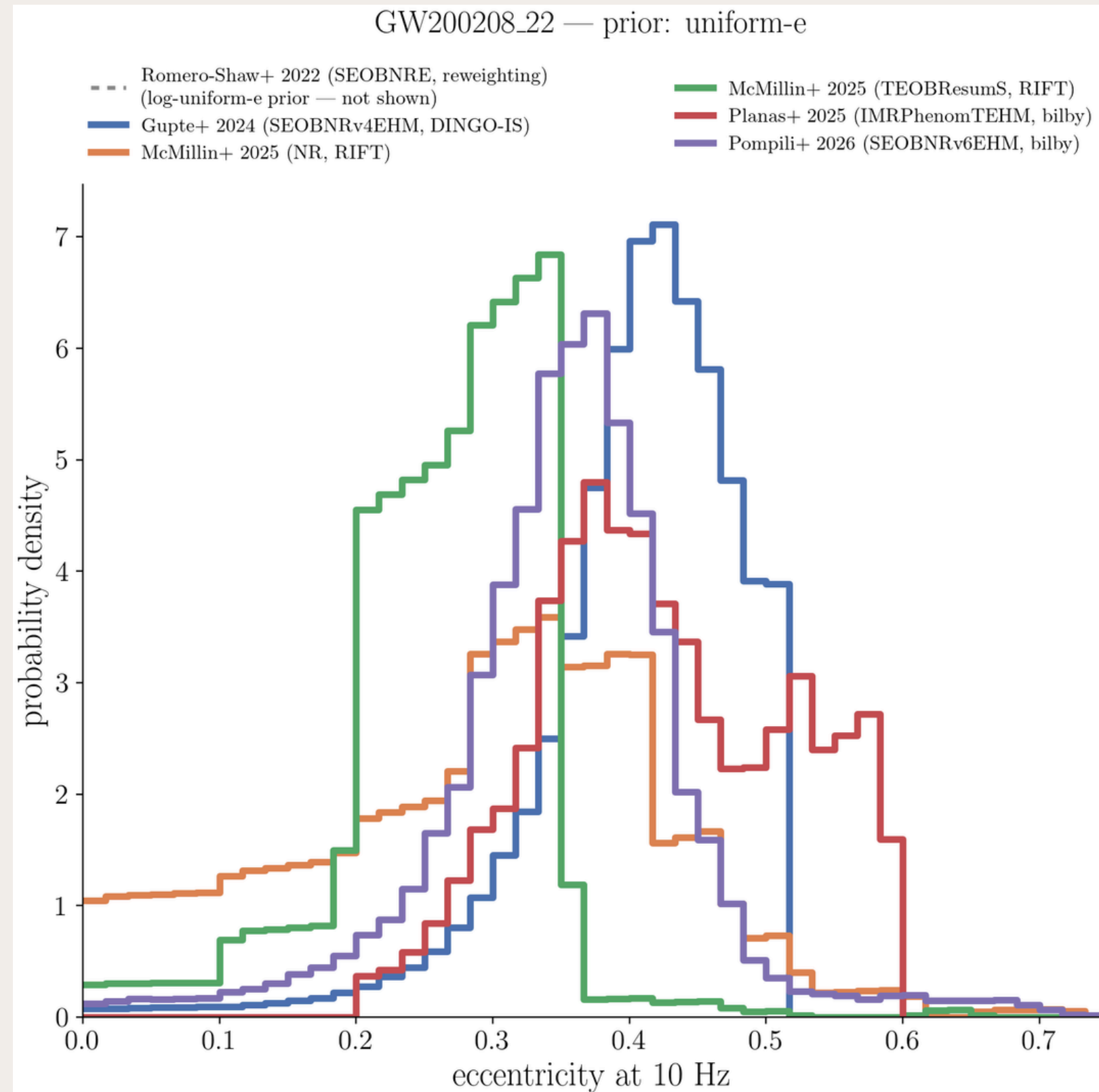
Noise Model



LVK Collaboration 2020

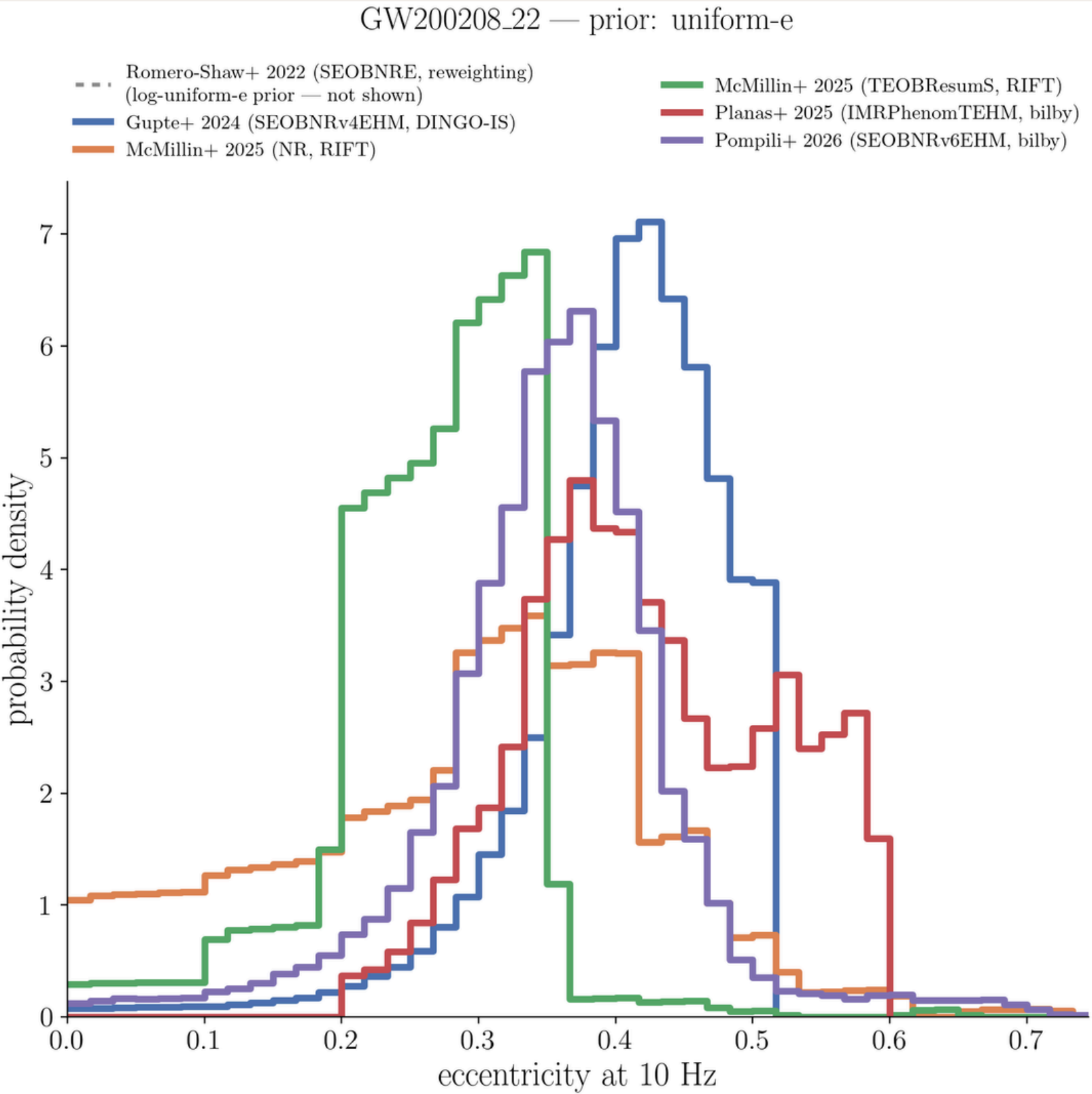
GW200208_22

This is a ~66 solar mass BBH with 14 cycles in band.
Bayes factor ~ 17, 3, — (uniform, log-uni, expert prior)



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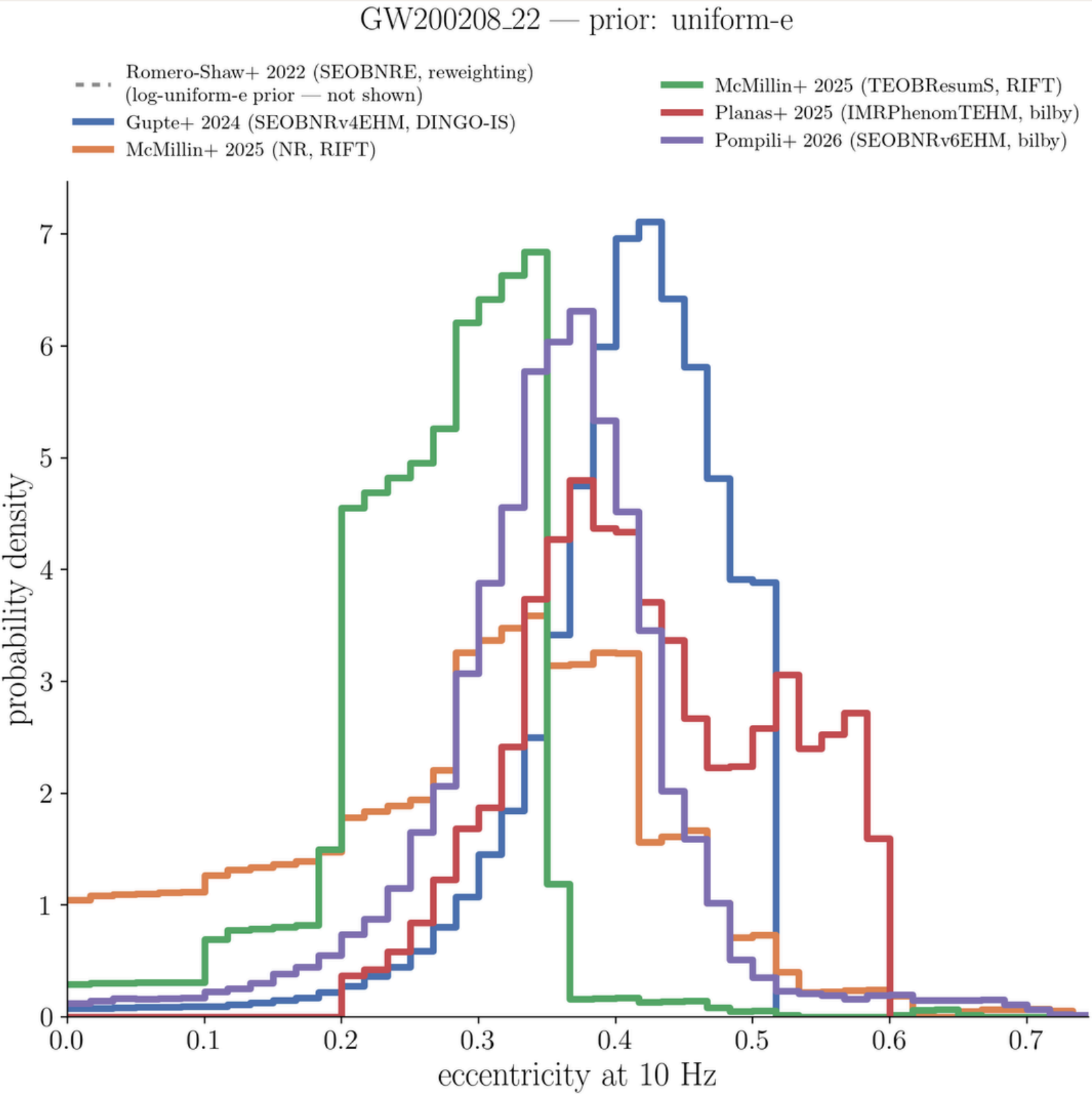


BUT IS IT AN EVENT?

LVK 2021 found that the FAR is 5 per year. The p-astro is 70%.

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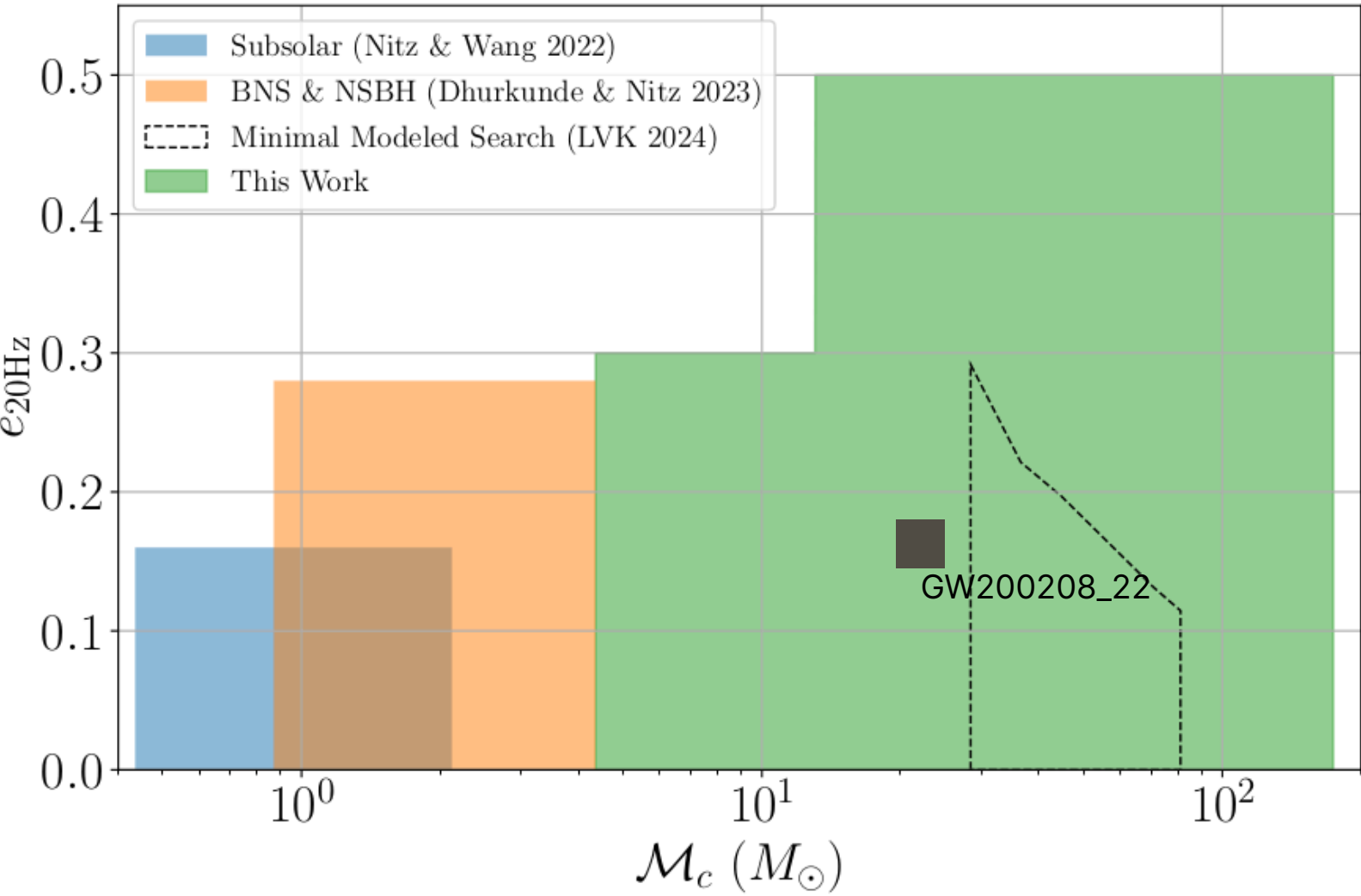
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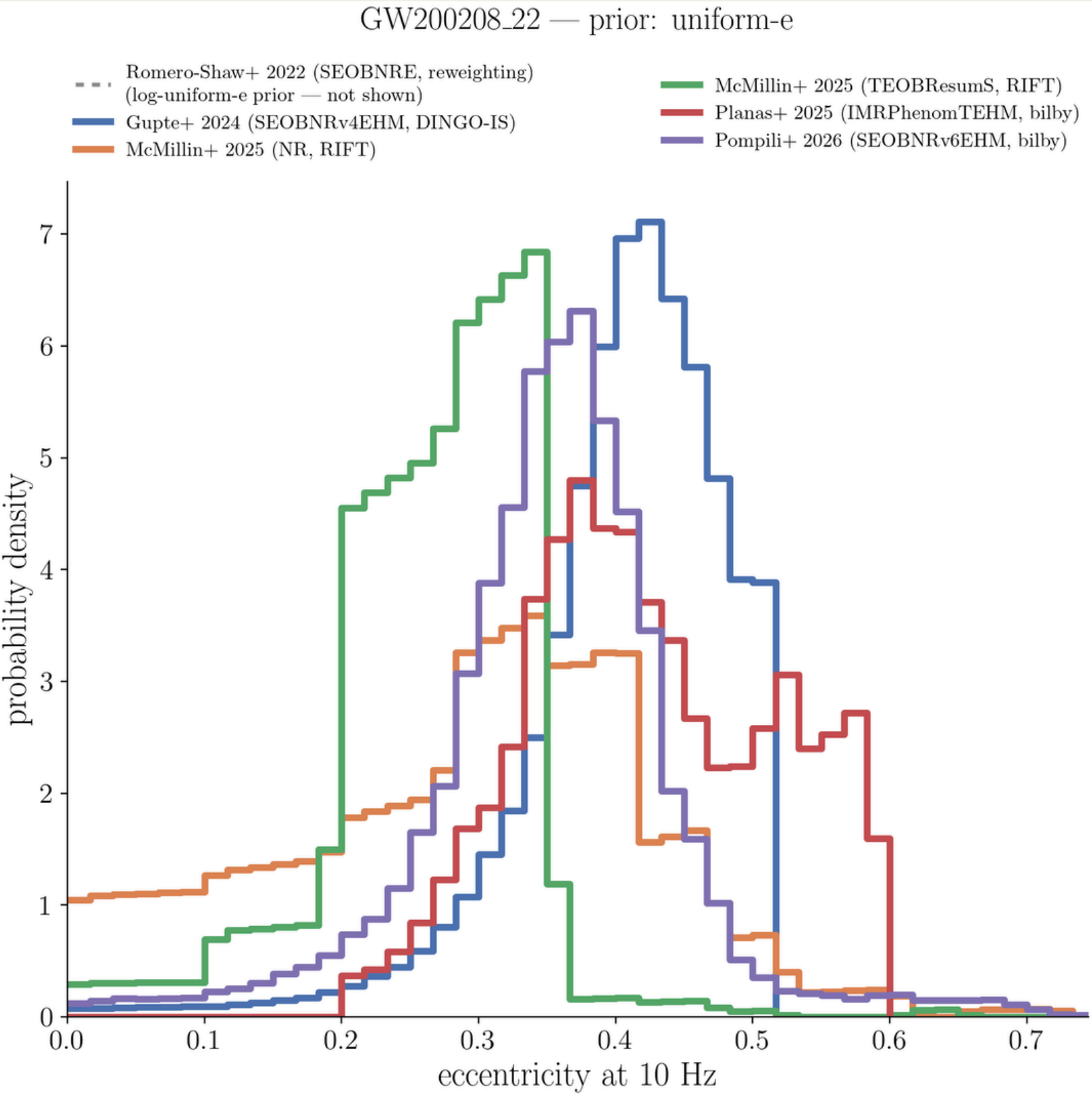
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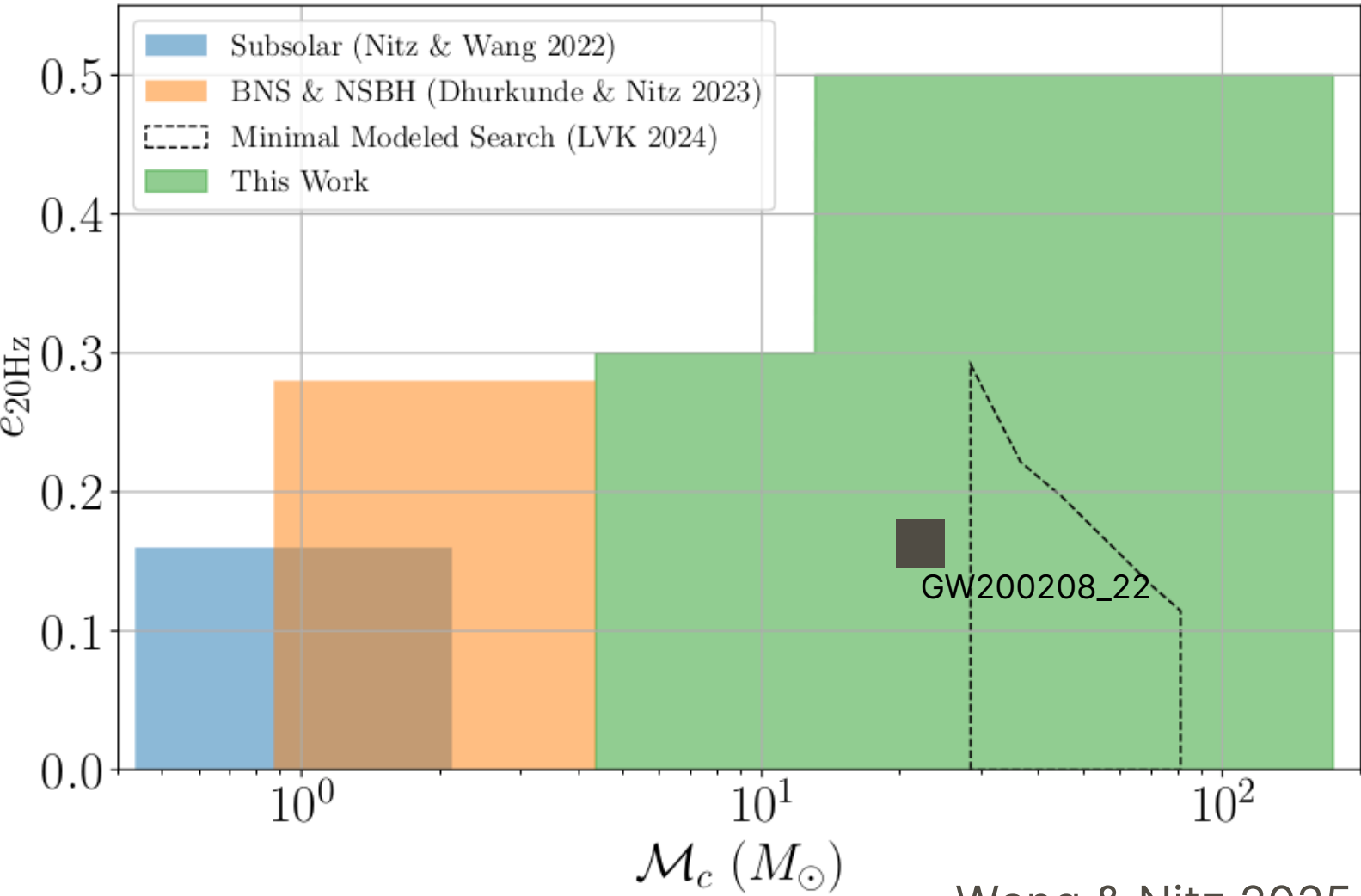
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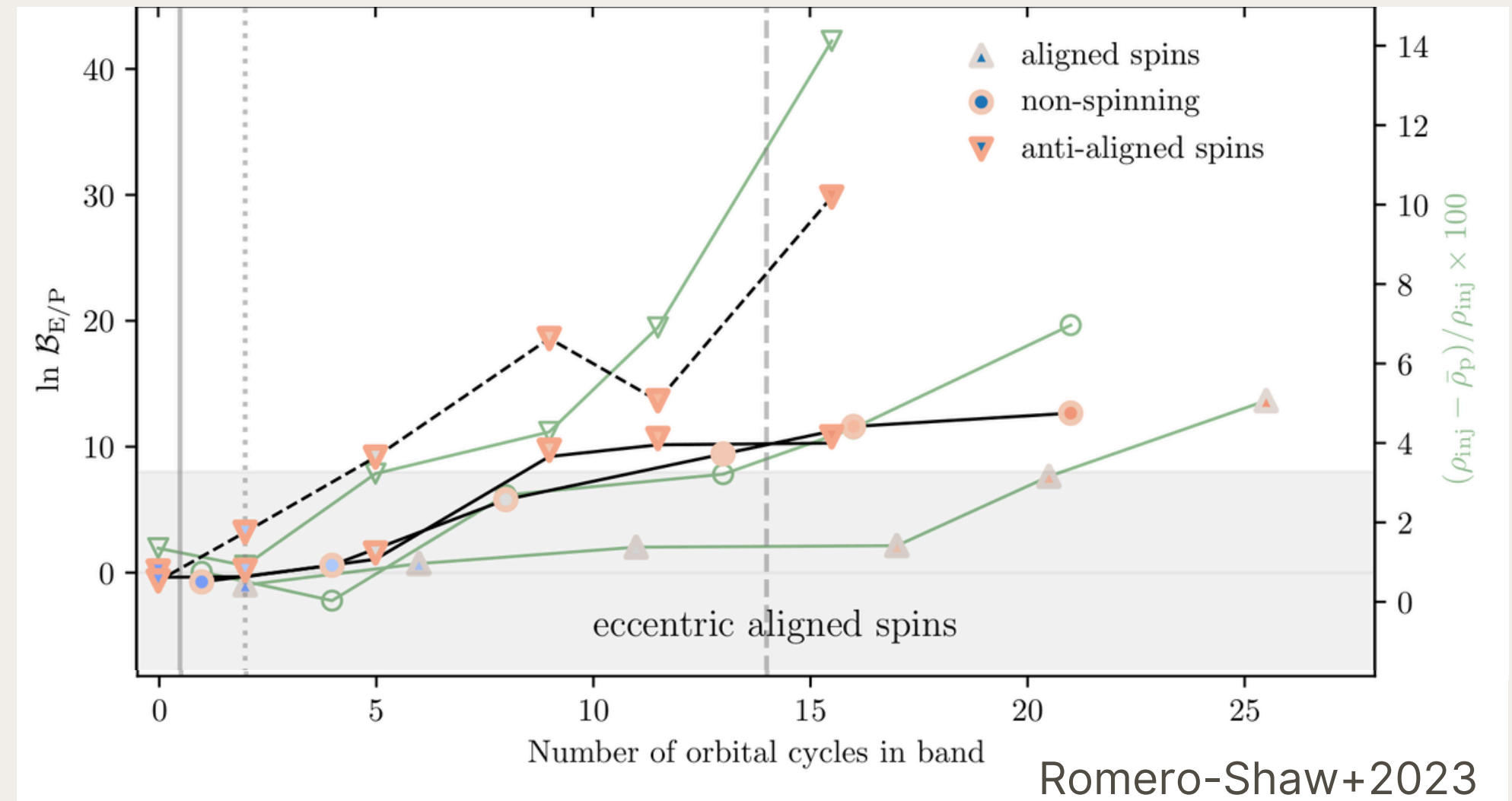
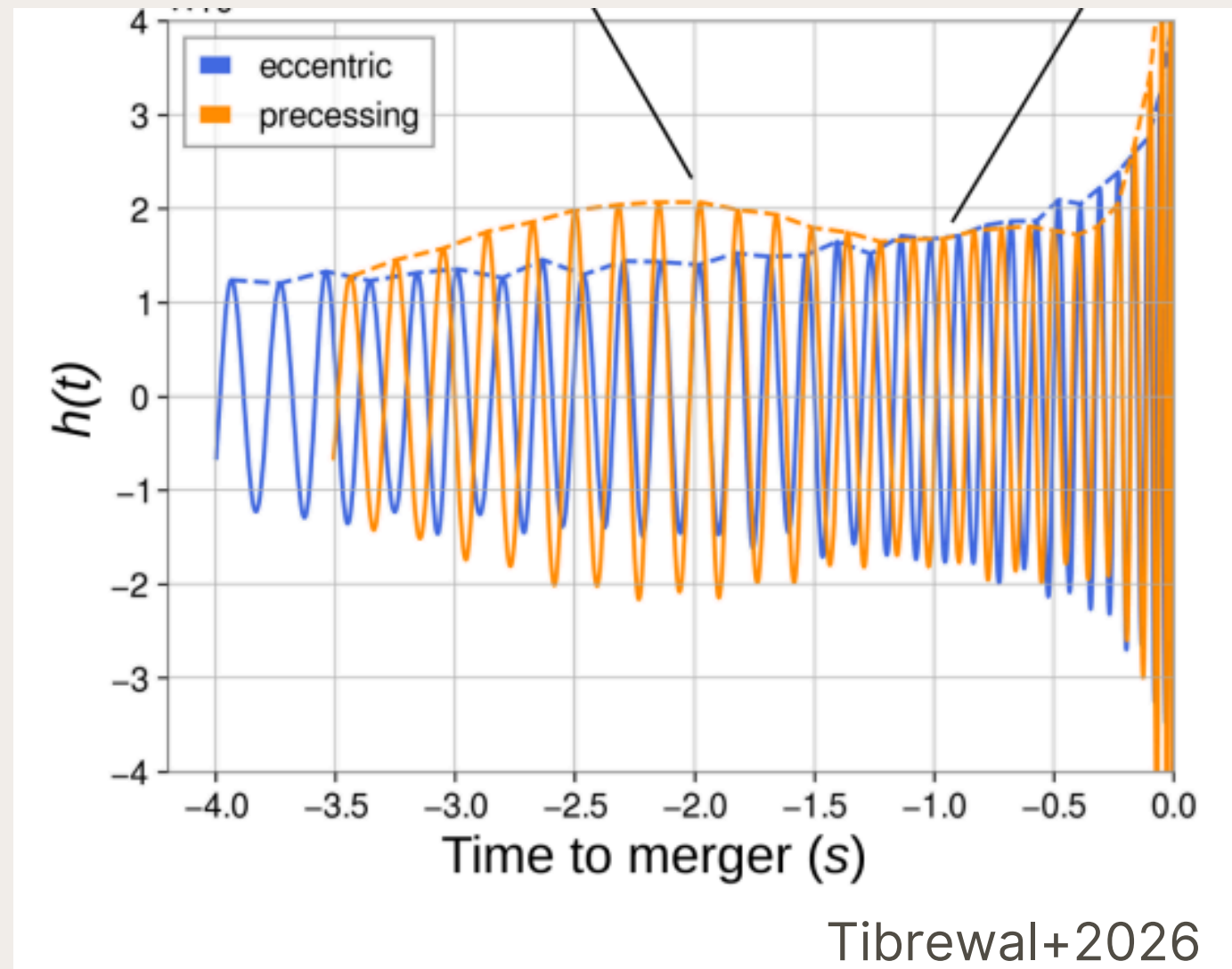
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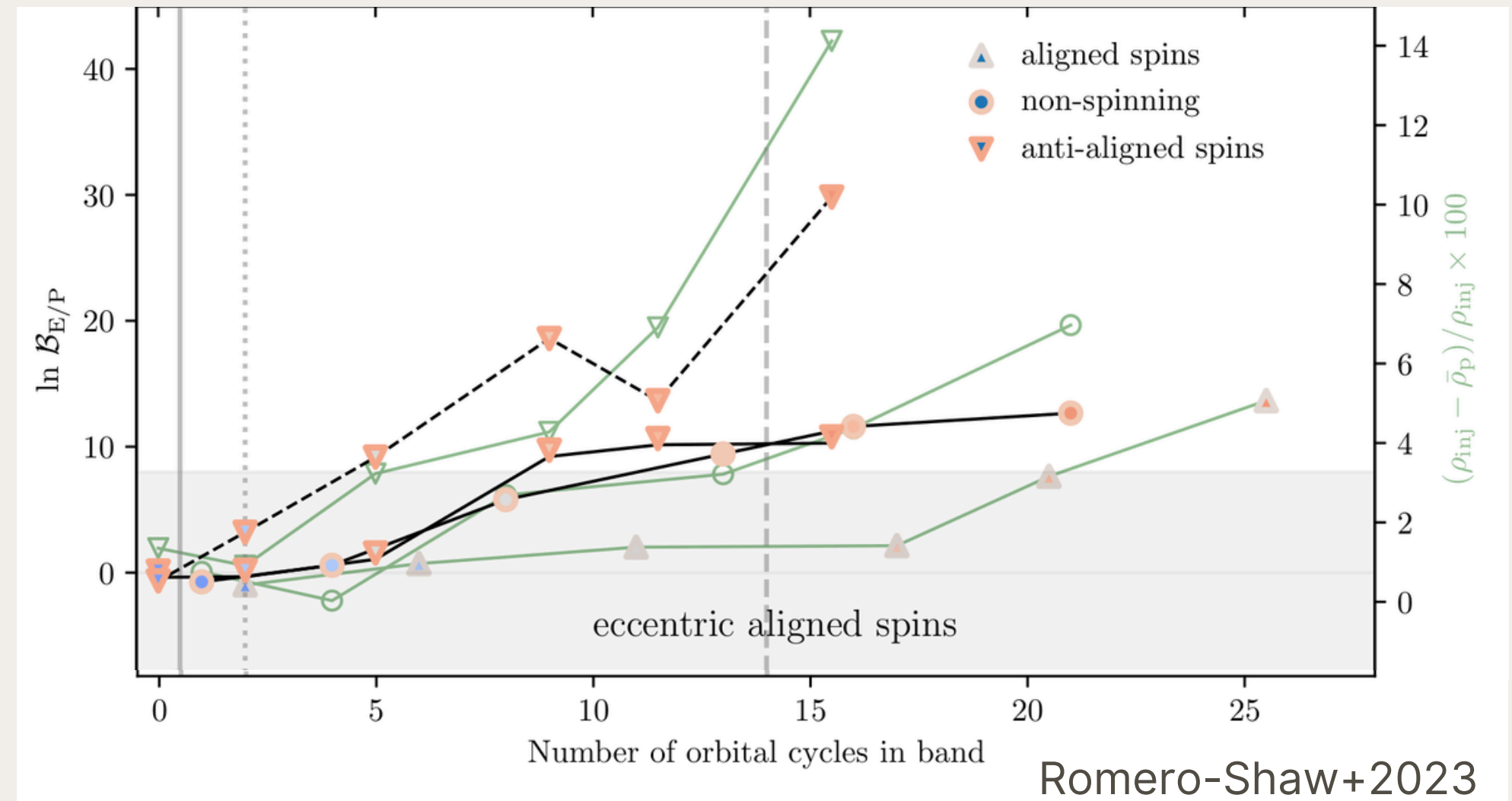
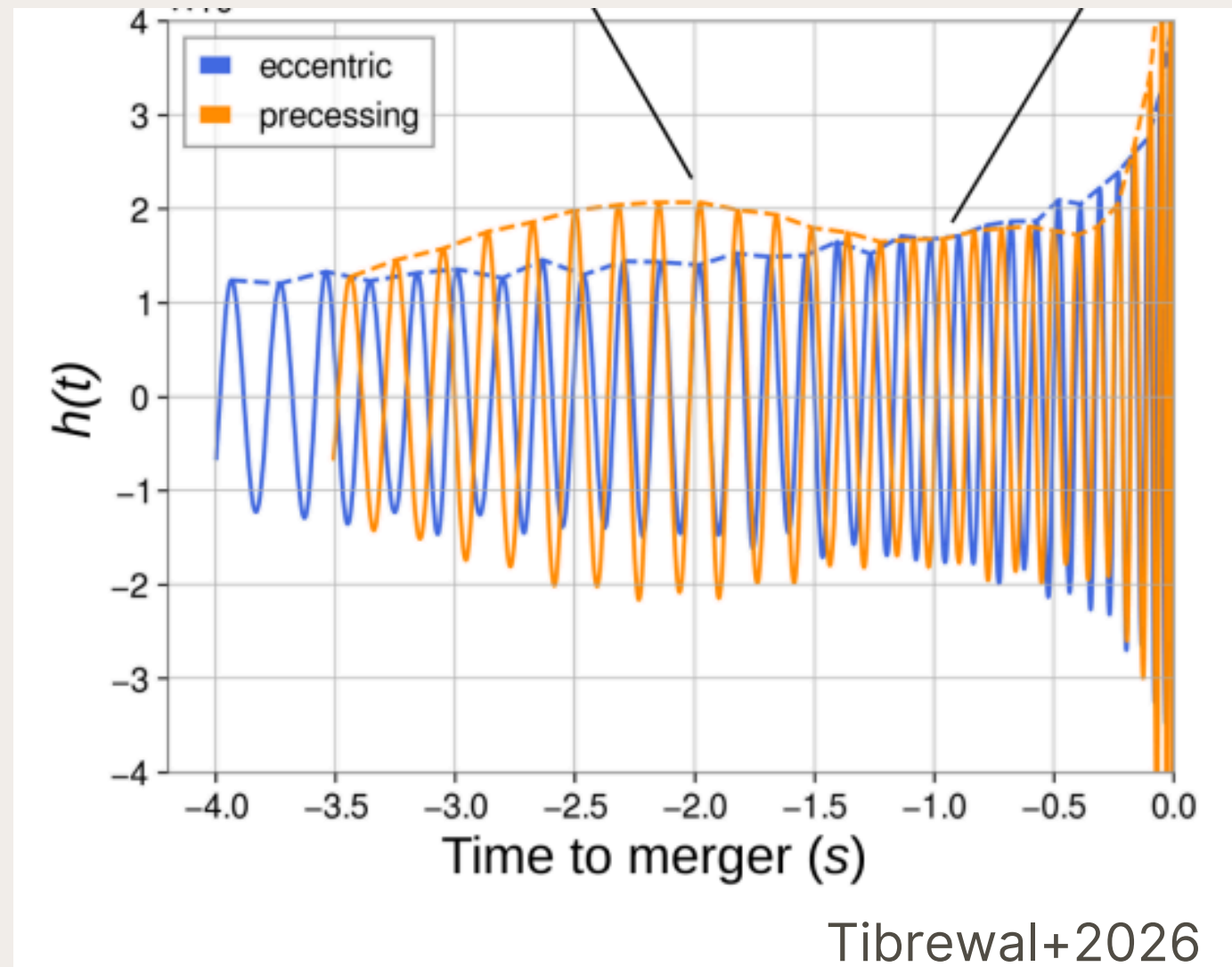
AN ASIDE ON DEGENERACY & WAVEFORM LENGTH

The Bayes factors on the previous slides compared the **eccentric aligned-spin vs the quasi-circular spin-precessing hypothesis**. (Bustillo+ 2021, Romero-Shaw+2023, Divyajyoti+ 2025, Tibrewal+ 2026).



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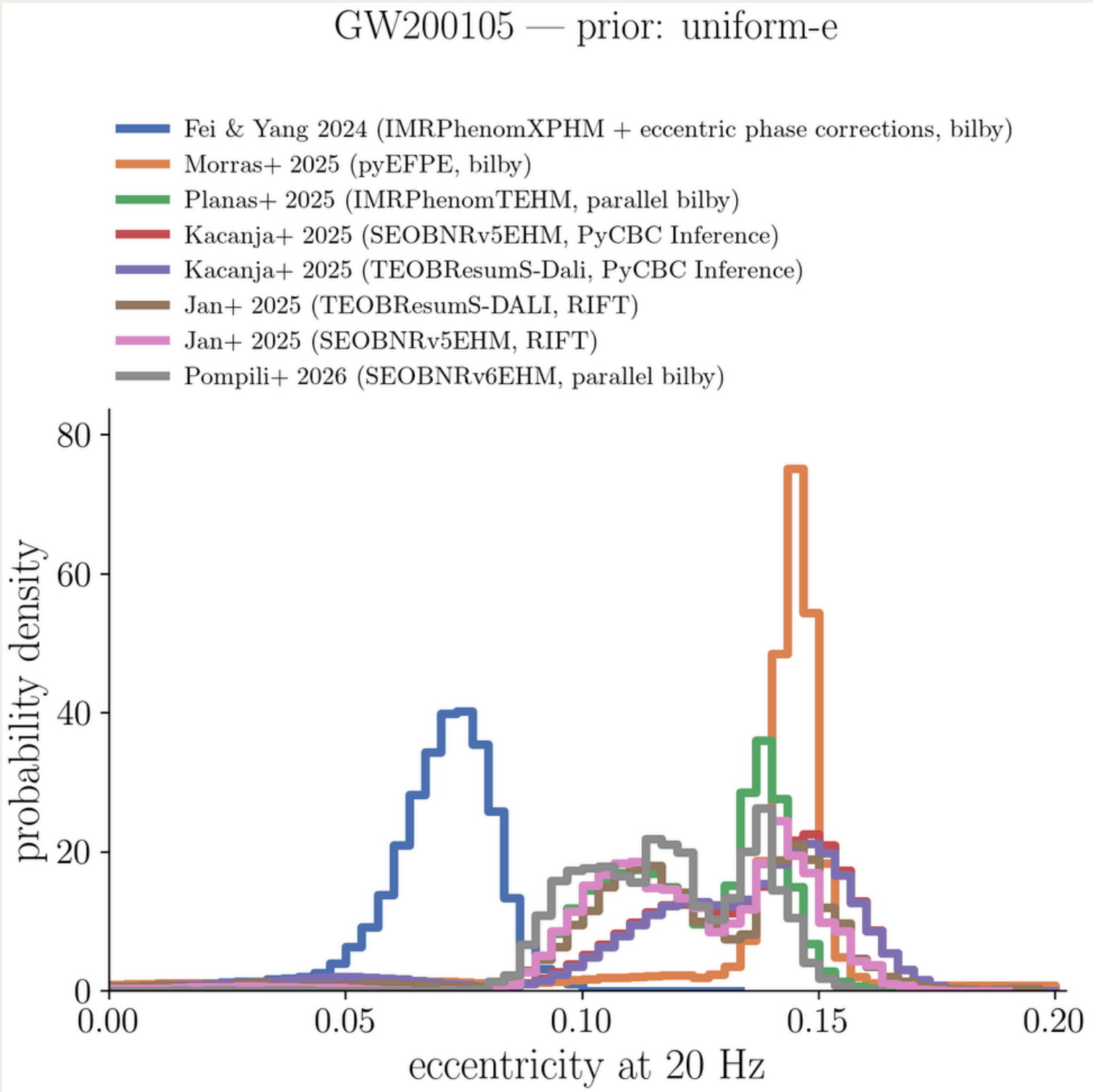


This is a statement on $p(\text{confusion} \mid \text{number of cycles})$. It is **not** a statement about $p(\text{confusion} \mid \text{number of cycles, data})$.

Consequence: if two observations have the same Bayes factor but one is shorter it does not mean that the longer one is more trustworthy.

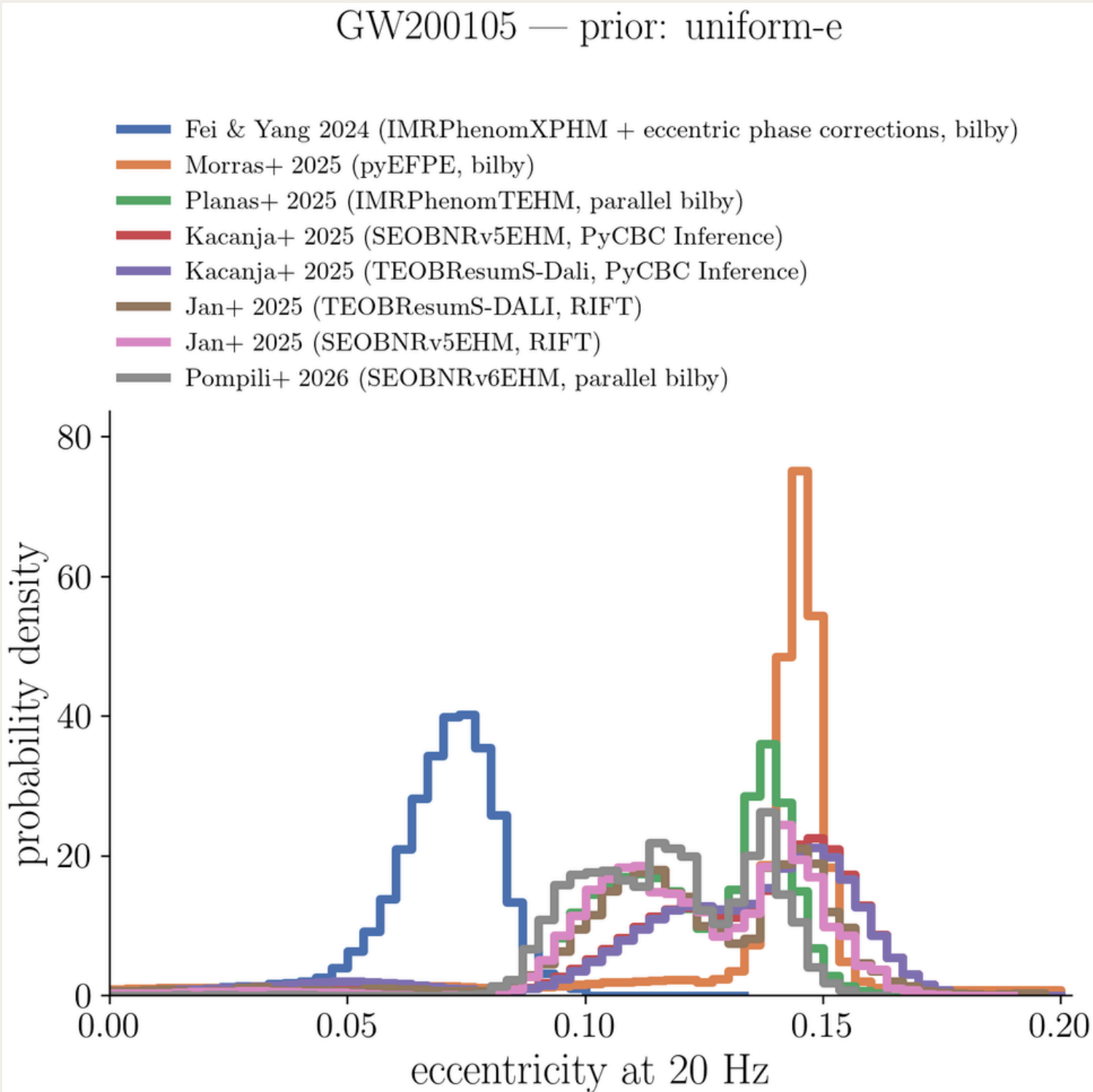
GW200105 – AN NSBH?

It was reported that there were signatures of eccentricity in a NSBH event GW200105. Note pyEFPE (Morras+ 2025) is inspiral only but **includes eccentricity and spin-precession!**



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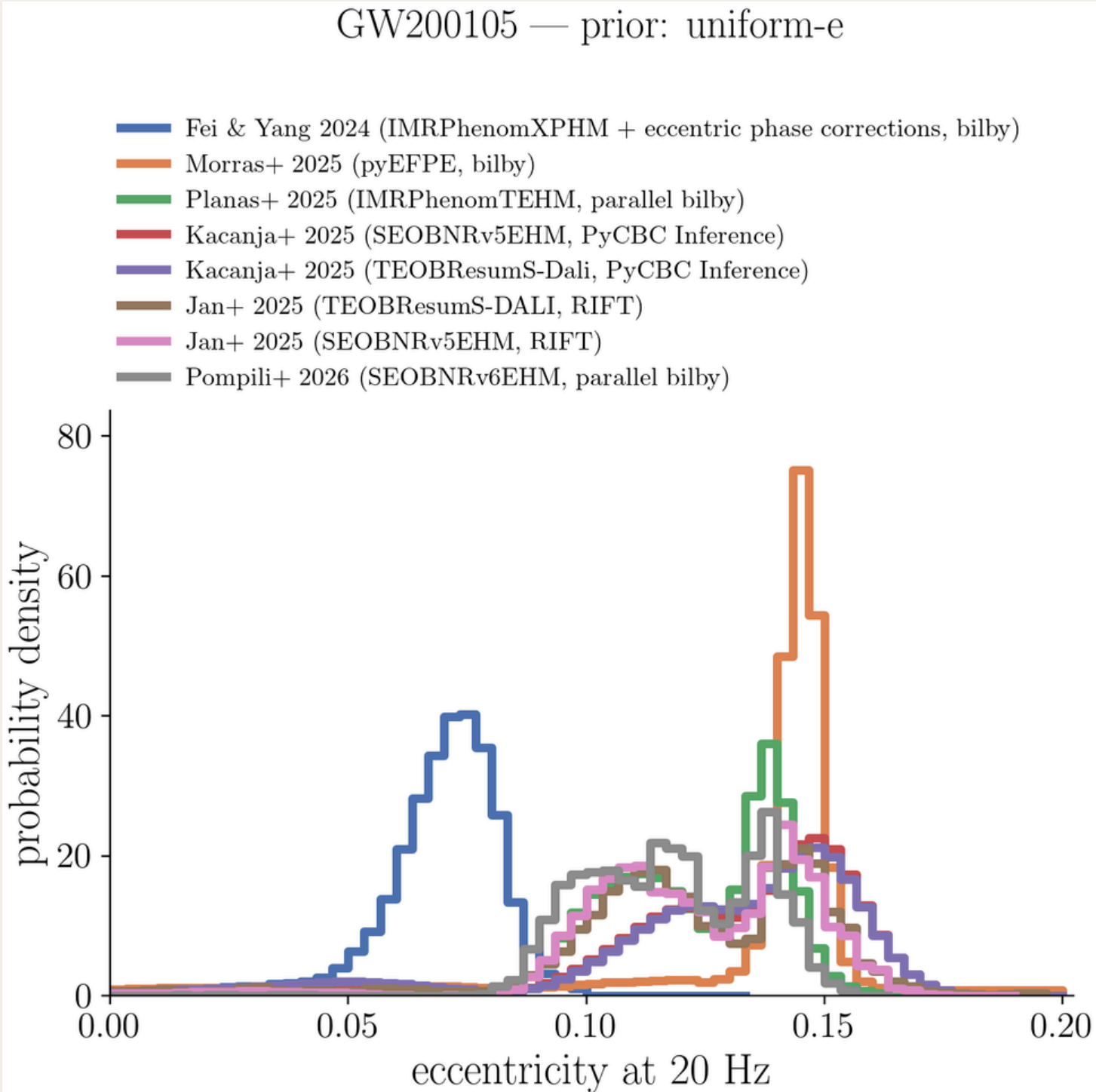
Criticism of this event was that GW200105 was reported as a low-significance single detector trigger (**FAR of 1 per 2.8 years**, (LVK 2021).

Davies & Harry 2022 showed with updated methods the **FAR is 1 per 50,000 yrs**. Phukon+ 2025 further showed that eccentric templates also decrease the FAR. The glitch also has negligible impact (Morras+ 2025)

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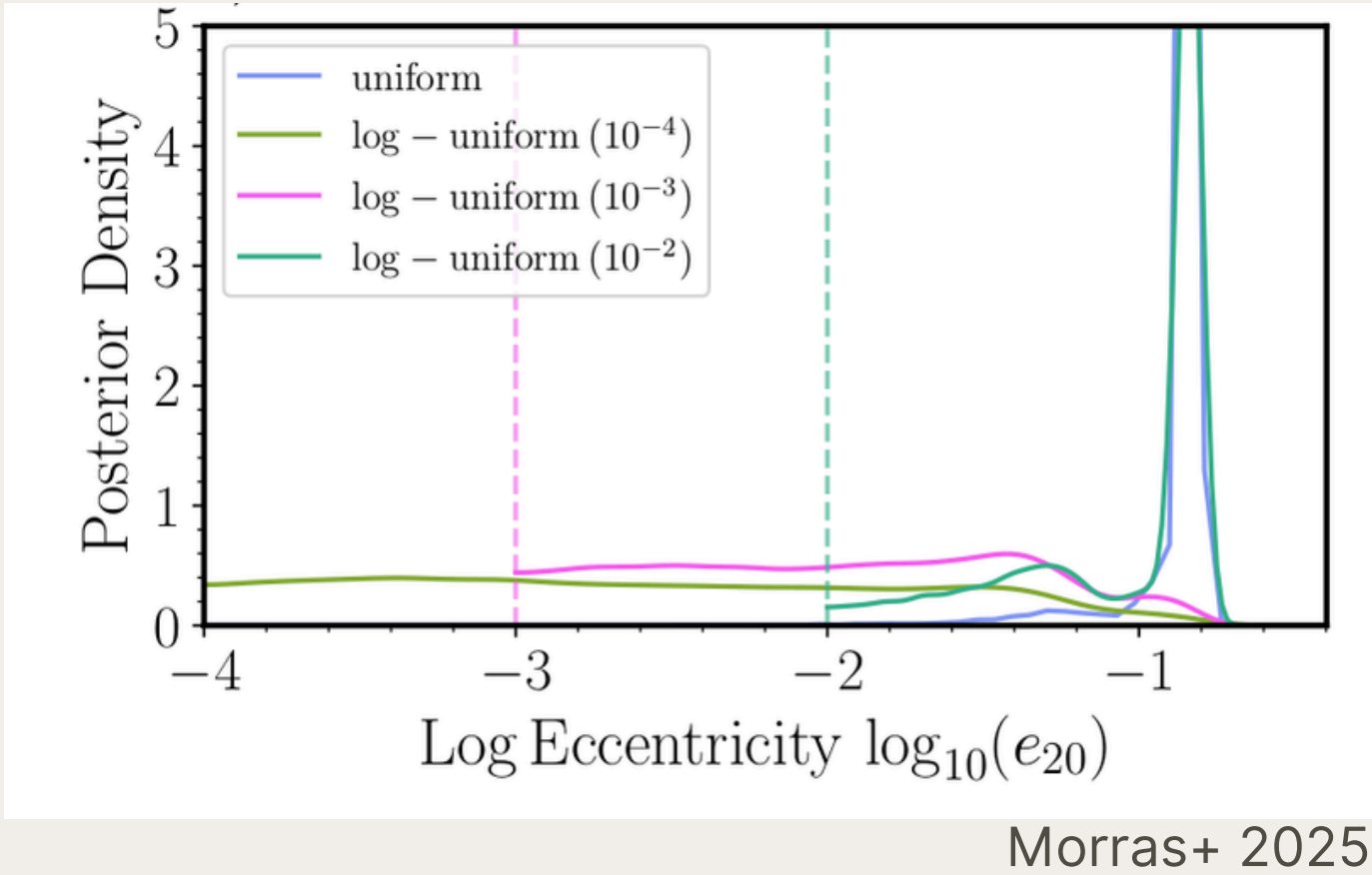
Bayes factor ~ 17, 1.3, __ (uniform, log-uni, expert prior).
See also Tiwari+ 2025



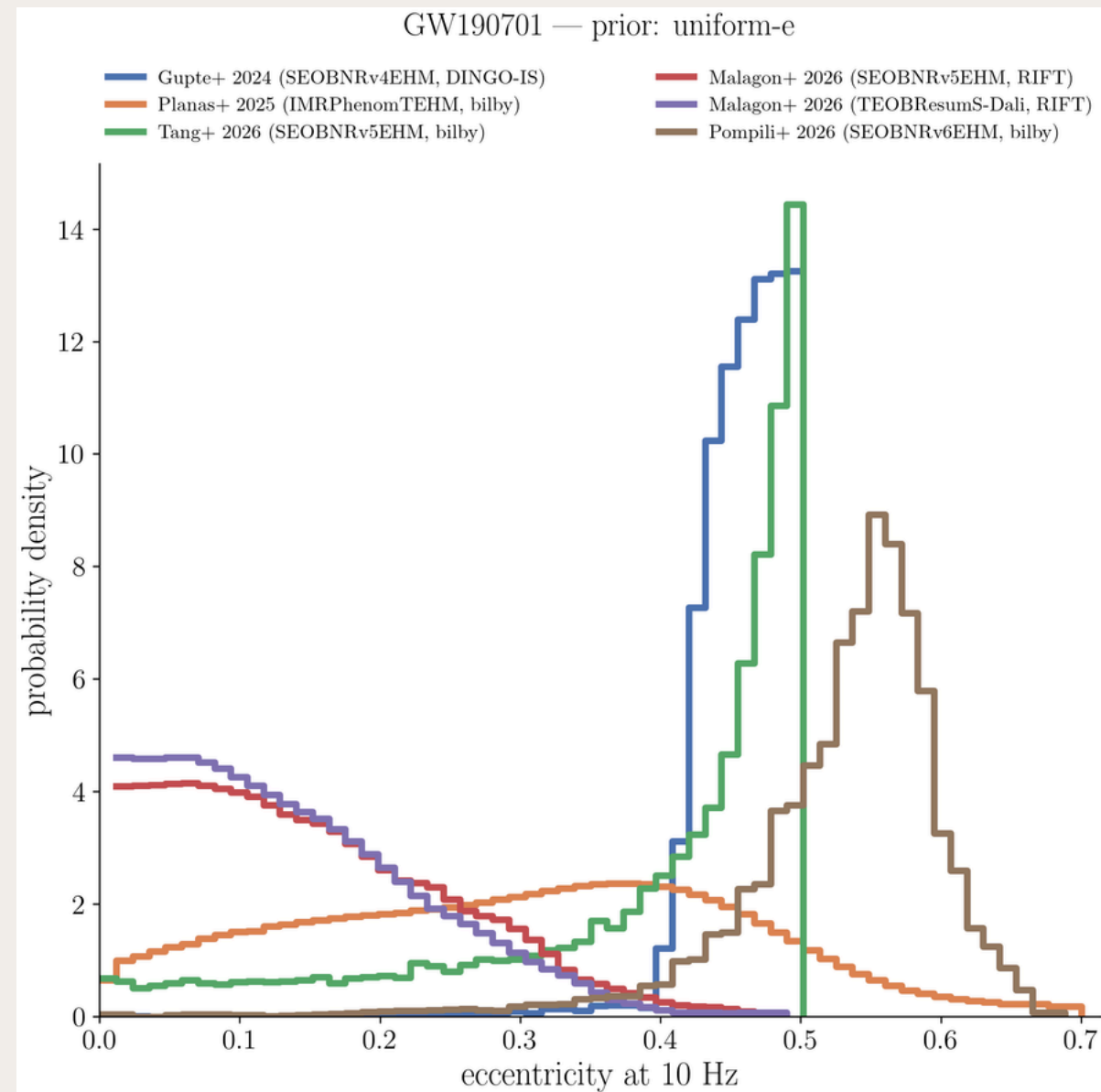
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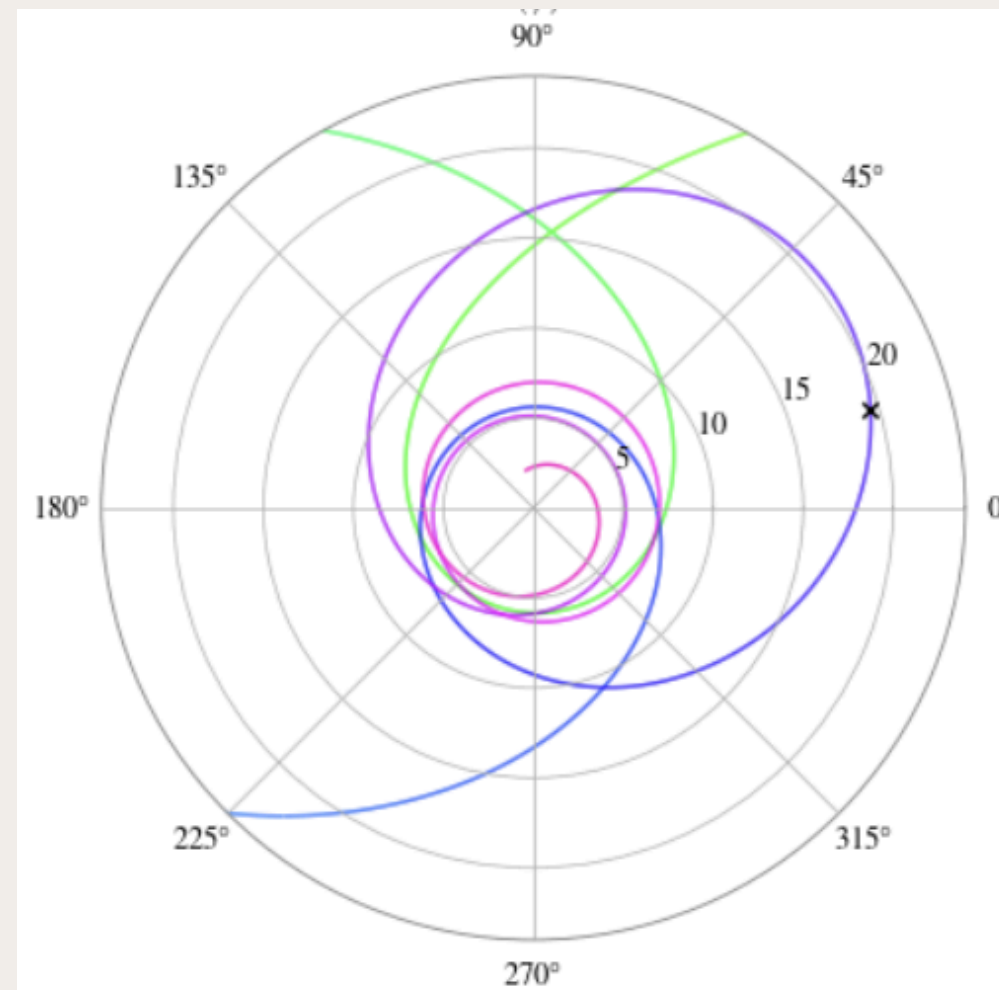
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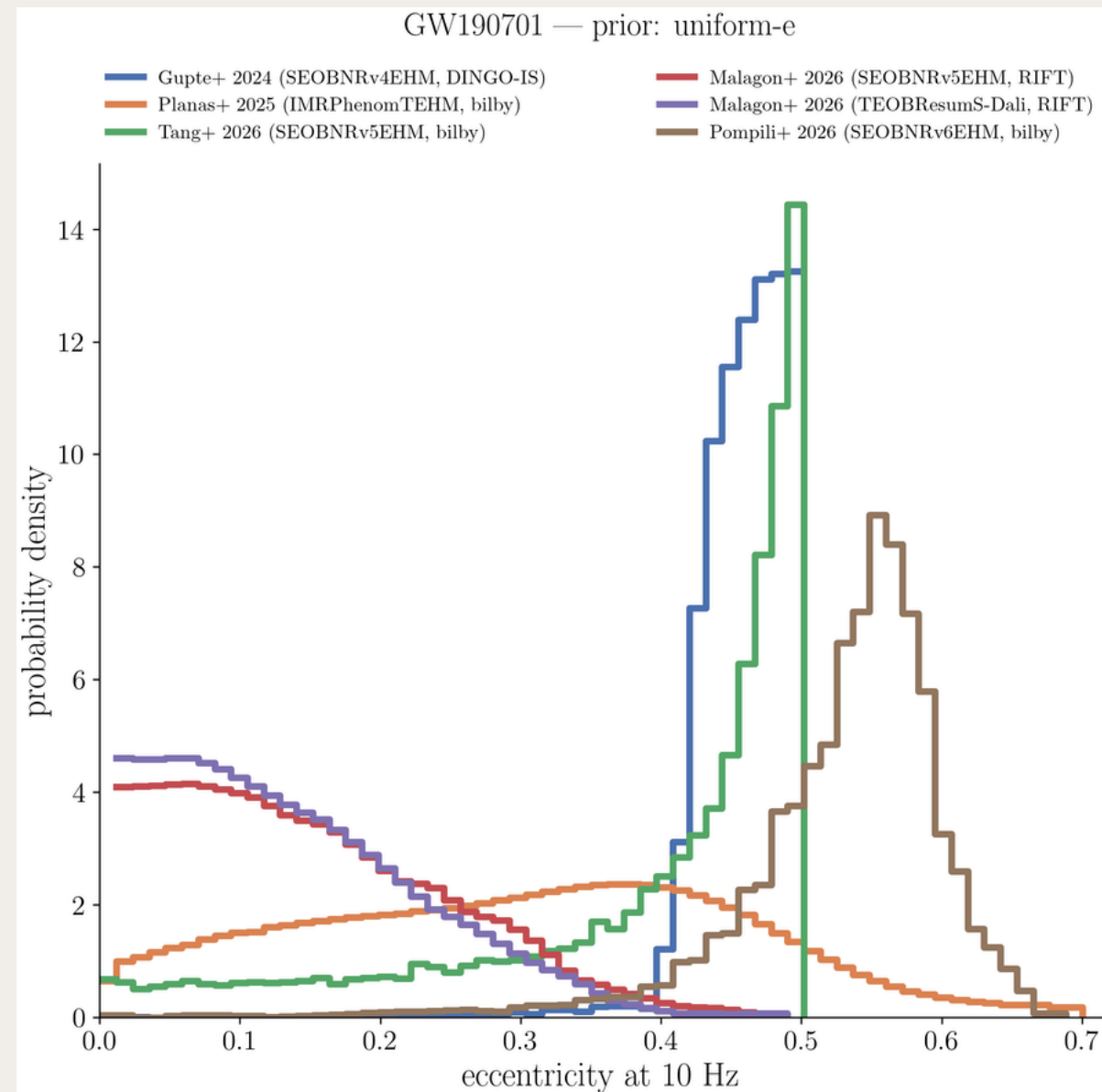


Orbital trajectory

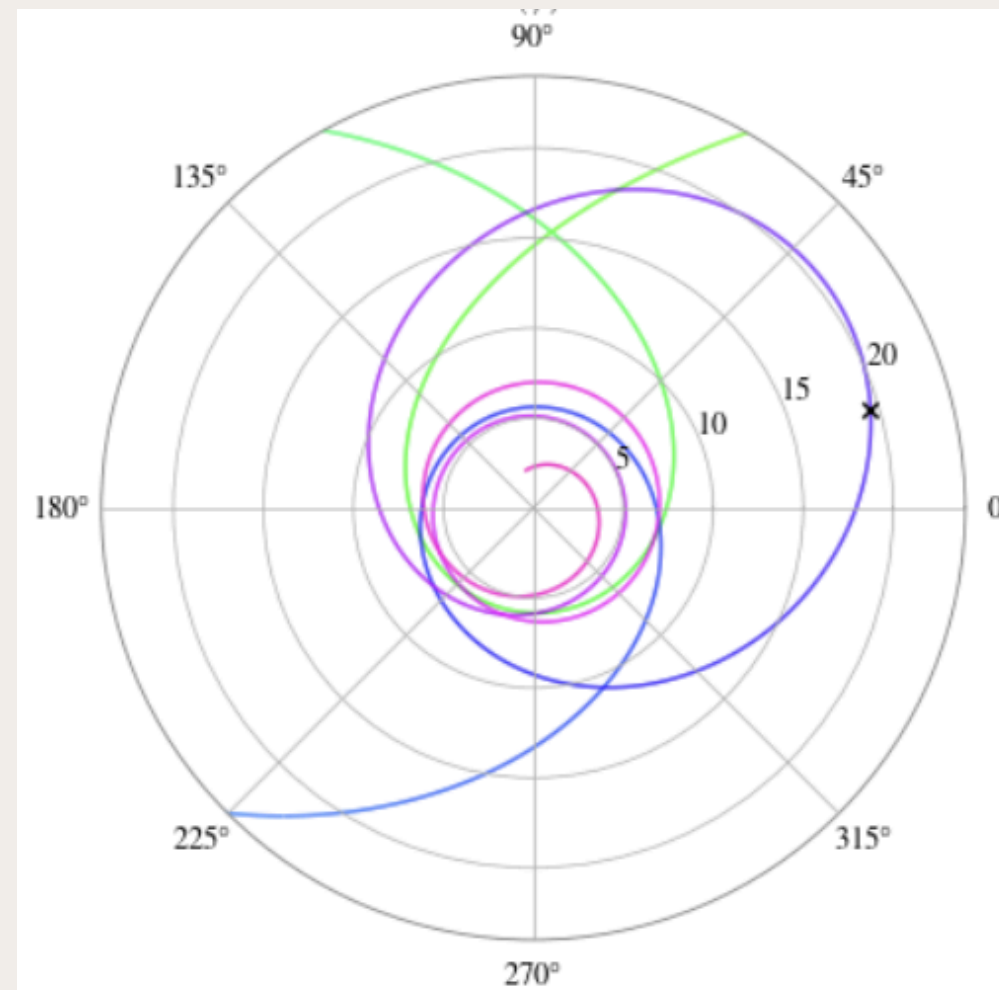


There was a scattered light glitch in LIGO Livingston during the detection. Therefore, the “BayesWave” (Cornish+ 2015) glitch mitigation algorithm was employed to subtract the glitch.

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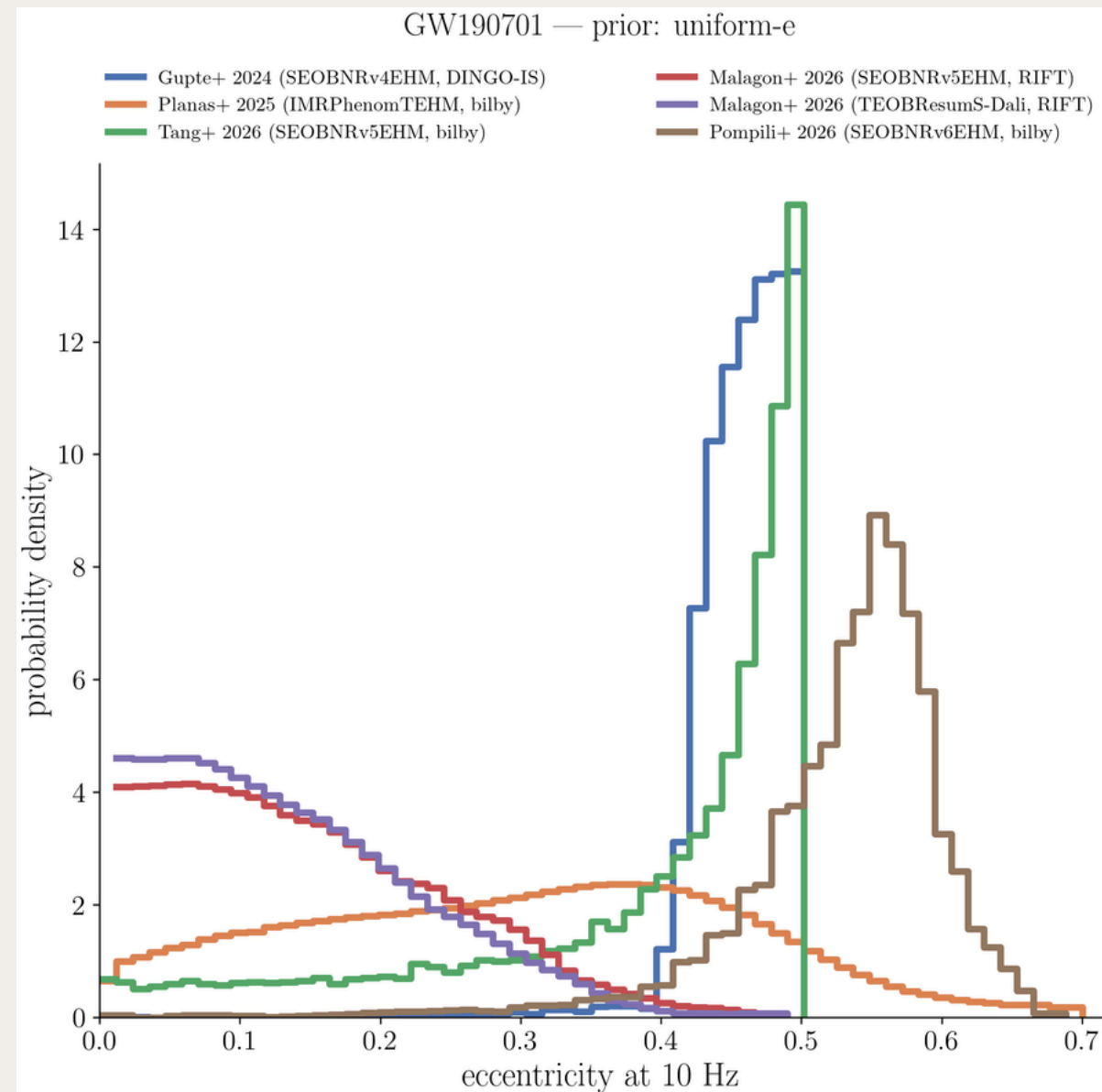


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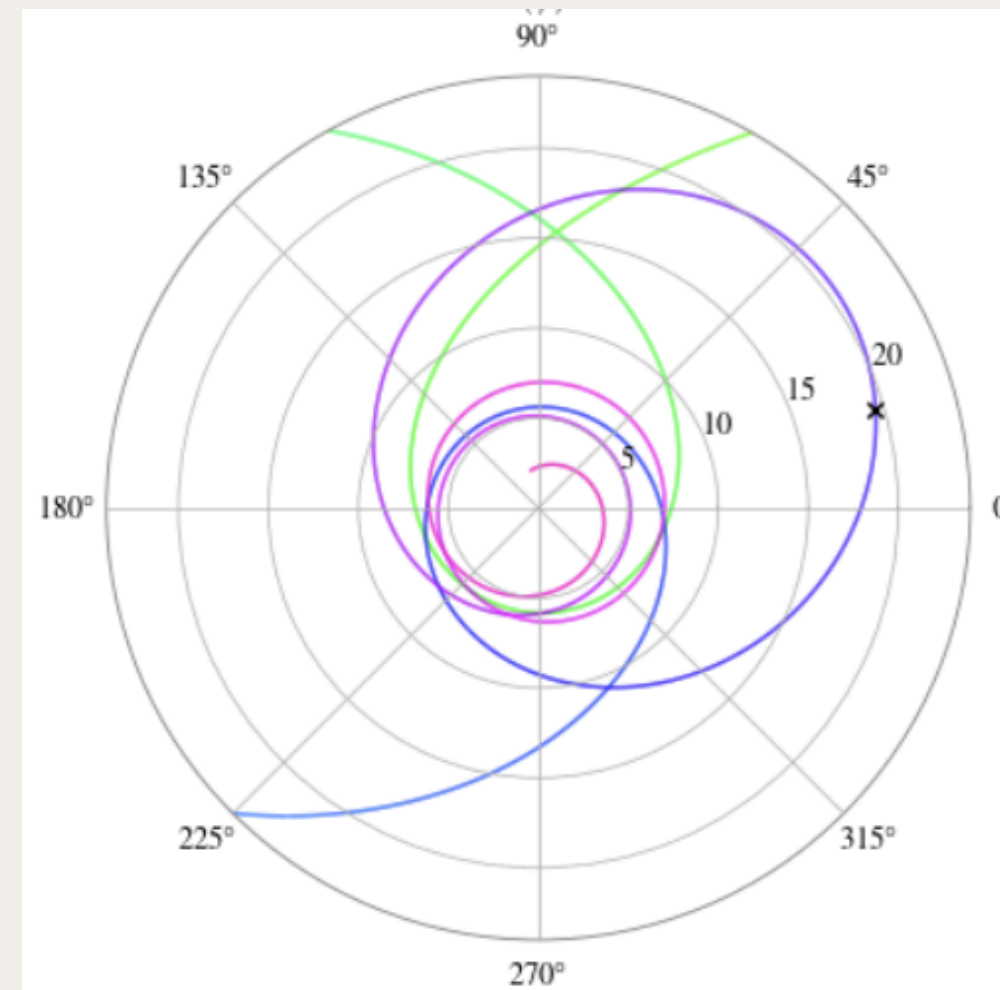
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> This event had a high Bayes factor 407, 51, — (uniform, log-uni, expert prior)

GW190701



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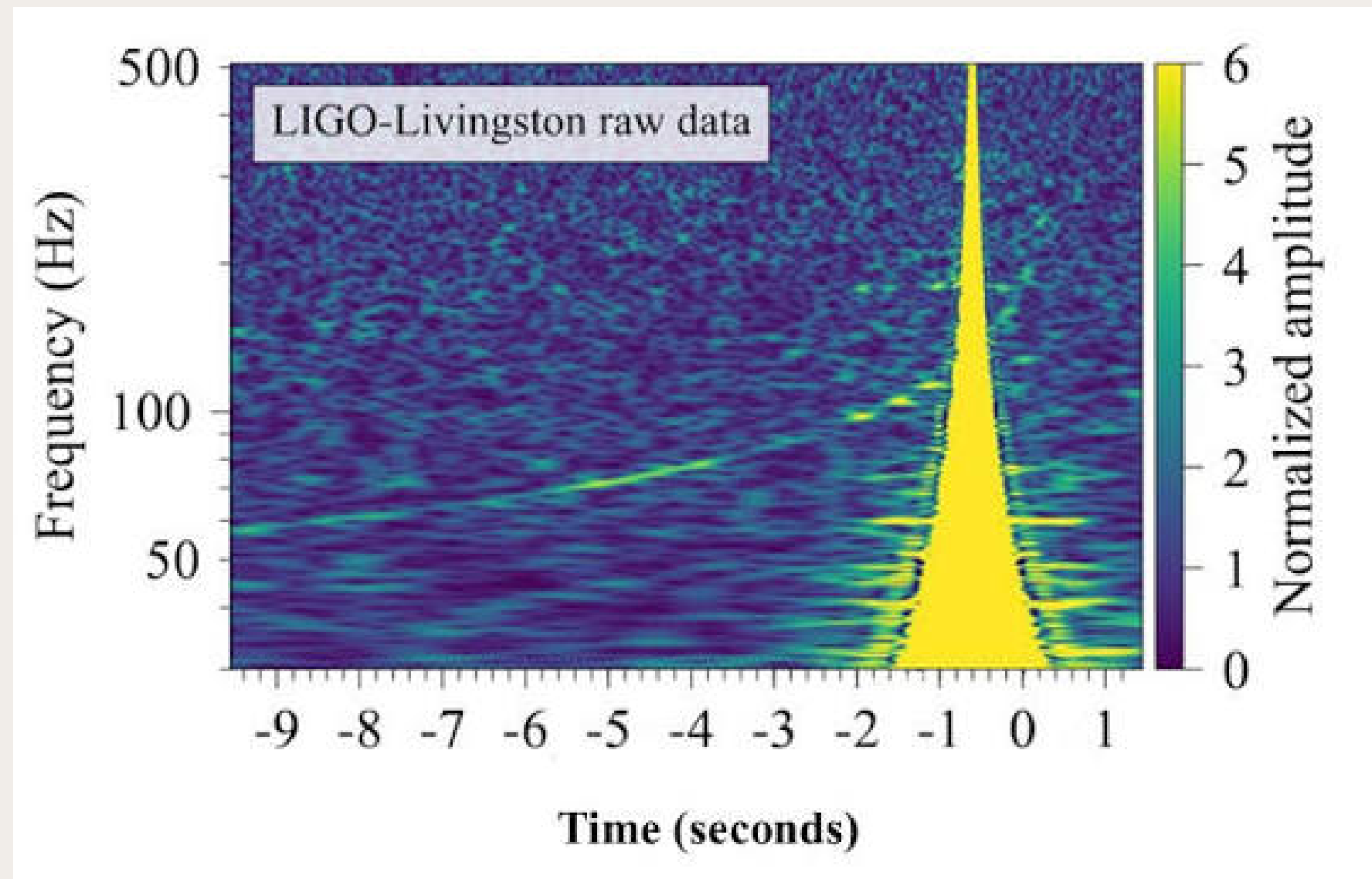
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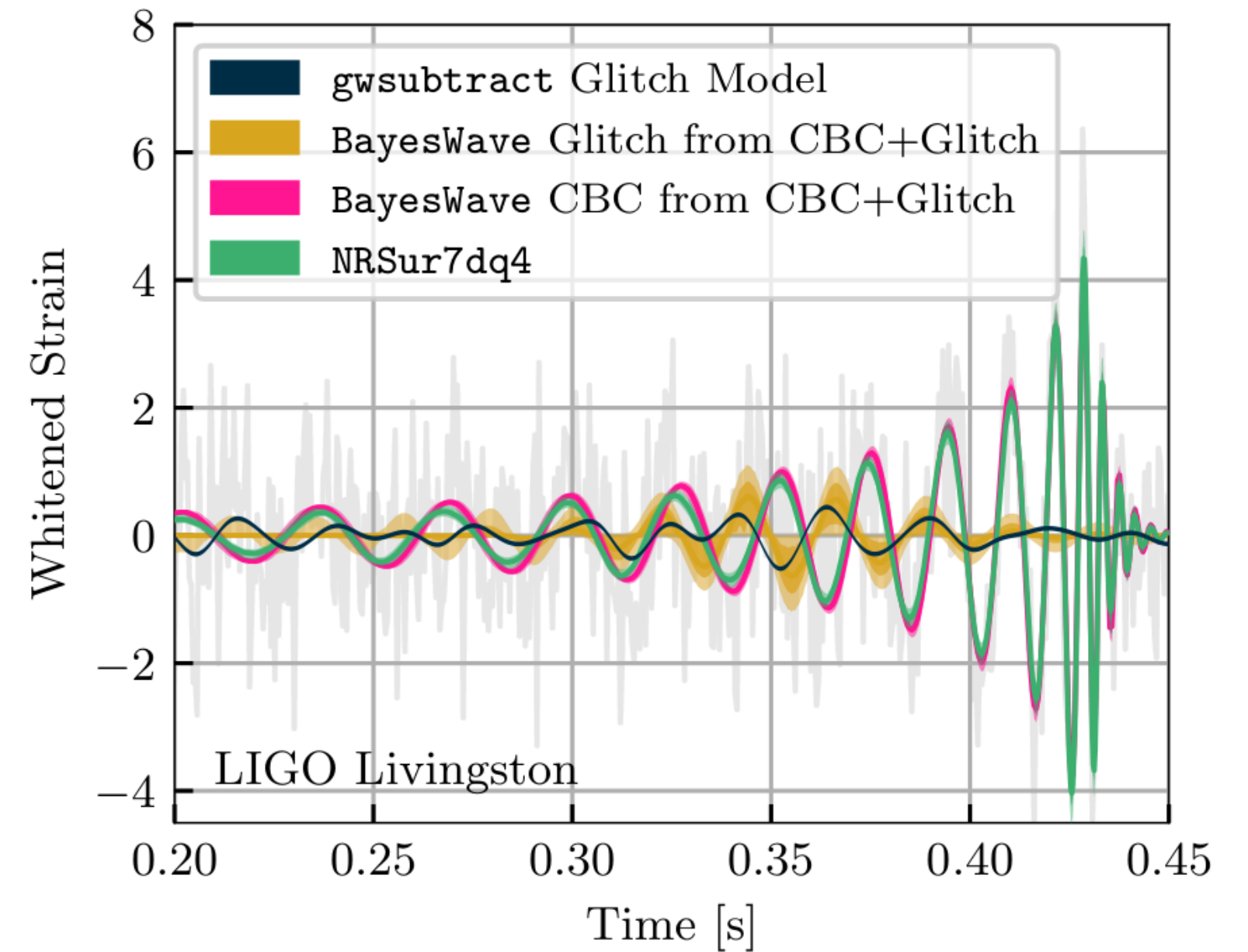
HOW DOES GLITCH MITIGATION WORK?



LVK Collaboration 2017

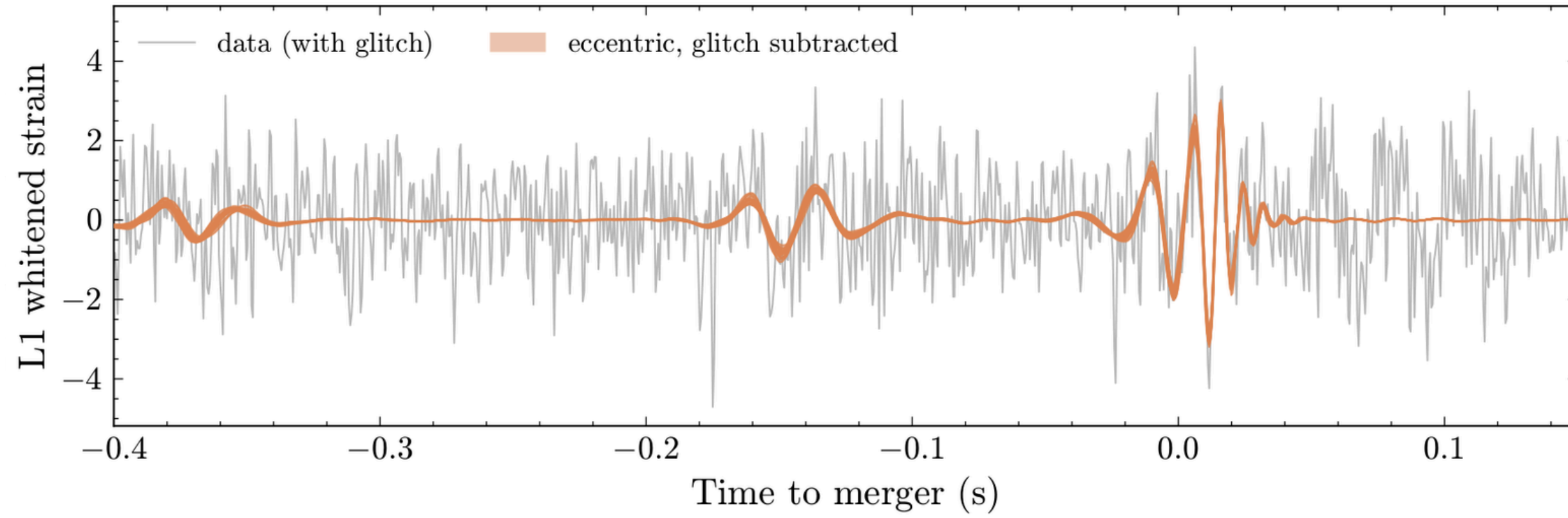
HOW DOES GLITCH MITIGATION WORK?

- > How does “BayesWave” work?
- > Fit the signal coherently with a waveform model
- > Fit the glitch incoherently with sine-Gaussian wavelets
- > **Note this is not waveform model agnostic.**



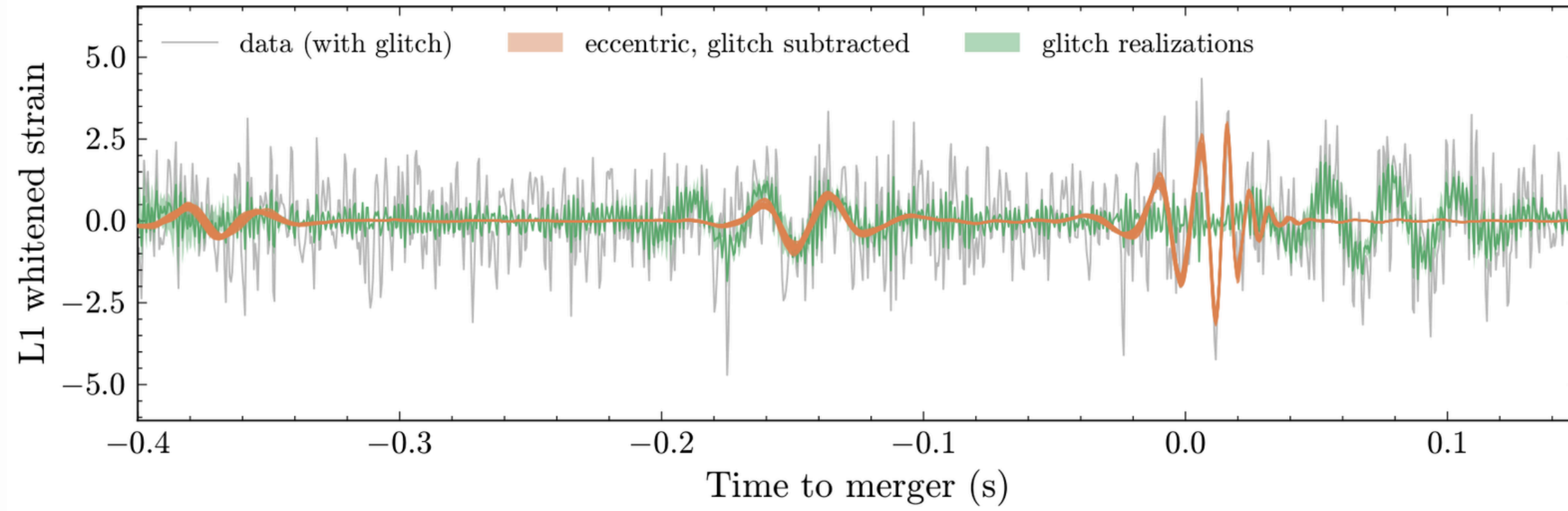
Payne+ 2022

GW190701

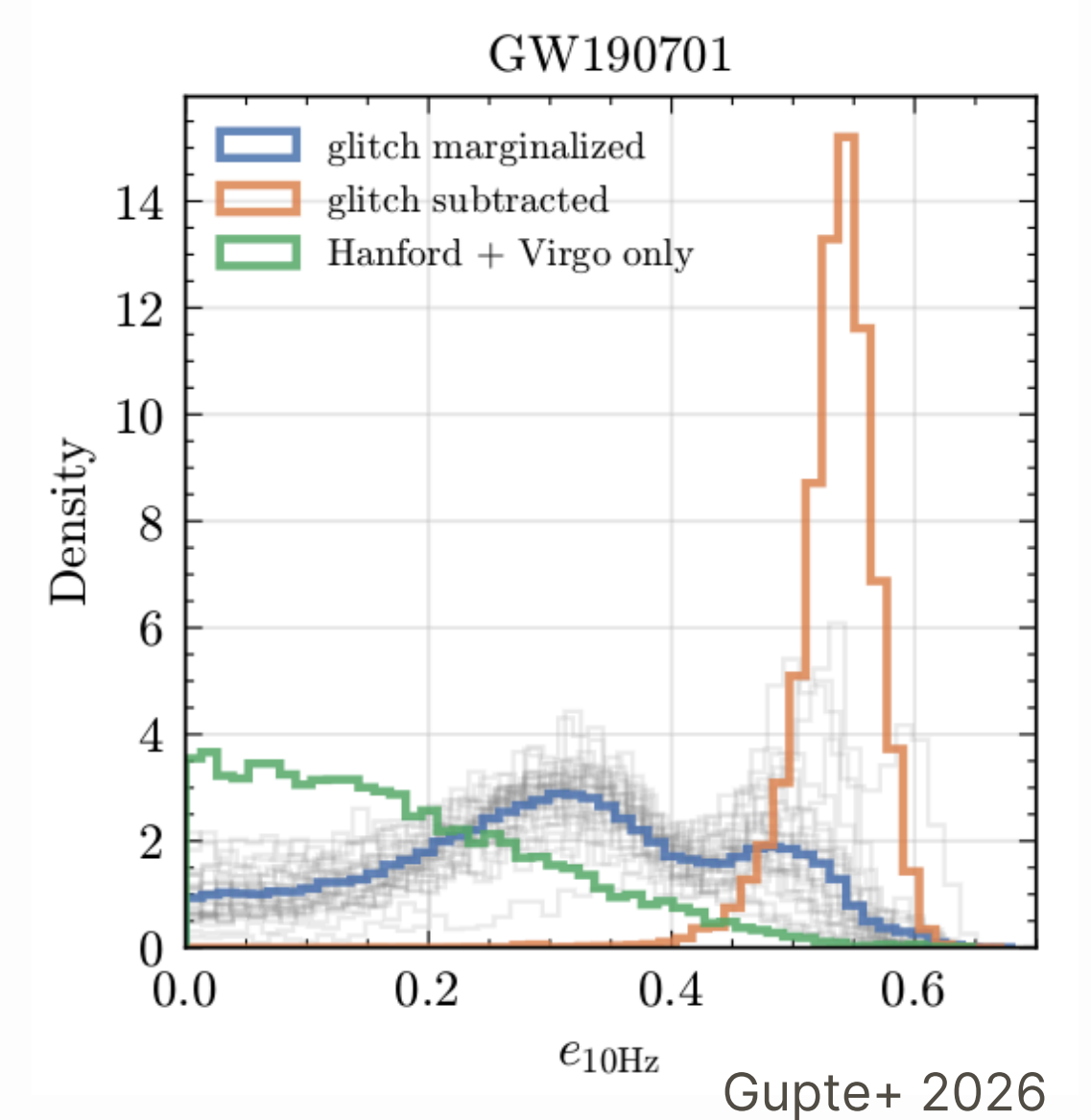
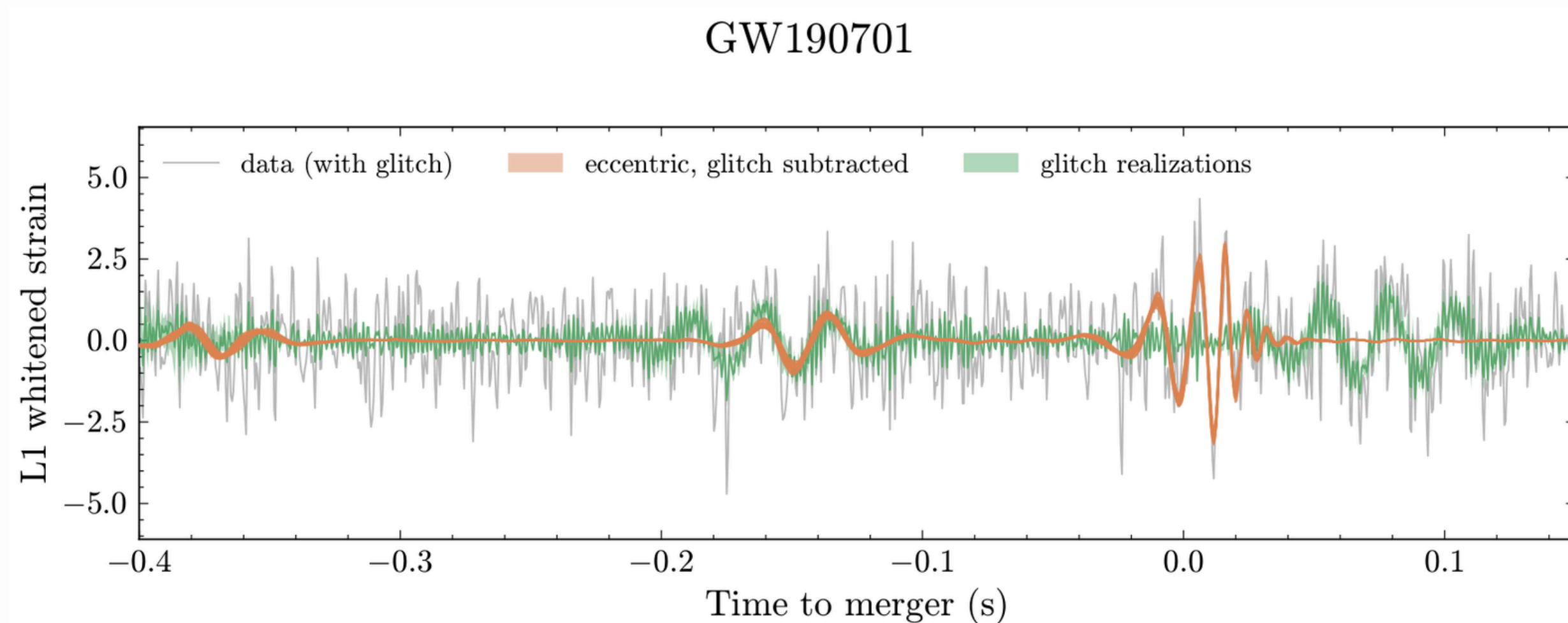


GW190701 - GLITCH

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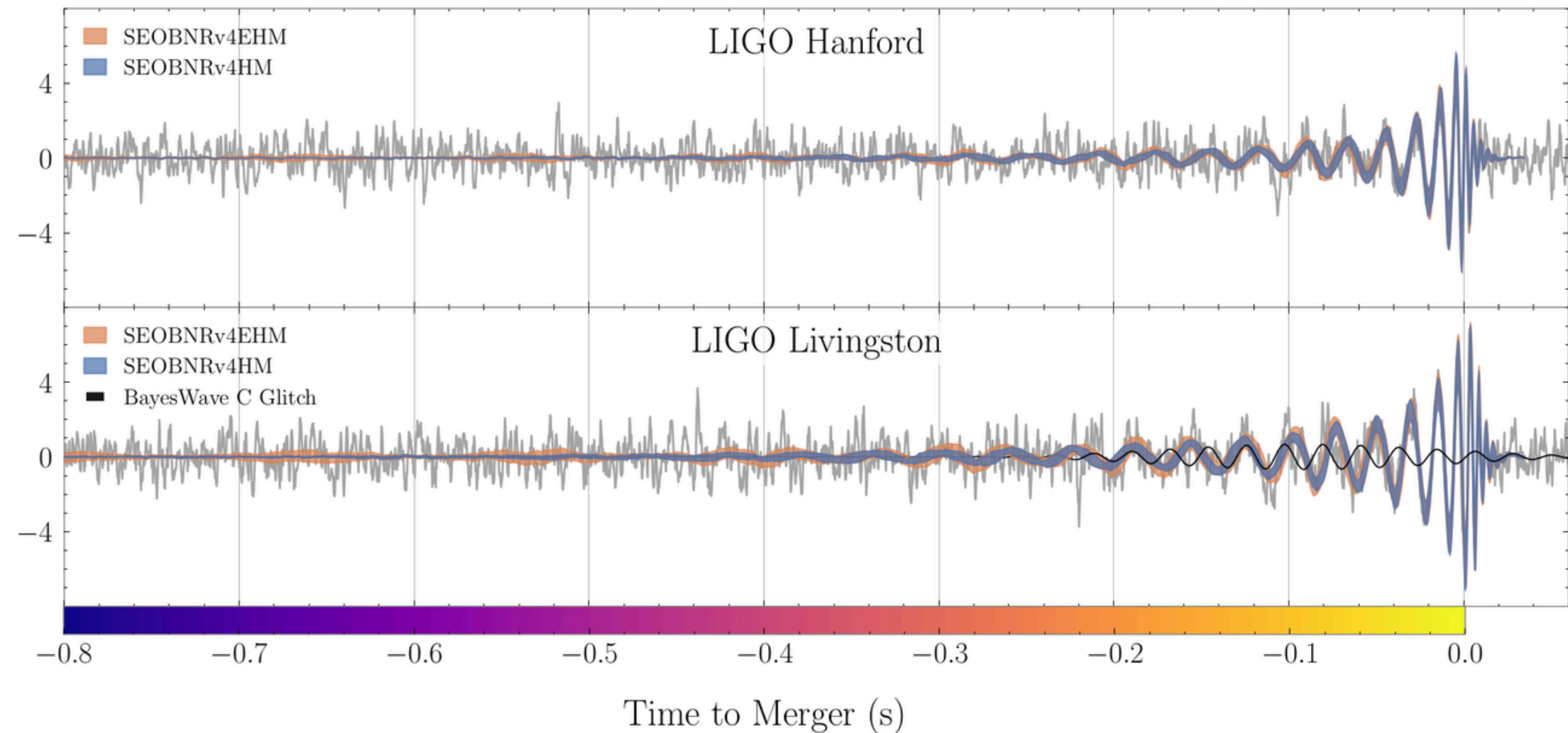
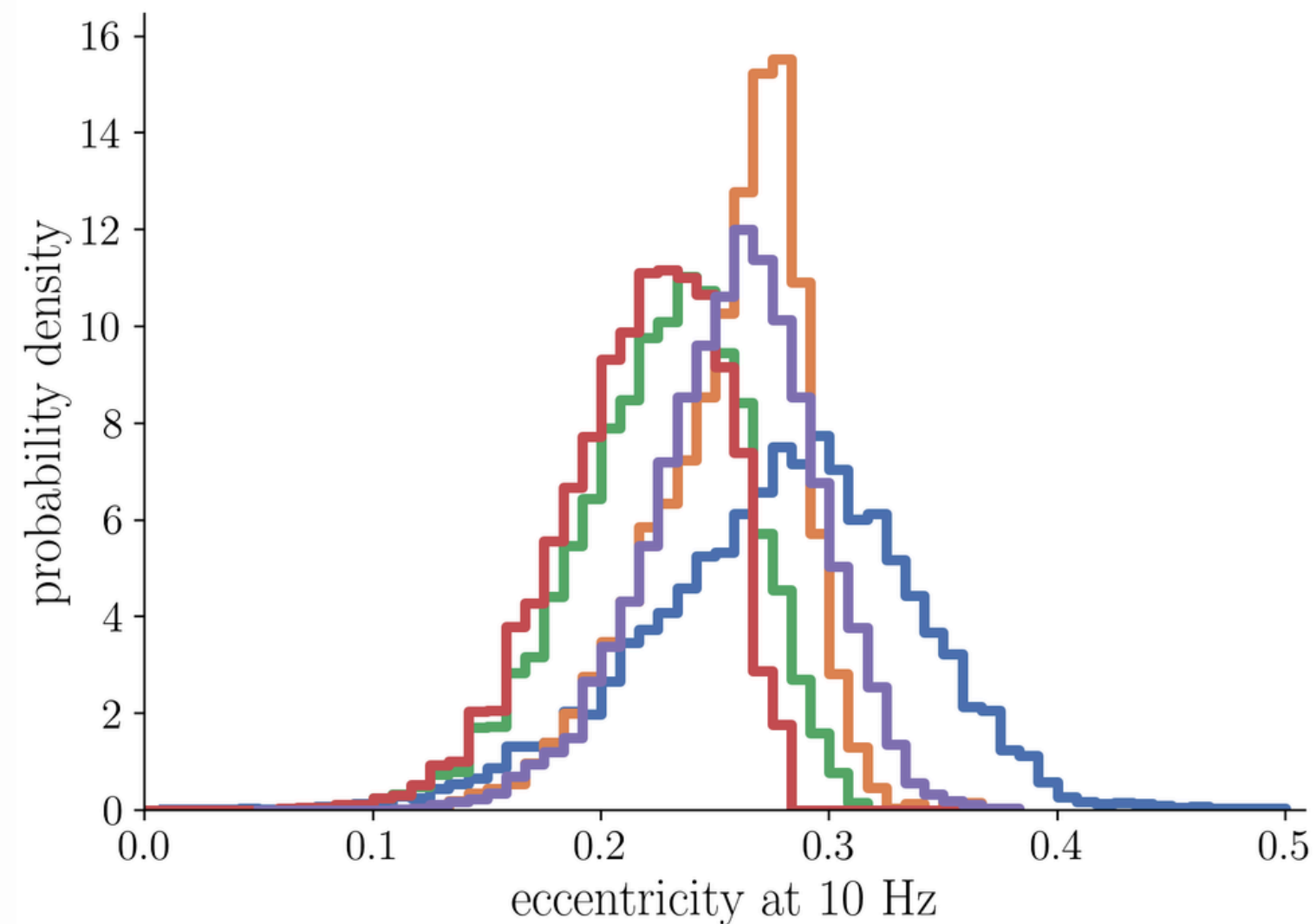
- > Just using one draw of the BayesWave glitch mitigation can massively differ compared to marginalizing over the glitch uncertainty
- > The majority of the eccentric evidence comes from L1.
- > To what extent is the glitch fitting the eccentricity and the eccentricity fitting the glitch?

GW200129

> This is a 70 solar mass BBH. The Bayes factors are 10,000, 1,698, ___ (uniform prior, log uniform prior, expert prior). There is also a glitch in this event!

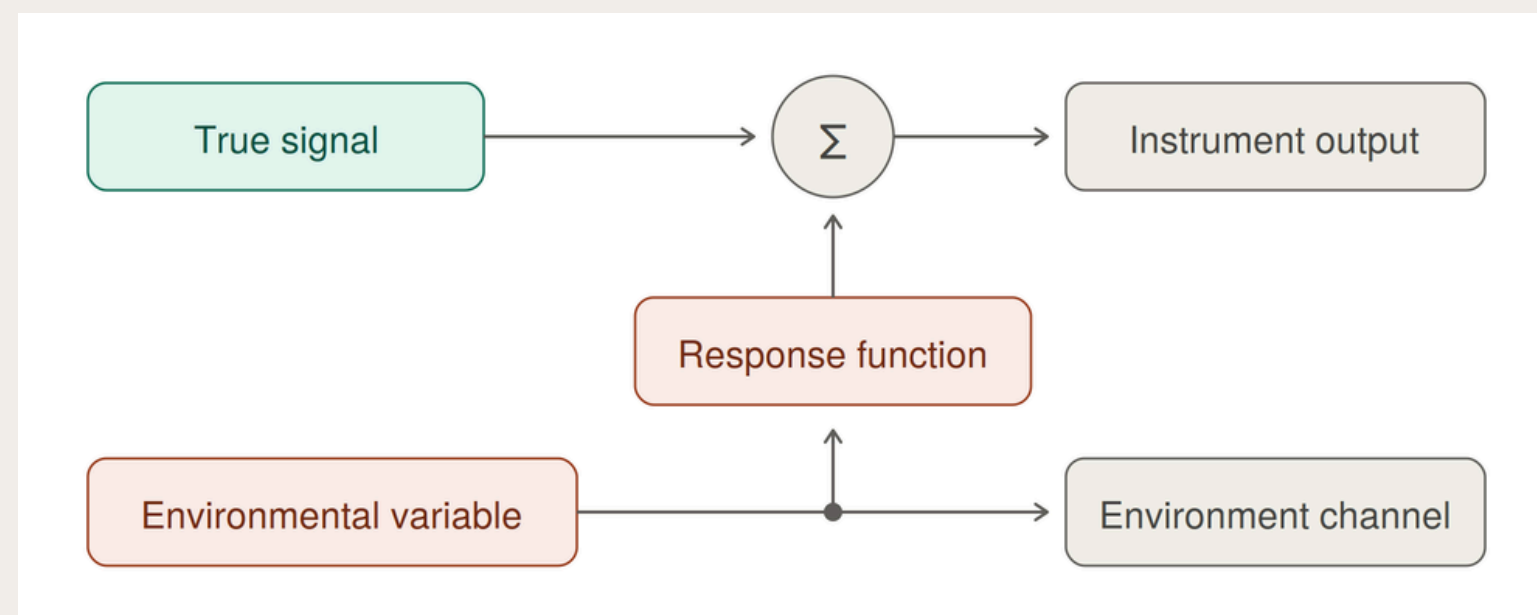
GW200129 — prior: uniform-e

- Gupte+ 2024 (SEOBNRv4EHM, DINGO-IS)
- Planas+ 2025 (IMRPhenomTEHM, bilby)
- Malagon+ 2026 (SEOBNRv5EHM, RIFT)
- Malagon+ 2026 (TEOBResumS-Dali, RIFT)
- Pompili+ 2026 (SEOBNRv6EHM, bilby)

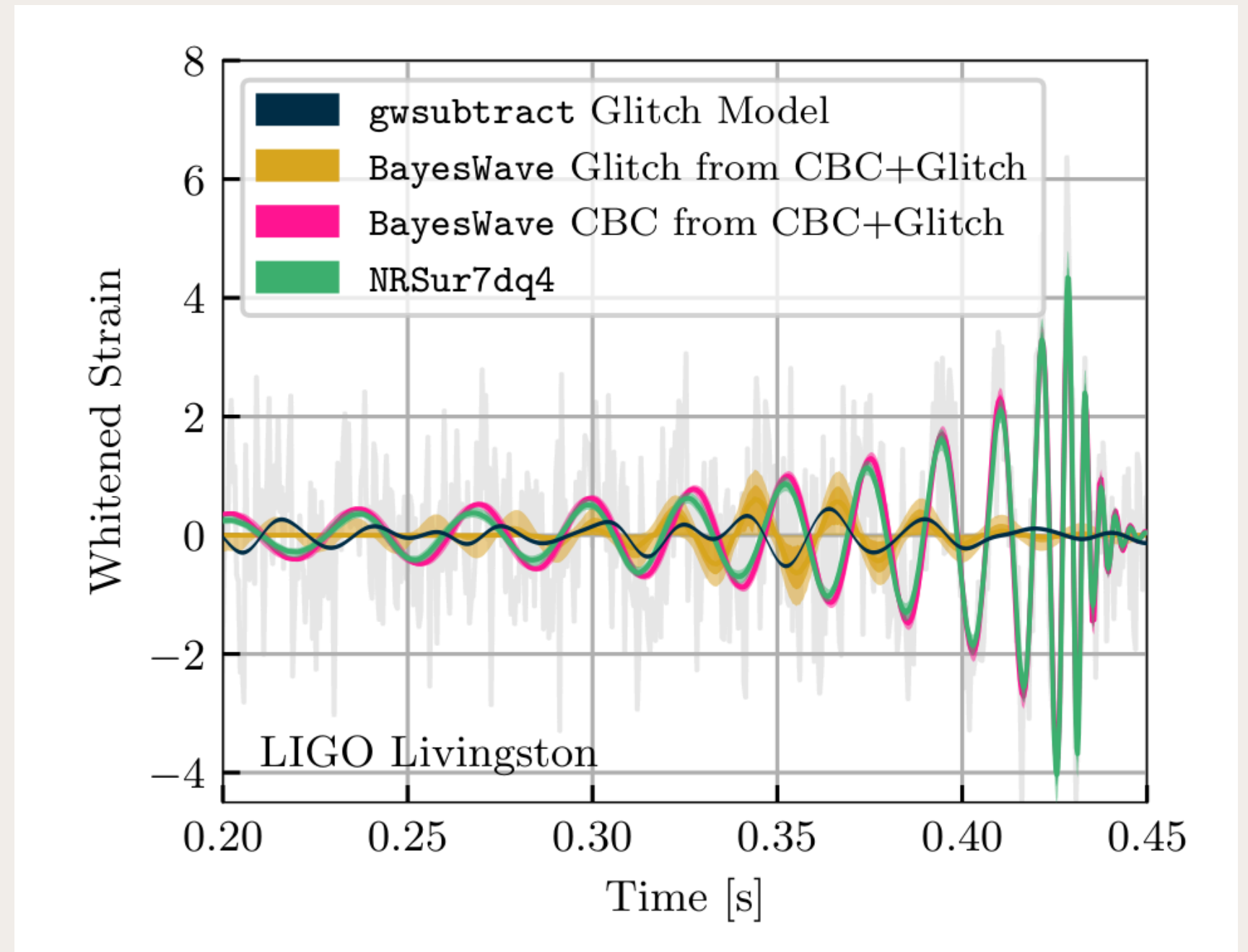


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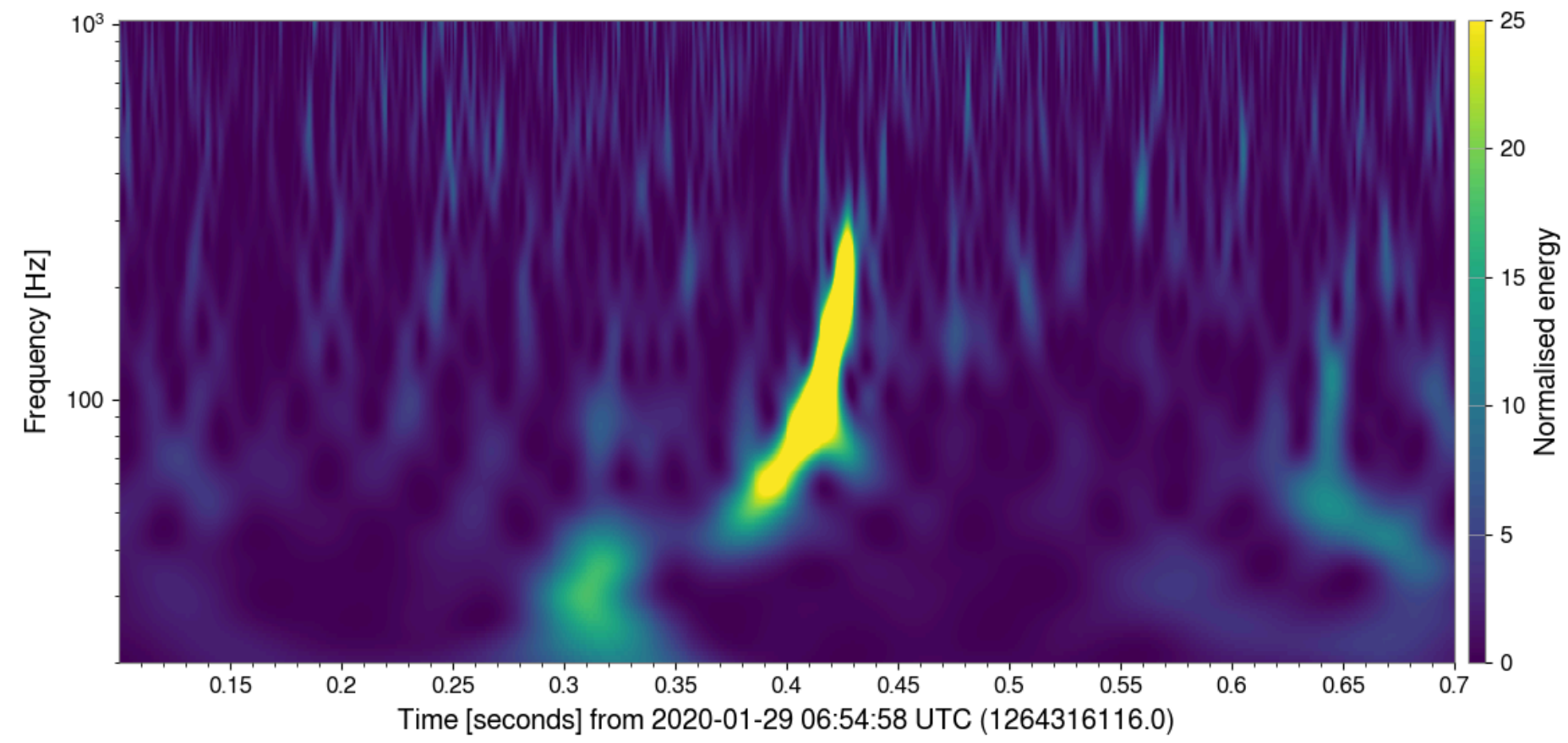
Adapated from Allen+1998



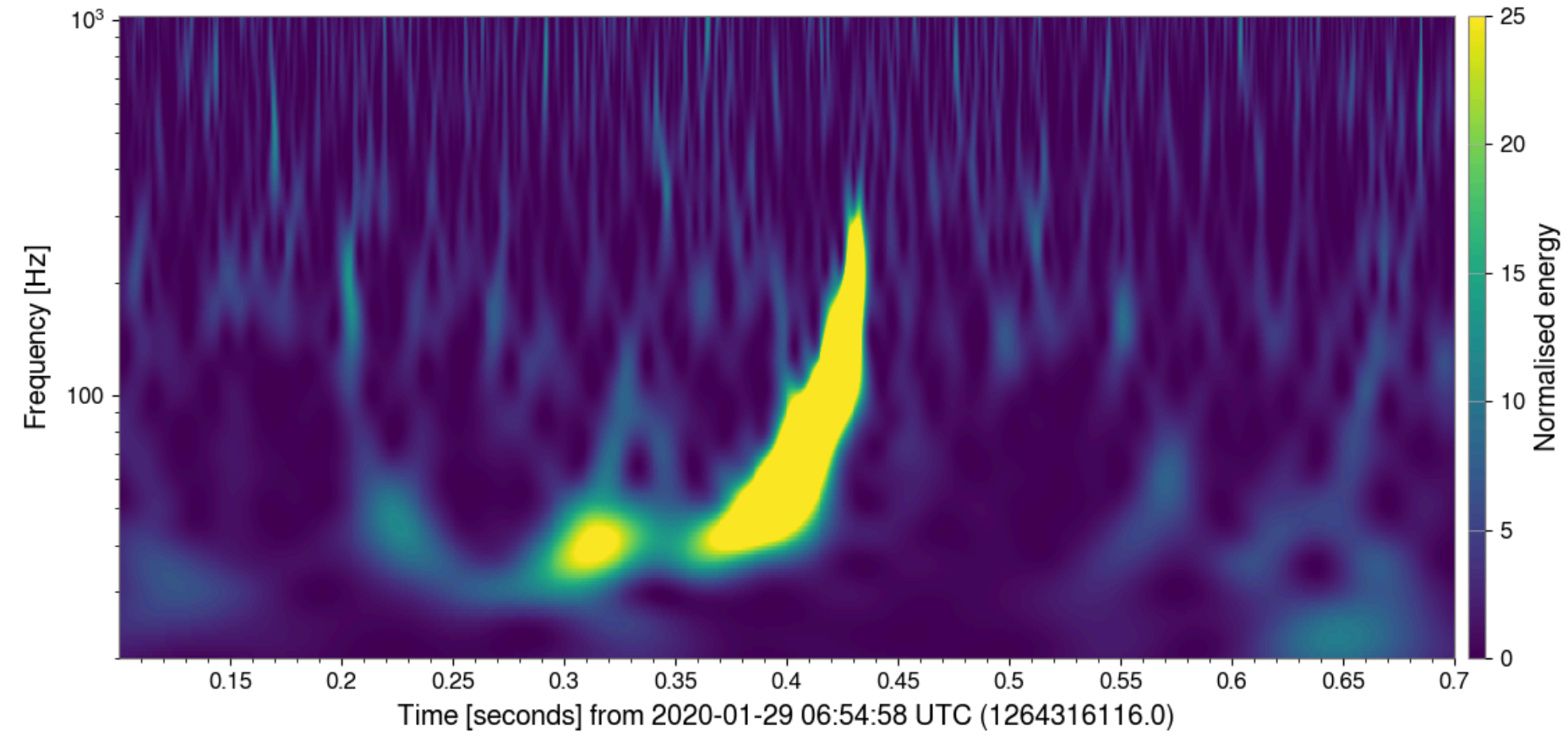
Payne+ 2022

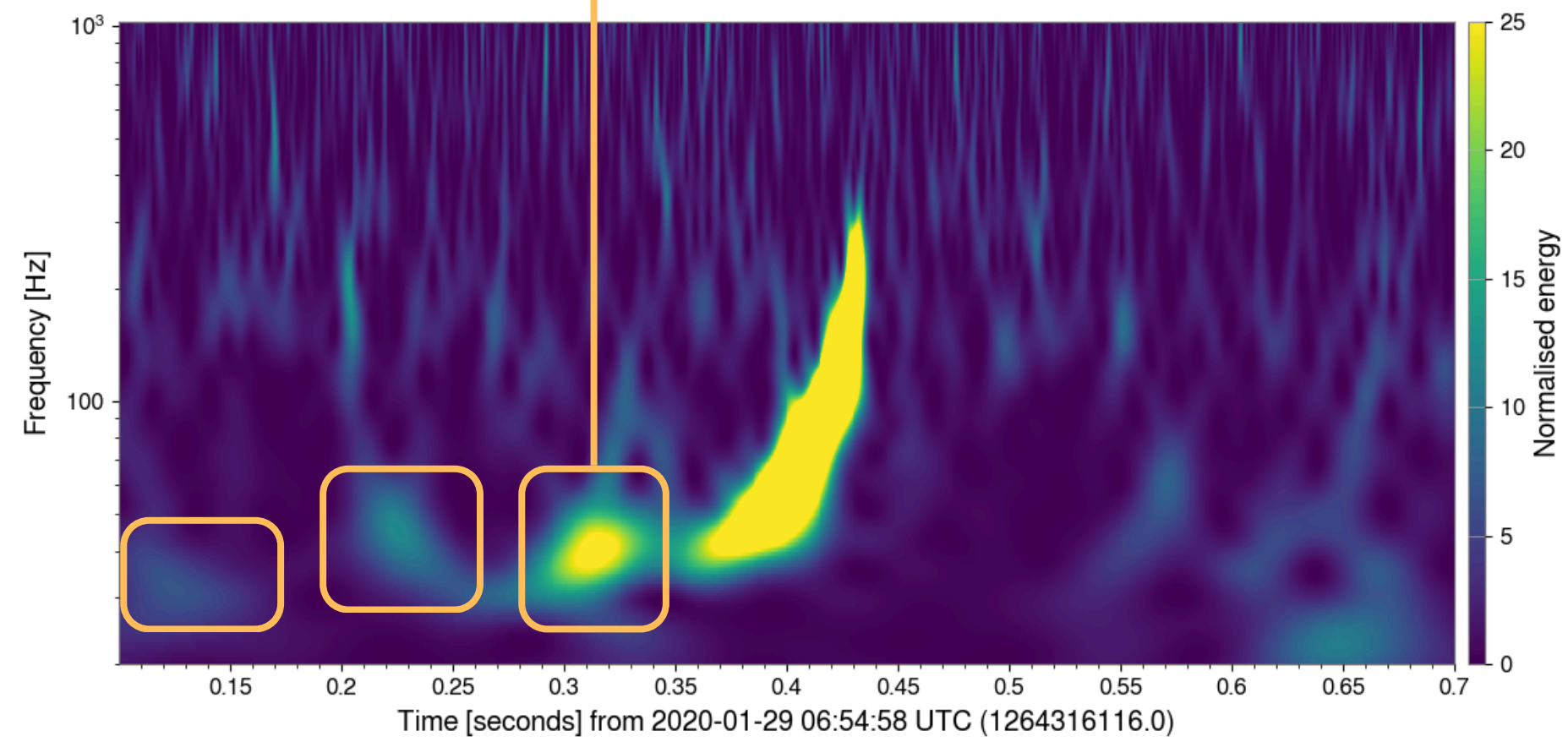
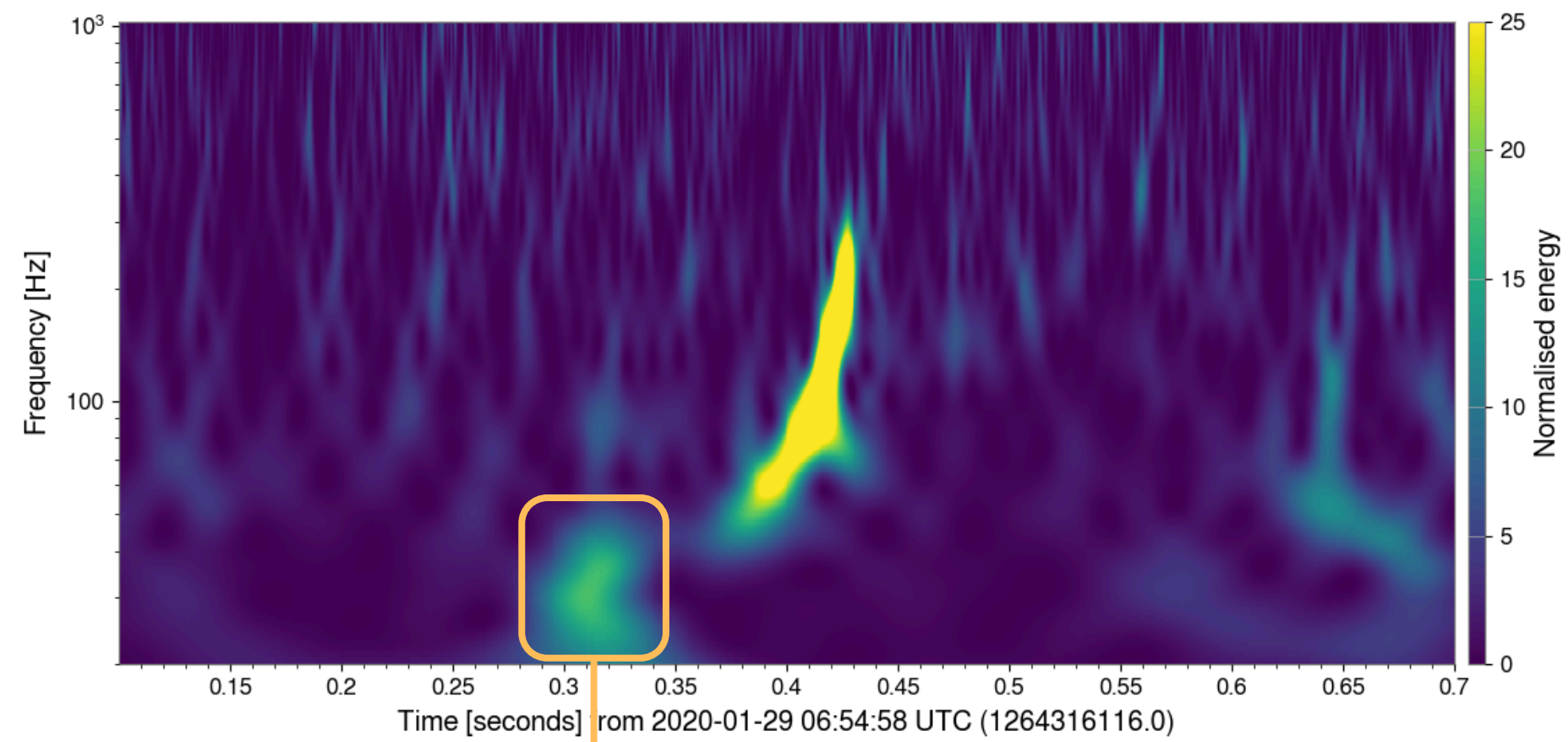
- > Use auxillary channels to deglitch (Allen+ 1999, Davis+ 2022, Macas+ 2024)?
- > **Note, this is waveform model agnostic.** It uses more information, but it doesn't output uncertainty.

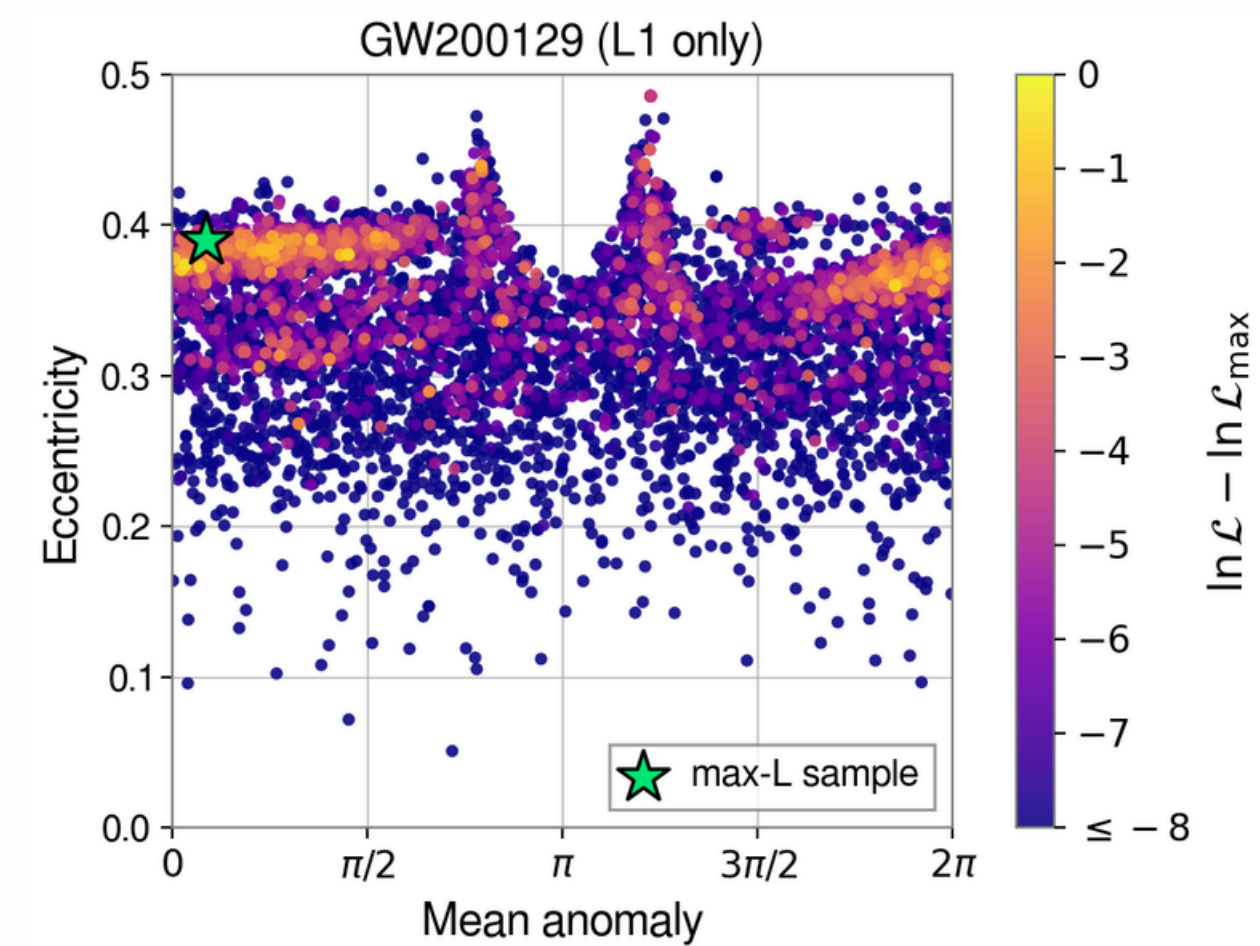
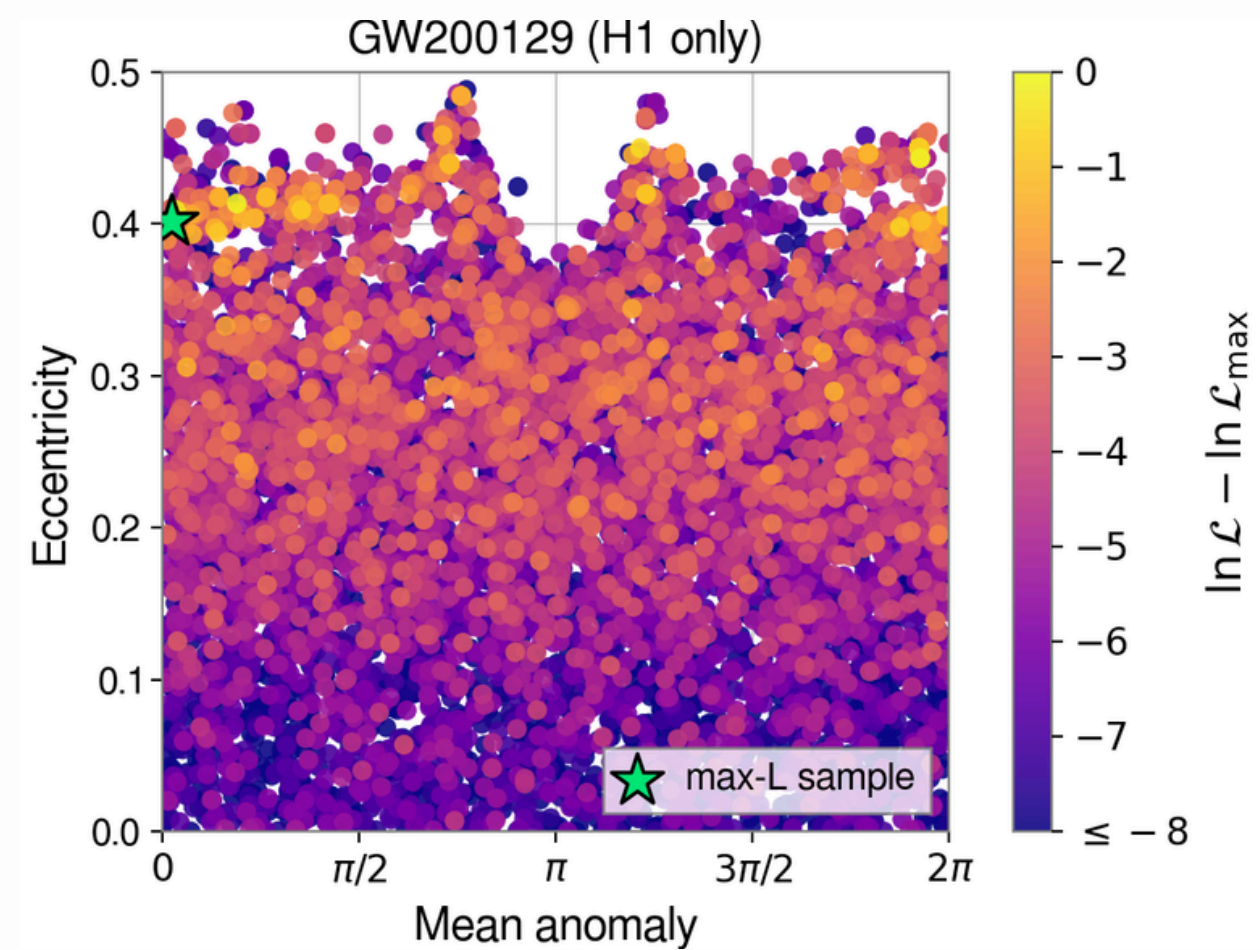
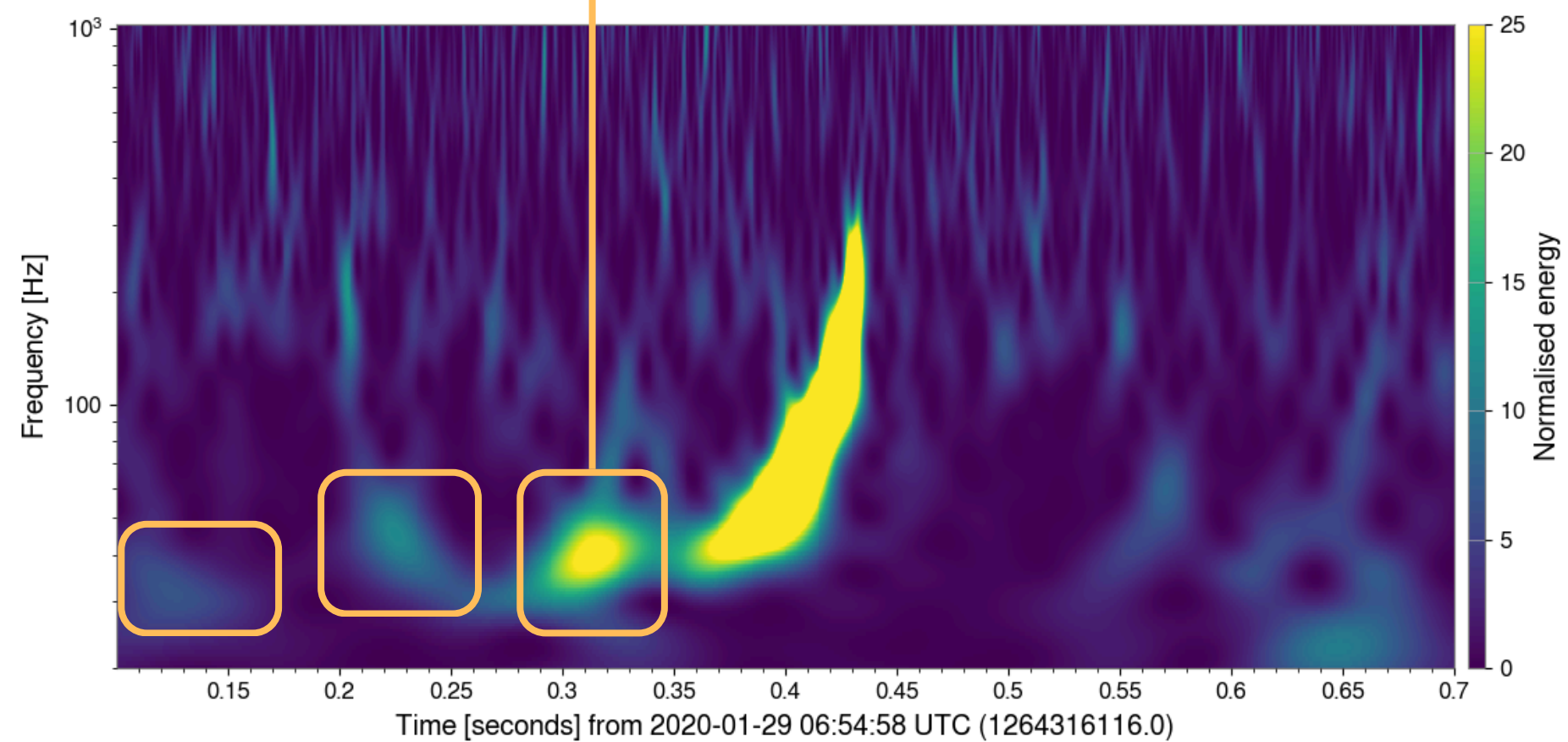
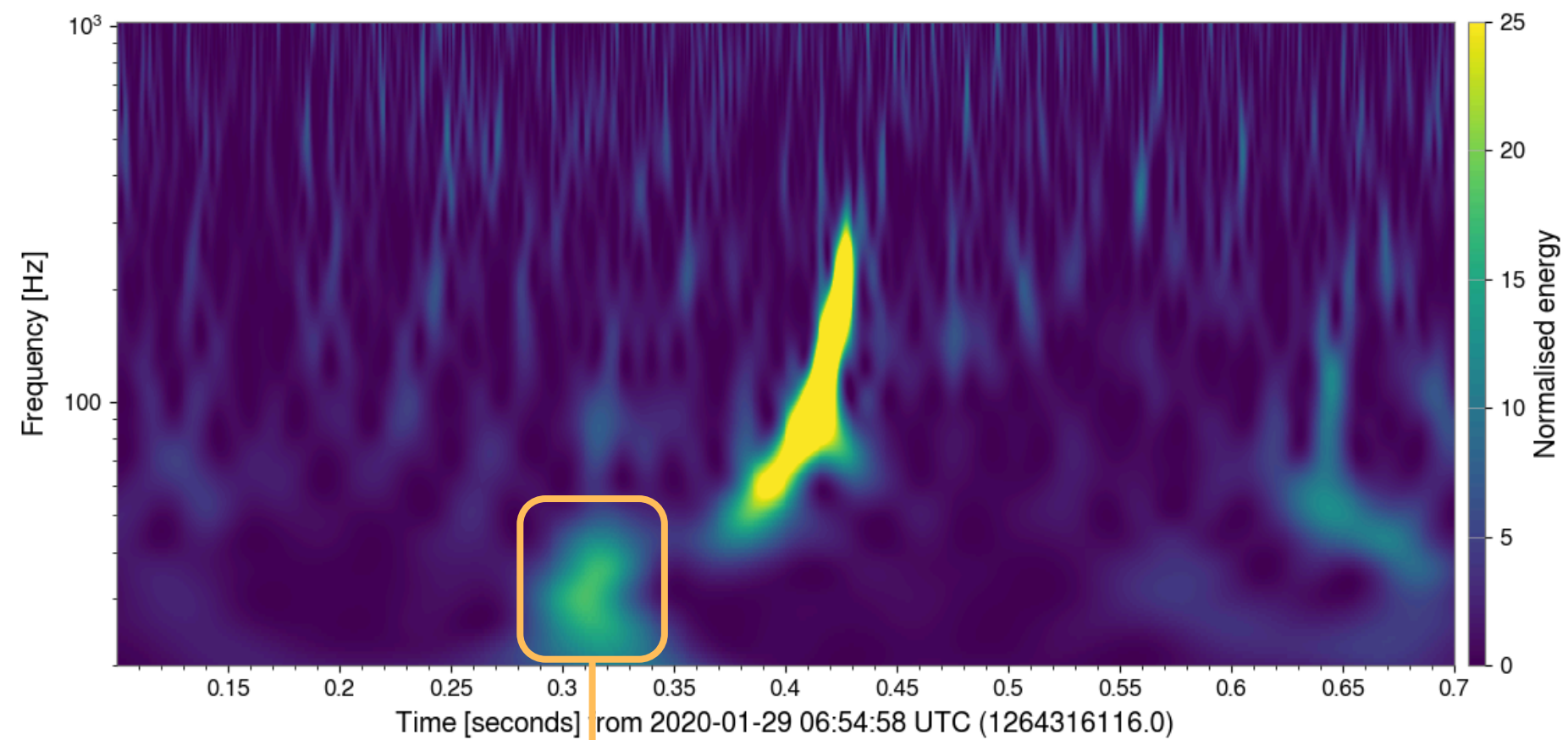
H1



L1





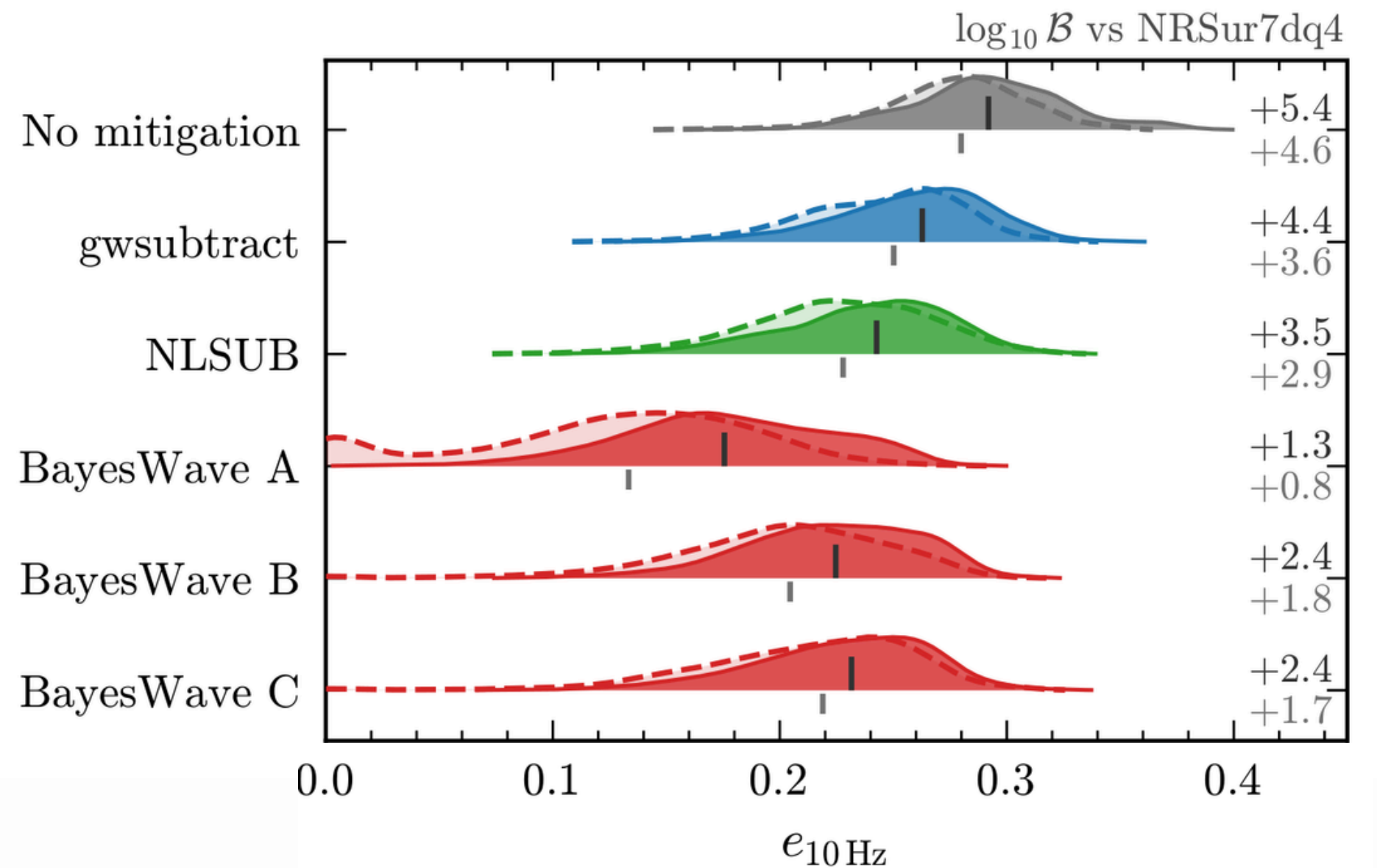


GW200129

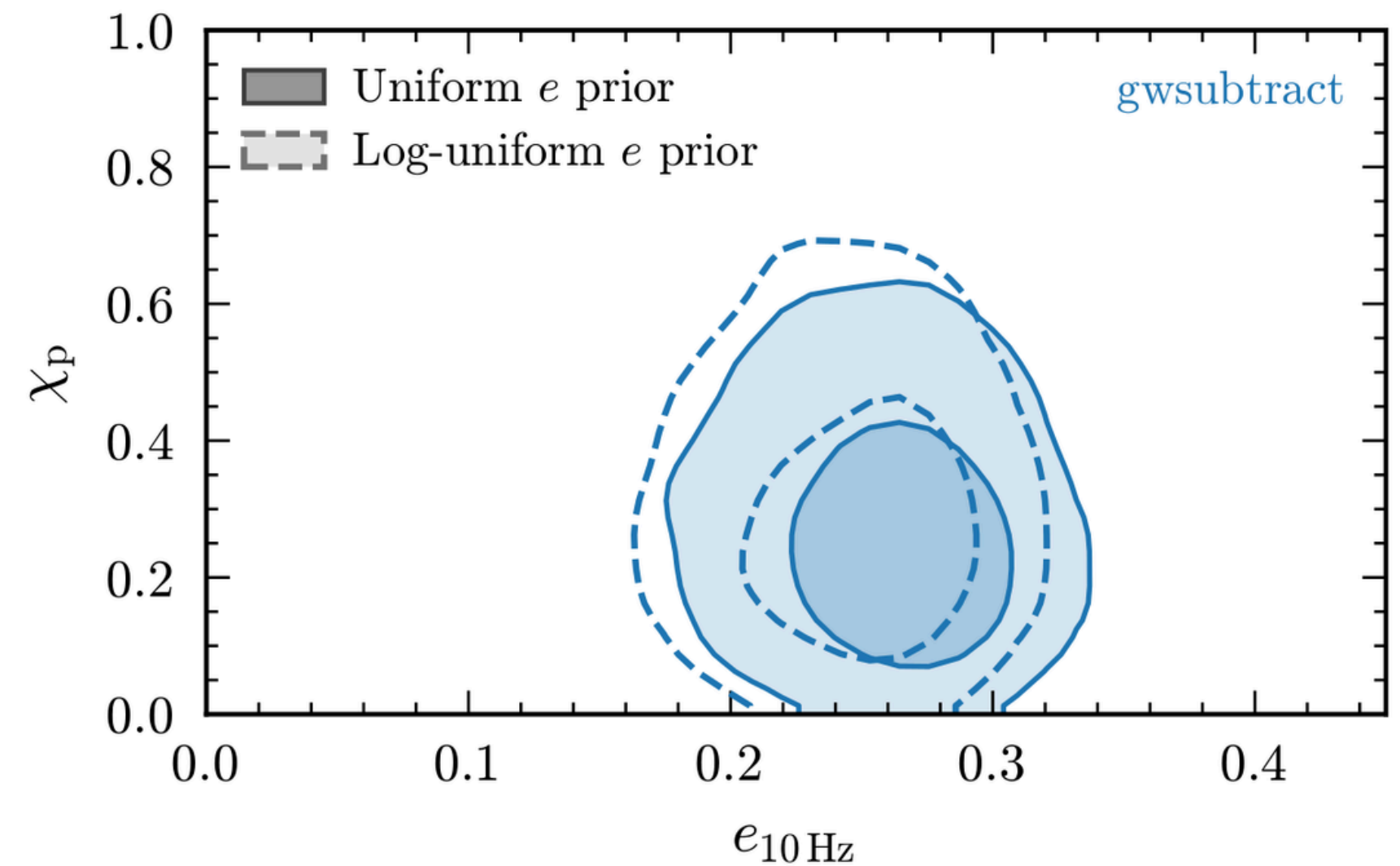
In a recent study Gamboa+ in prep. showed that when analyzed with a **eccentric and spin-precessing waveform model** (SEOBNRv6EPHM) GW200129 **gains even more evidence for eccentricity, but does not show evidence for precession.**

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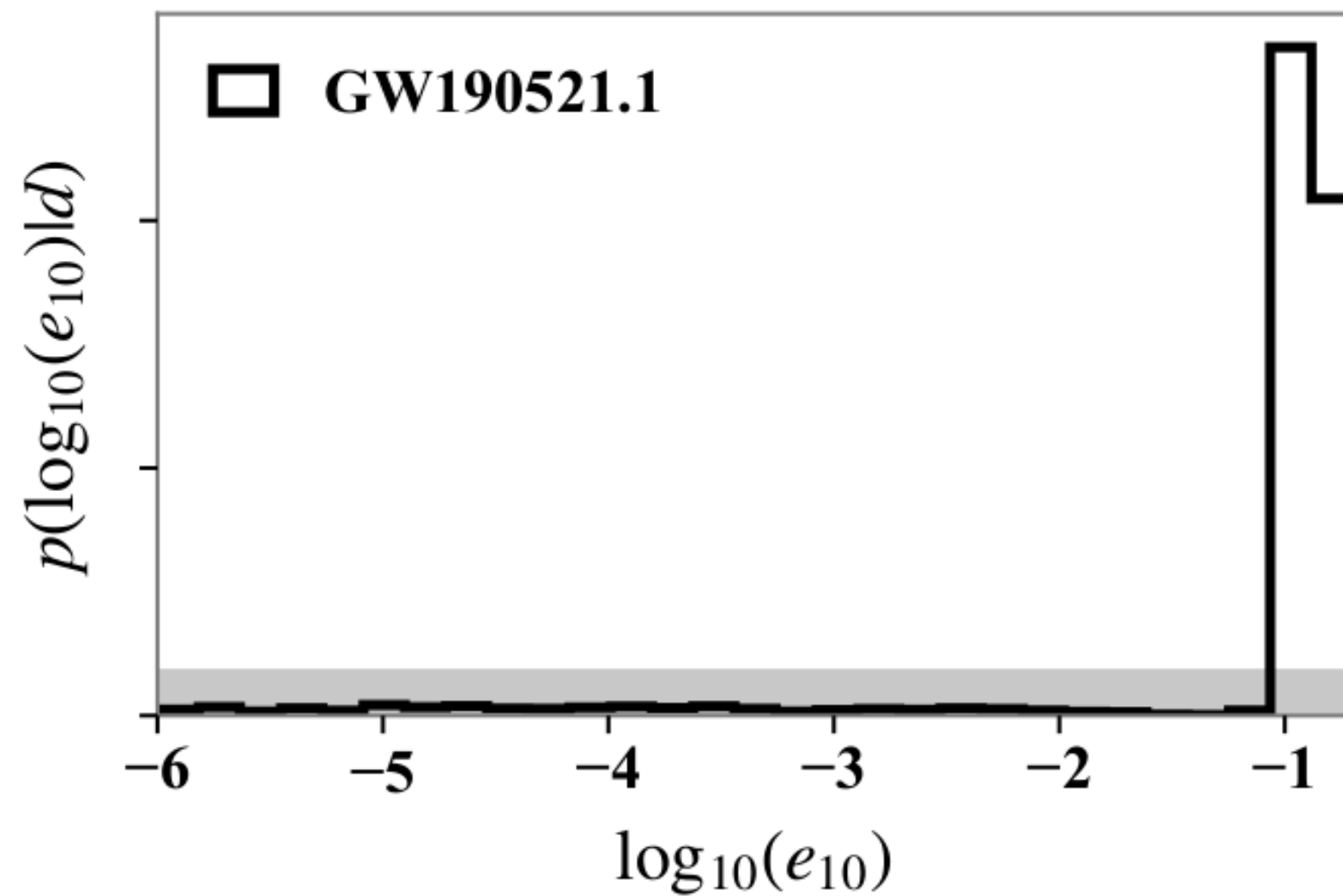
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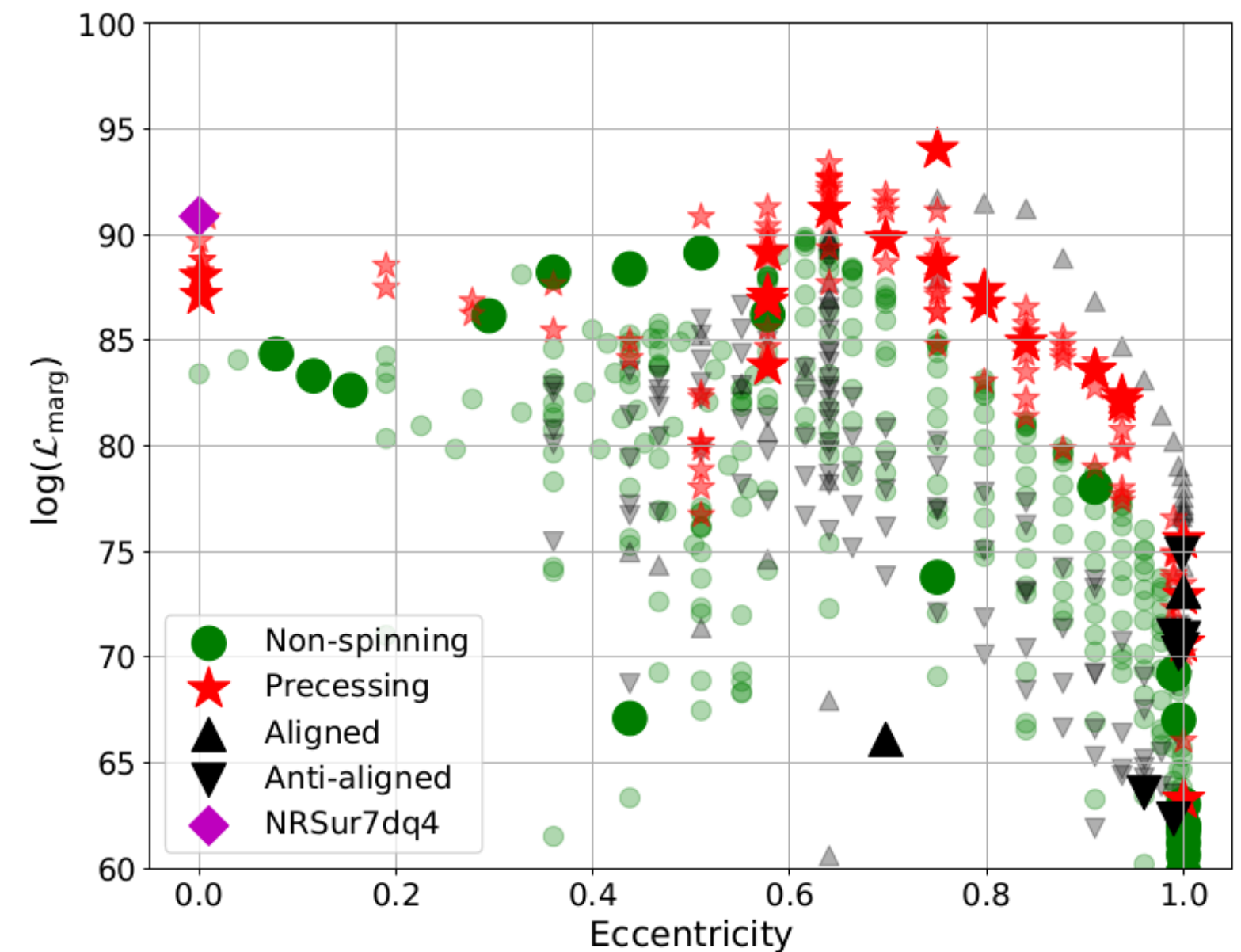
GW190521

> This is a ~146 solar mass BBH.
Bayes factors of 21 (log-uniform prior) from Romero-Shaw+ 2020



Romero-Shaw+ 2020

> There was also a study using NR simulations to compare the eccentric likelihood

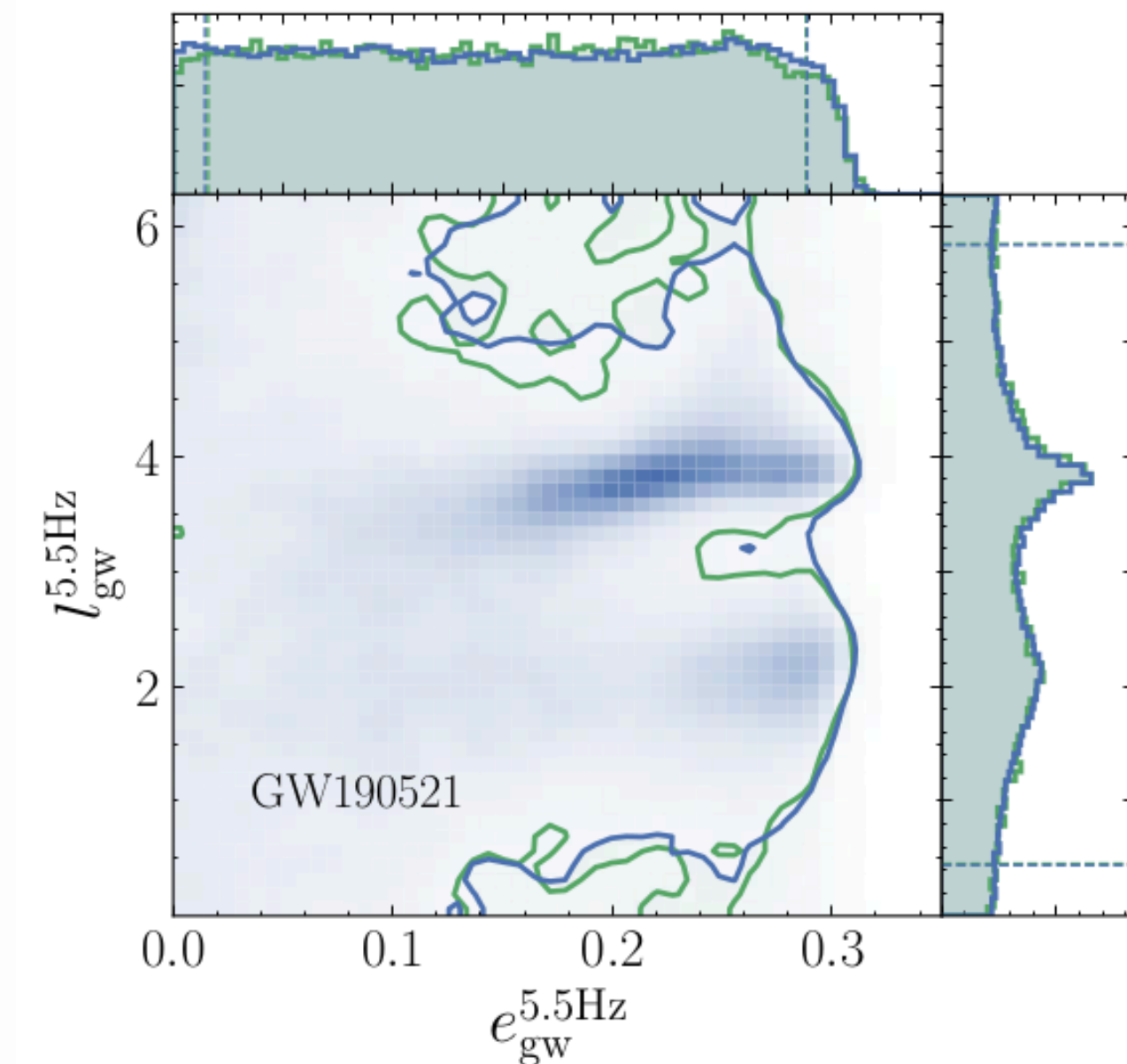


Gayathri+ 2020

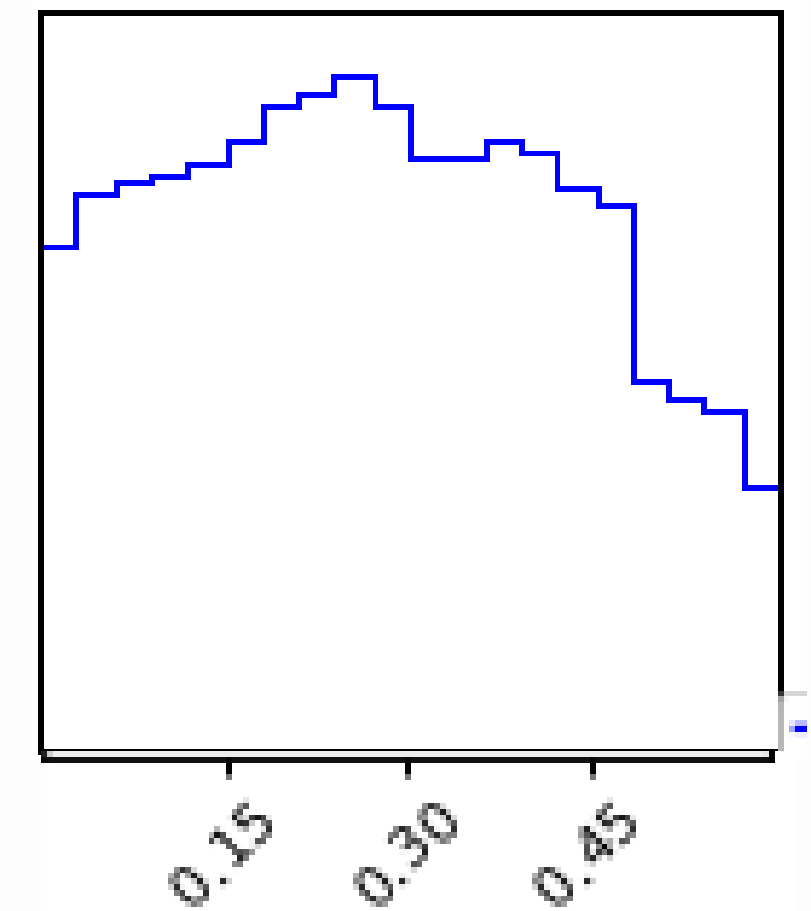
GW190521

> Problem is, subsequent papers were did not find eccentricity in GW190521 (Iglesias+ 2022, Ramos-Buades+ 2023, Gupte+2024, Planas+ 2025, Malagon+ 2026)

> Note, the waveform models used in these papers circularize towards merger



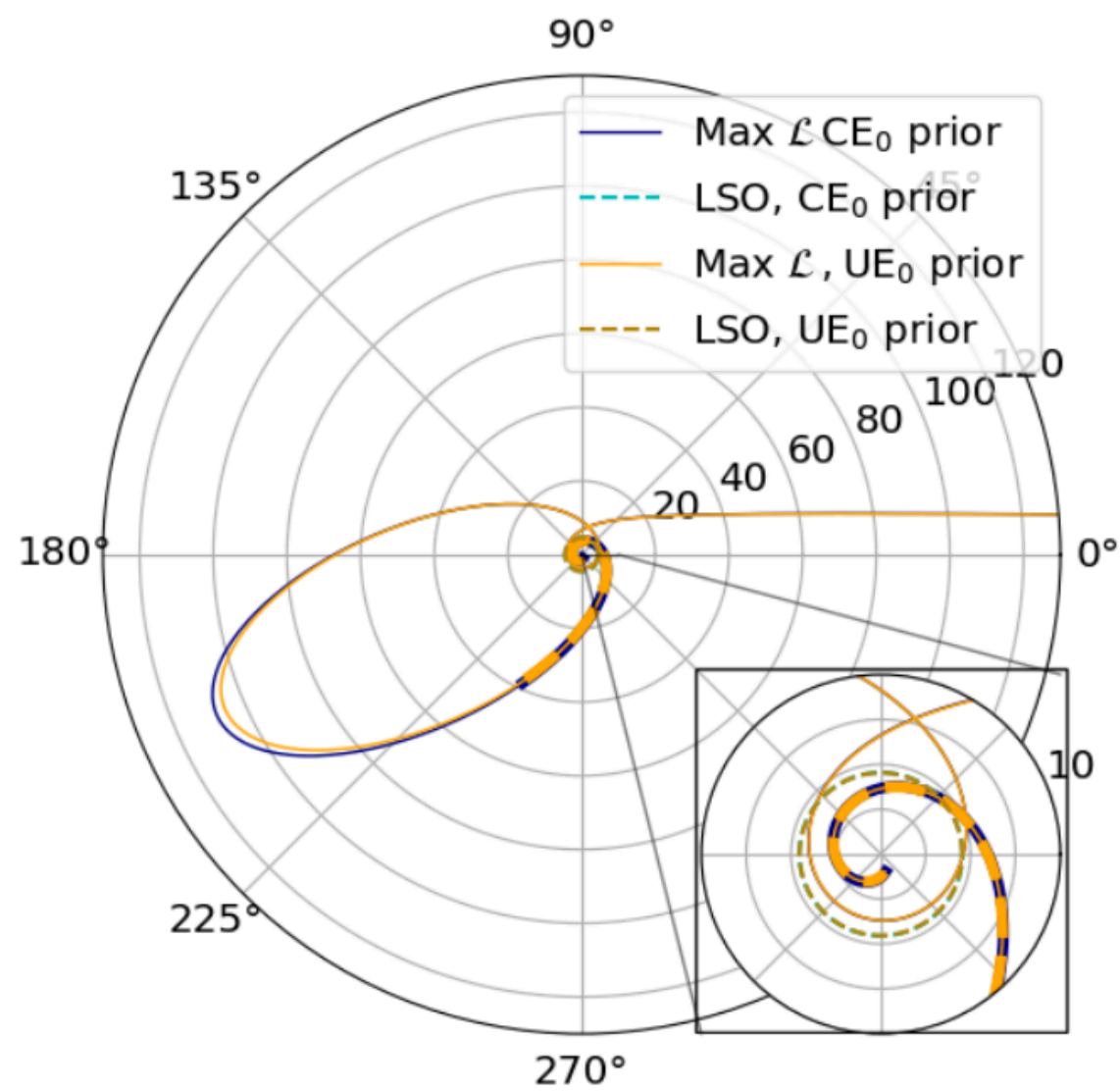
Ramos-Buades+ 2023



Iglesias+ 2022

GW190521 - HYPERBOLIC?

> Gamba+ 2021 proposed that GW190521 was a dynamical capture of two non-spinning black holes with Bayes factors of 4315

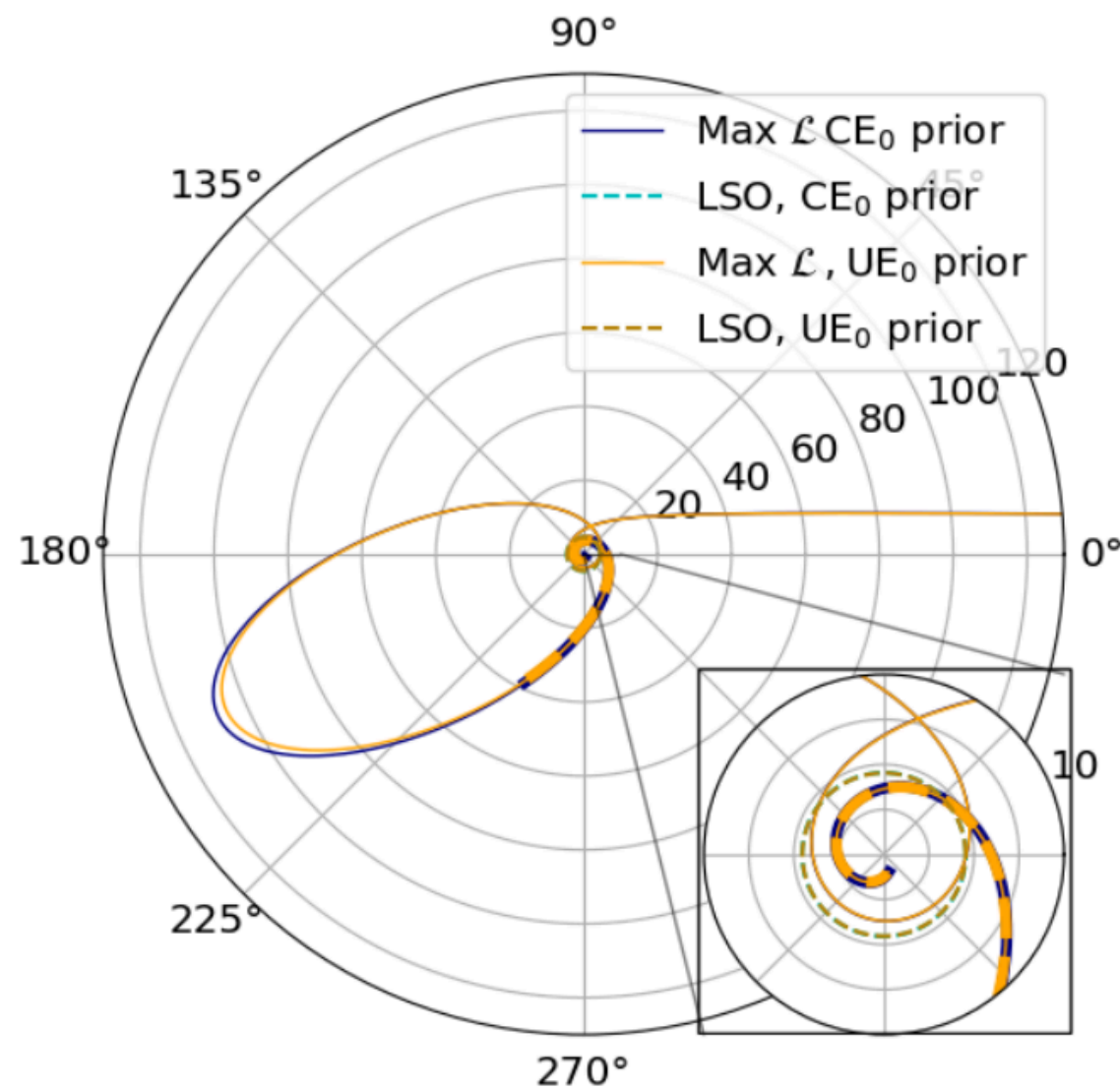


Gamba+ 2021

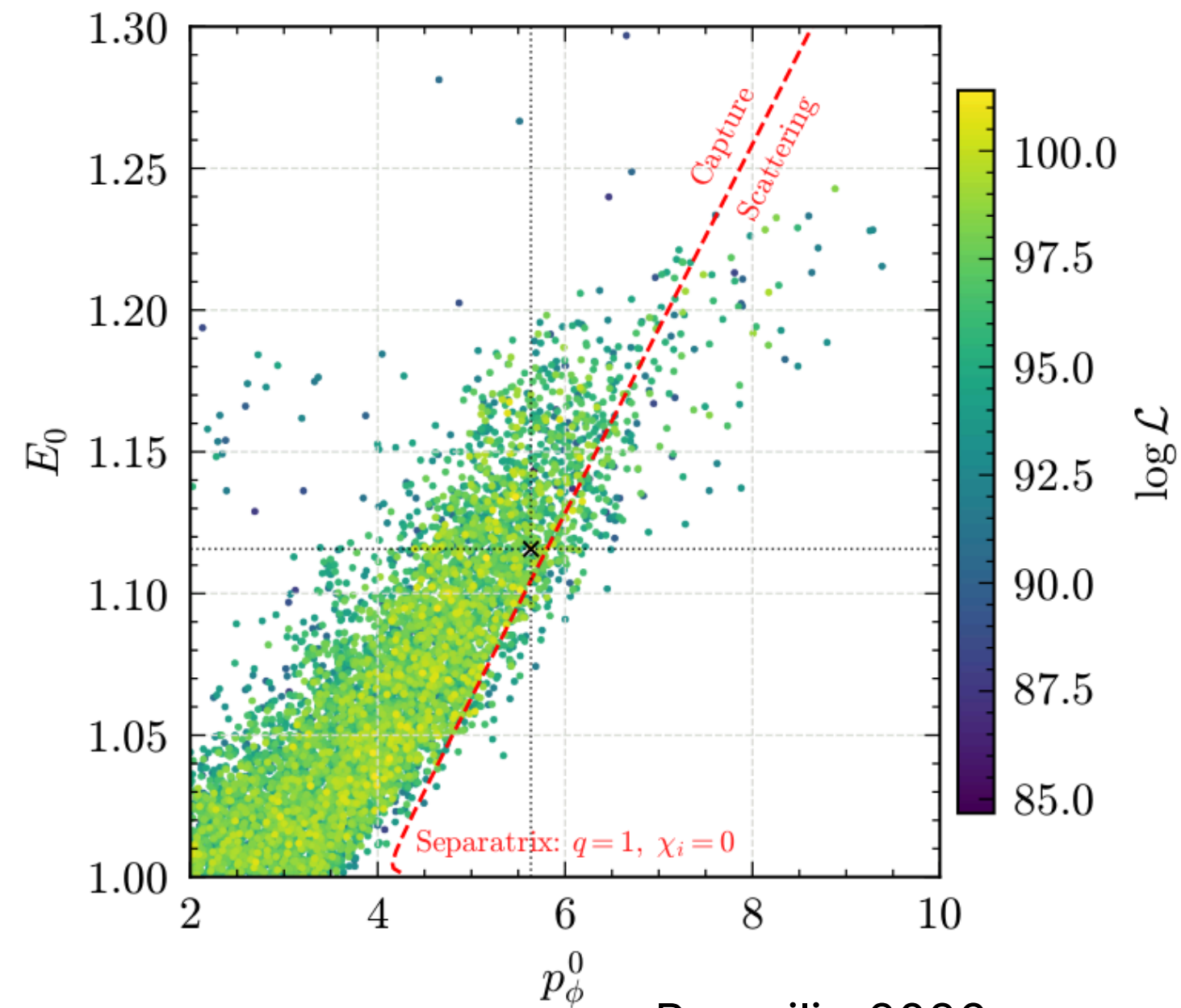
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> Recent works show that when using updated waveform models the Bayes factors become 4 (Pompili+ 2026 SEOBNRv6EHM) and 8 (Lange+ 2026 TEOBResumS-DALI)



Gamba+ 2021

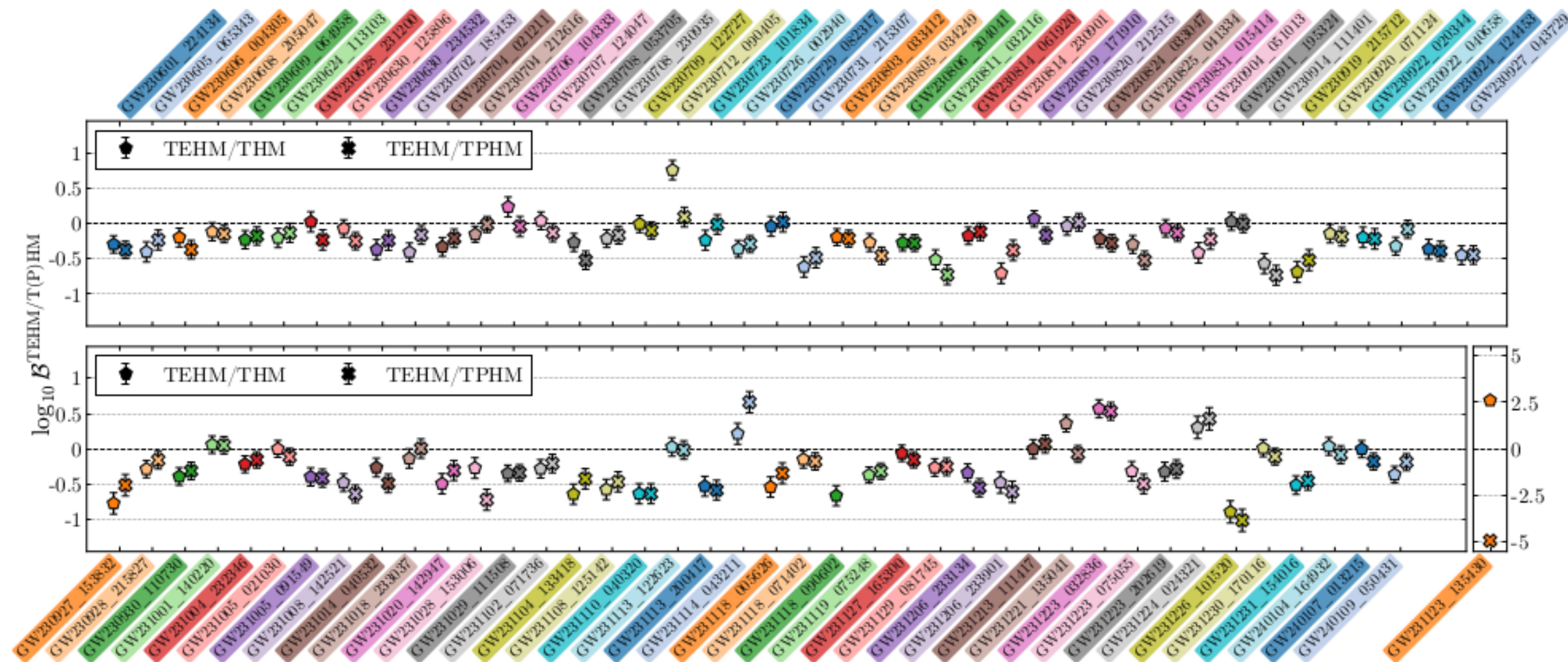


Pompili+ 2026

> These papers sample in the energy and angular momentum at infinity. This allows one to see the posterior compared to a fiducial separatrix

PLENTY OF OTHER EVENTS, THOUGH NONE REACH HIGH SIGNIFICANCE

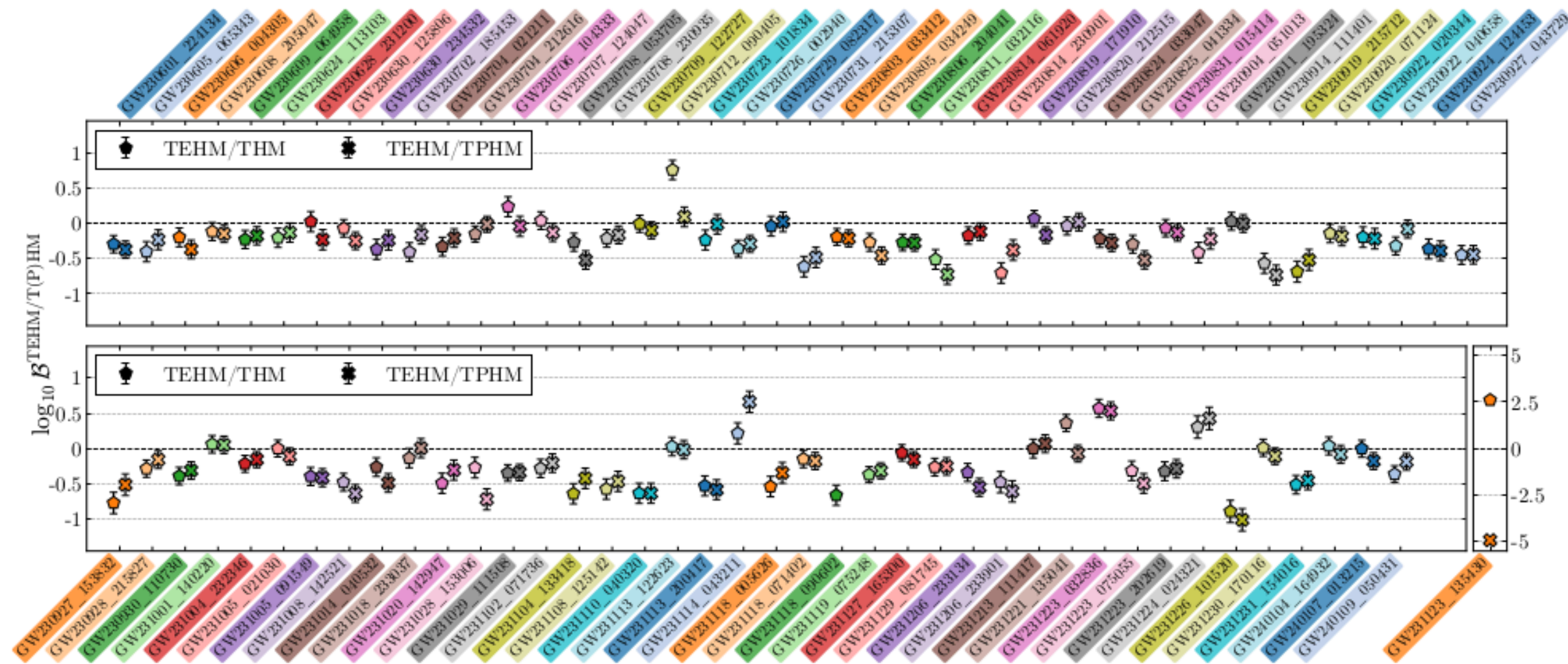
> Three independent catalogs (Xu+ 2025, Gupte+ 2026, Malagon+ 2026) have confirmed O4a has no promising eccentric event



Xu+ 2025

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Xu+ 2025

> That's not to say these events are unimportant! They still constrain the prior

> A few ones with “mild” eccentricity support are GW190620, GW191109, GW231223, GW230706, GW230712.

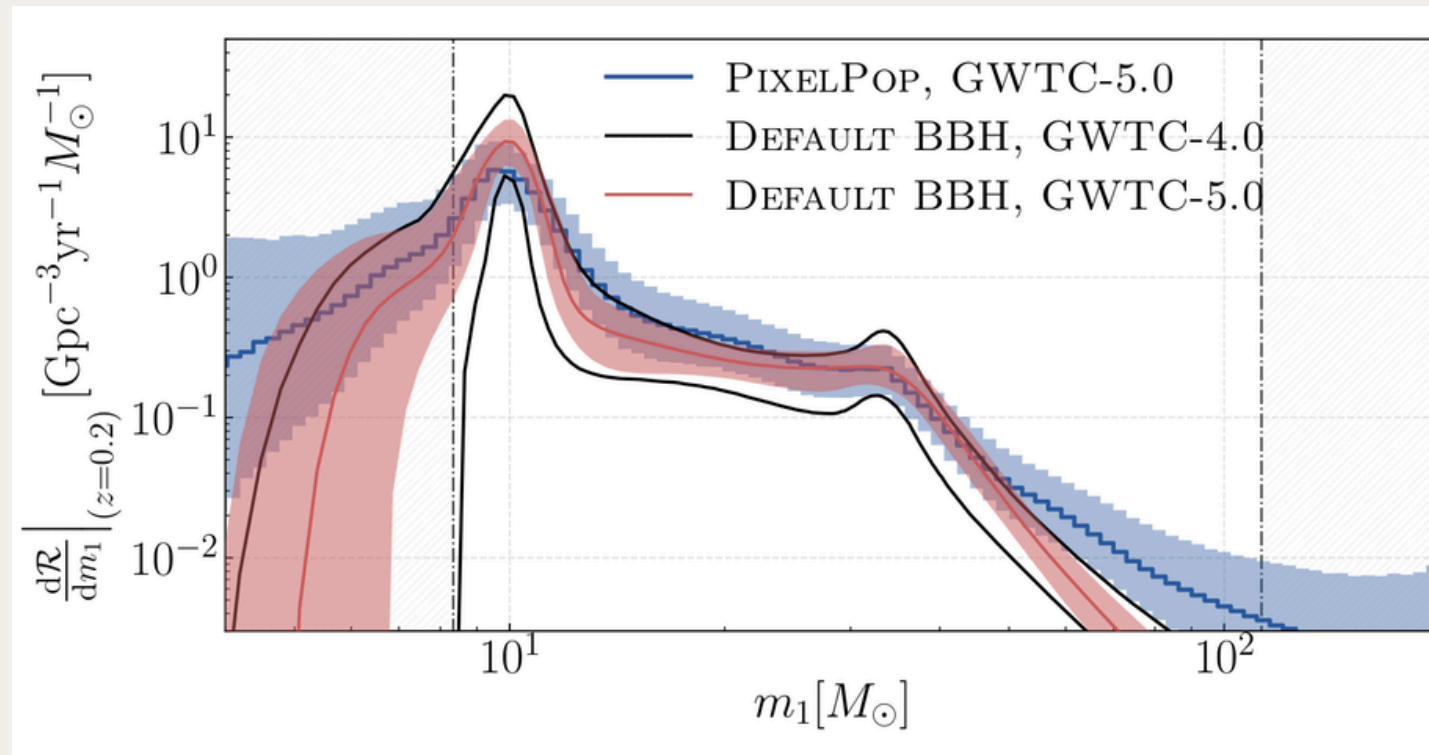
BUT WHAT IF OUR PRIORS ARE ALL WRONG?

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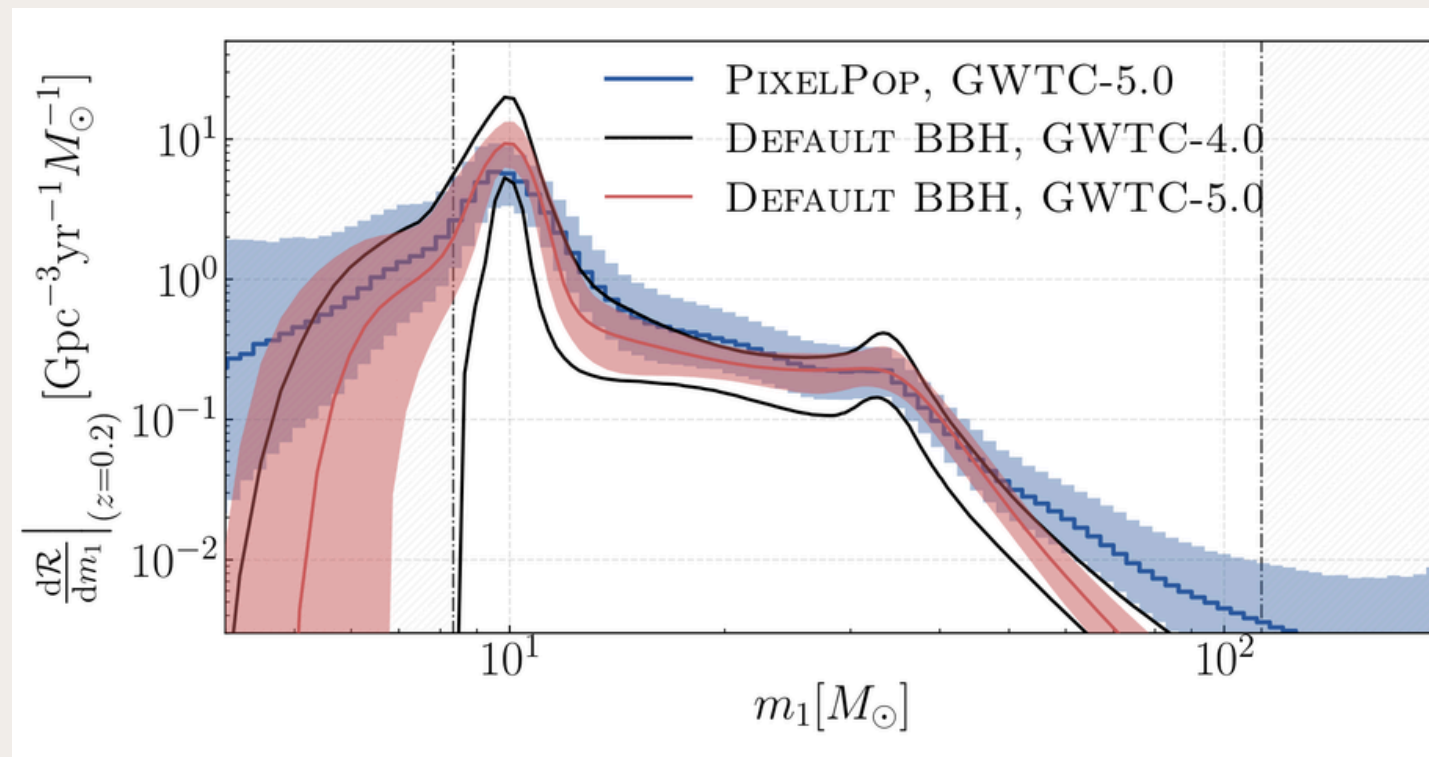


This [accounts for the look elsewhere effect](#). You are looking at a “global” property of the data. We can compare astrophysical predictions directly to this

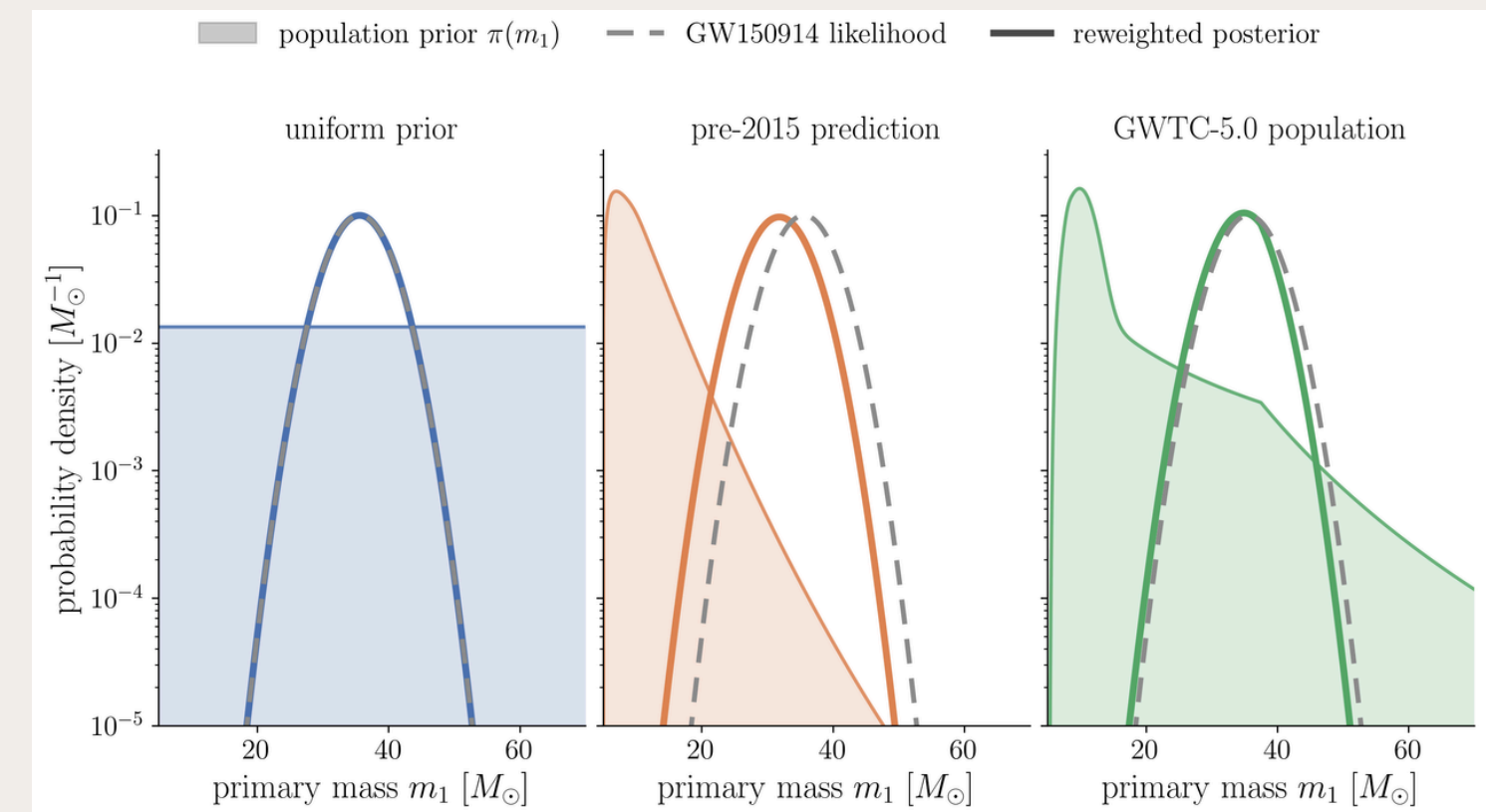
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Step 2: Update our belief about the individual events



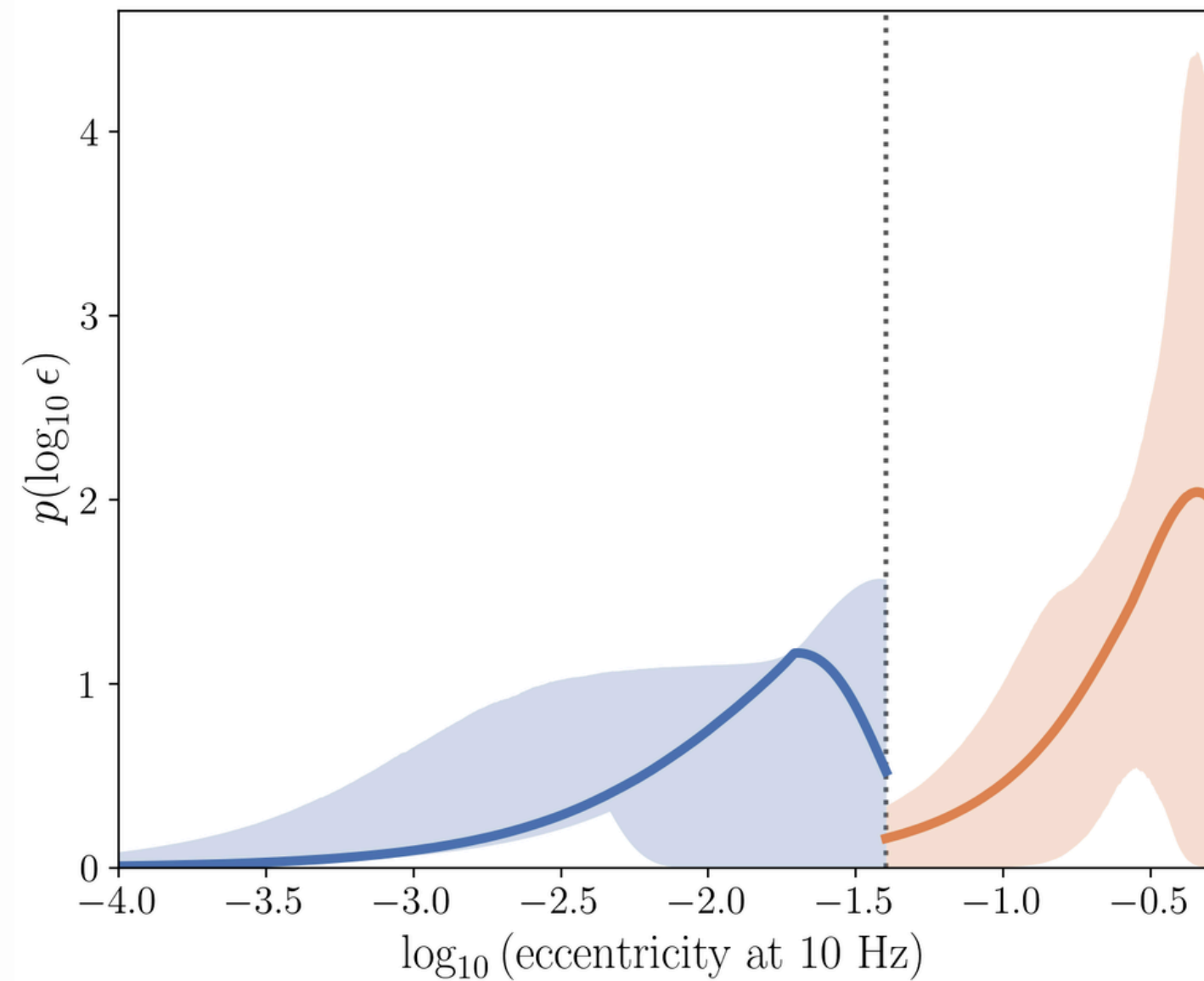
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By using data from other events, we can find $p(\text{eccentric} \mid \text{all GW data, our astrophysical priors})$. This **makes mildly eccentric events important** as they affect the population!

STUDIES LOOKING AT THE POPULATION PROPERTIES

Zeeshan+ 2026 defined a mixture between two Gaussians as a population model.

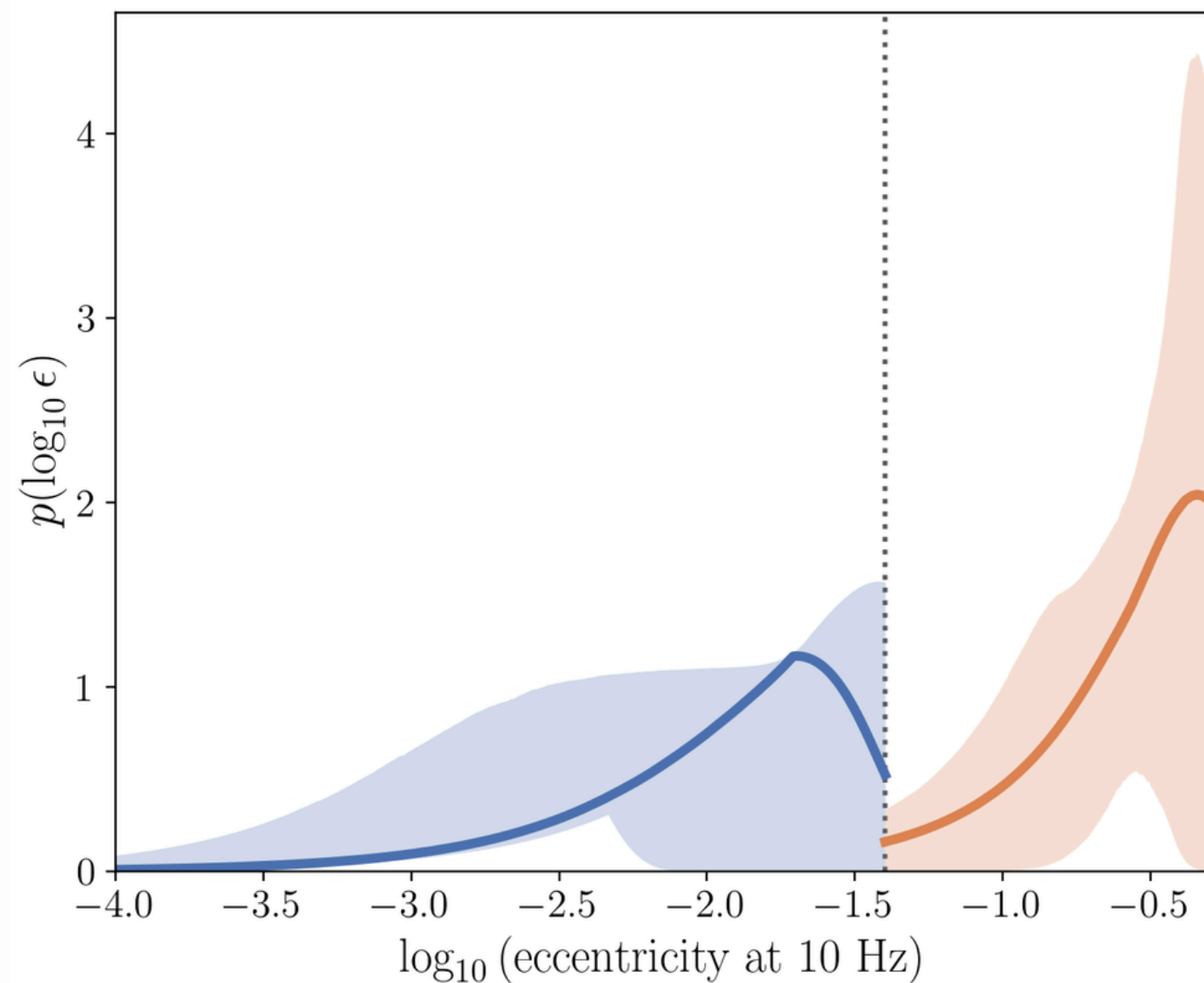
$$\pi(\epsilon) = (1 - \zeta_{\epsilon}) \cdot p(\epsilon \mid 0, \sigma_{\text{circ}}) + \zeta_{\epsilon} \cdot p(\epsilon \mid \mu_{\epsilon}, \sigma_{\epsilon}).$$



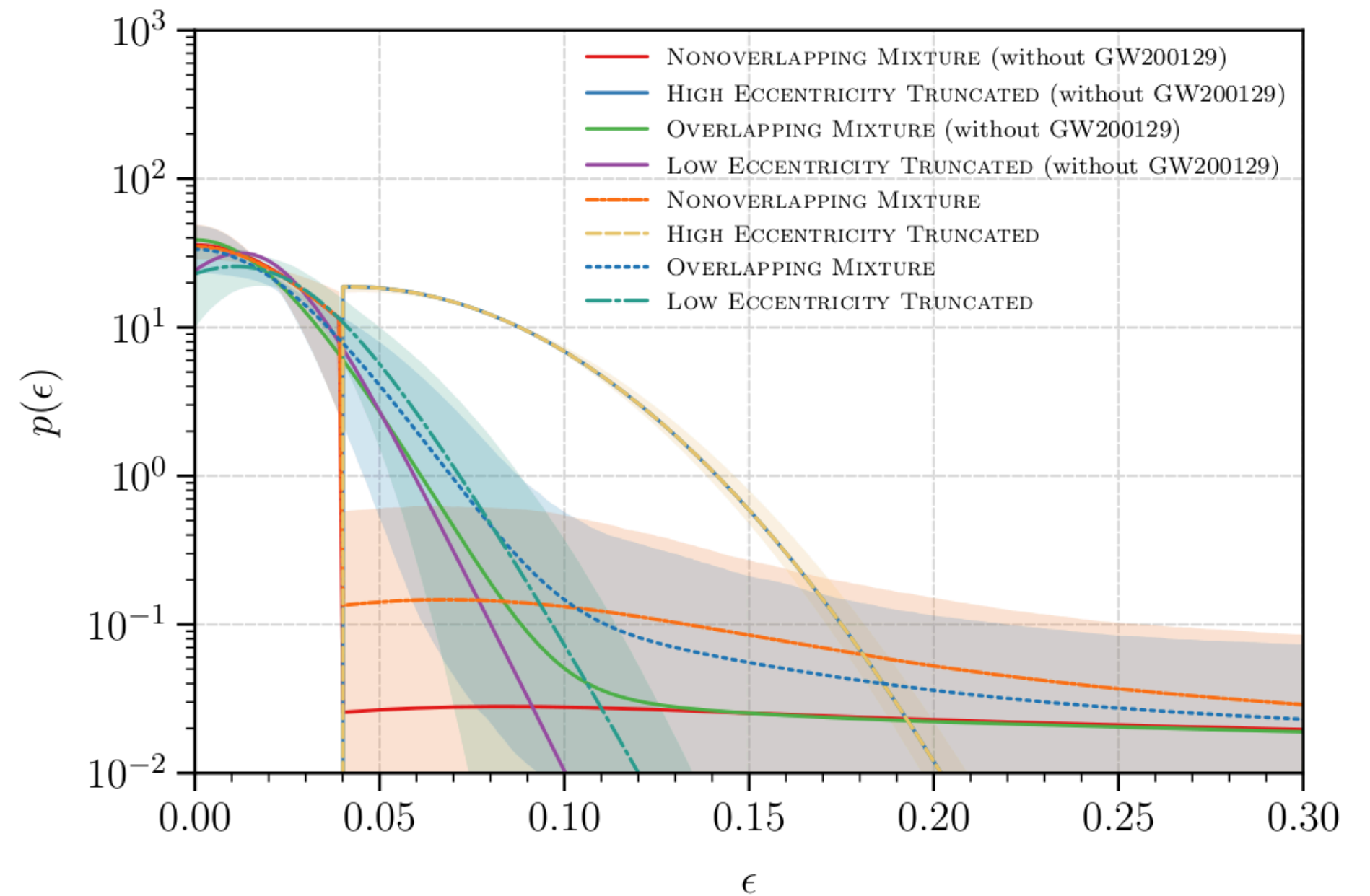
STUDIES LOOKING AT THE POPULATION PROPERTIES

Zeeshan+ 2026 defined a mixture between two Gaussians as a population model.

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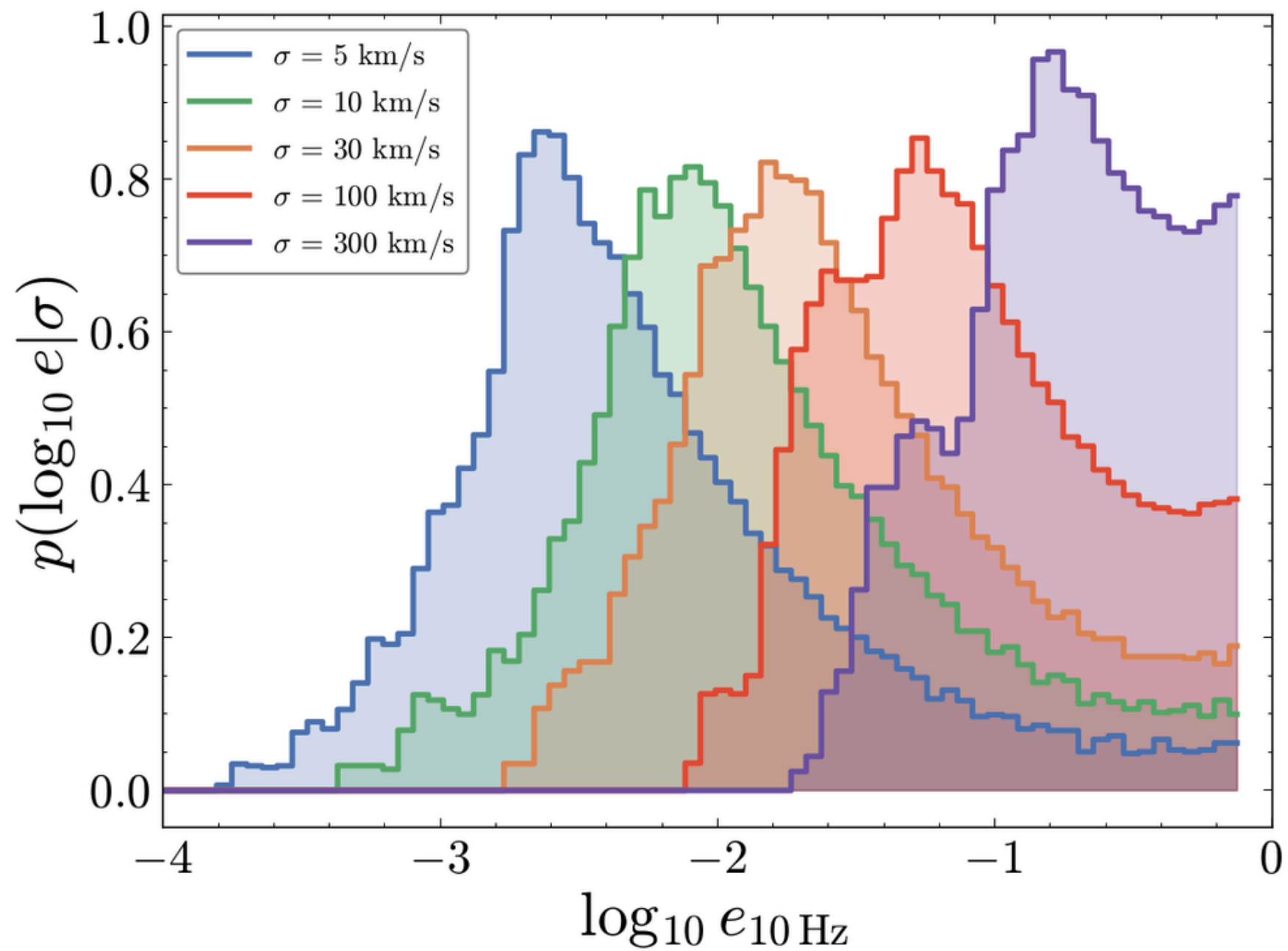


With a FAR cutoff of 1 per year, **<5% of binaries have eccentricity > 0.04**. When not including GW200129 this goes to <2%.



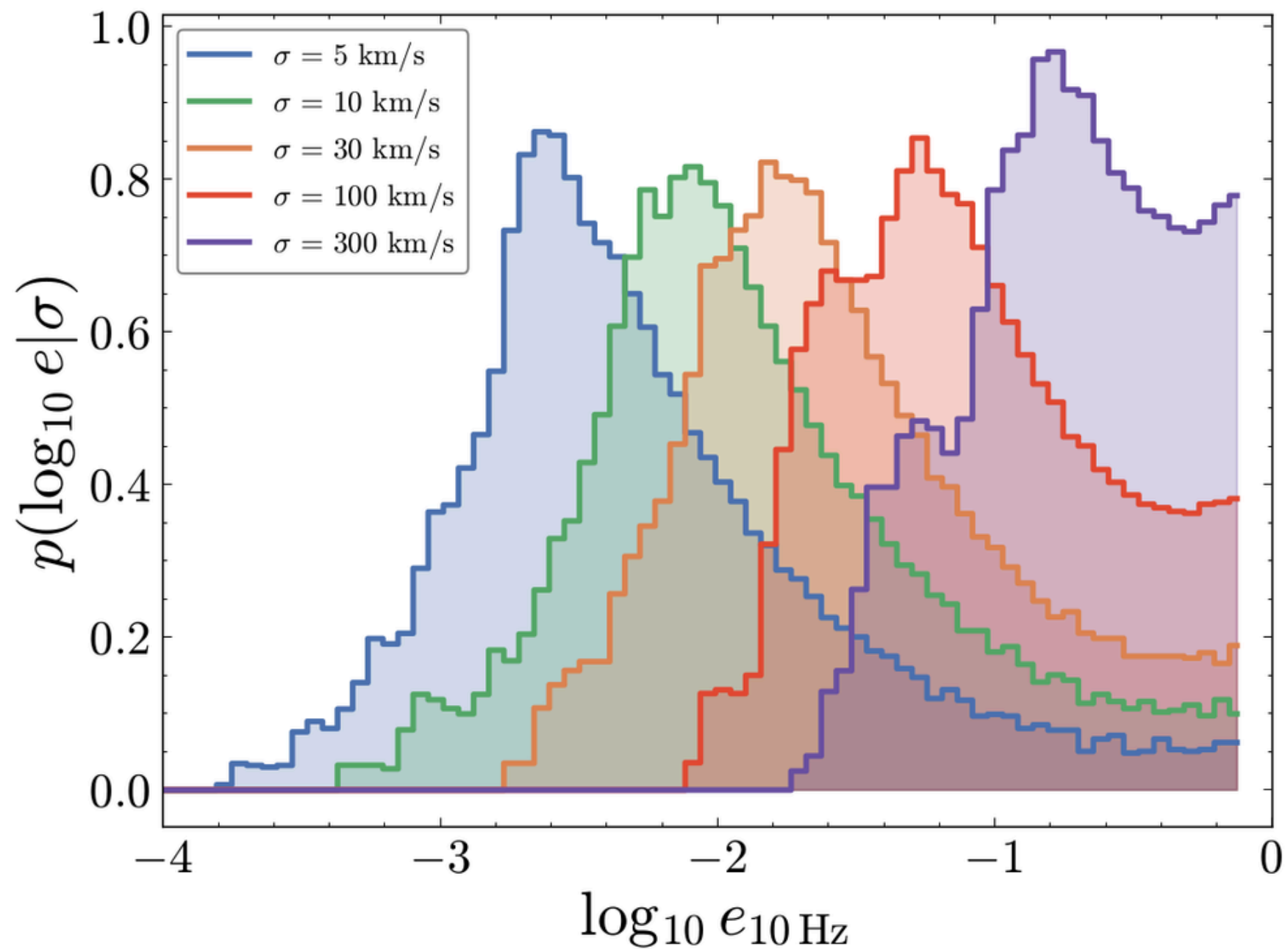
A PHYSICS INFORMED POPULATION MODEL

The distribution of eccentricities expected from dense stellar clusters with a dispersion, σ , due to BH captures



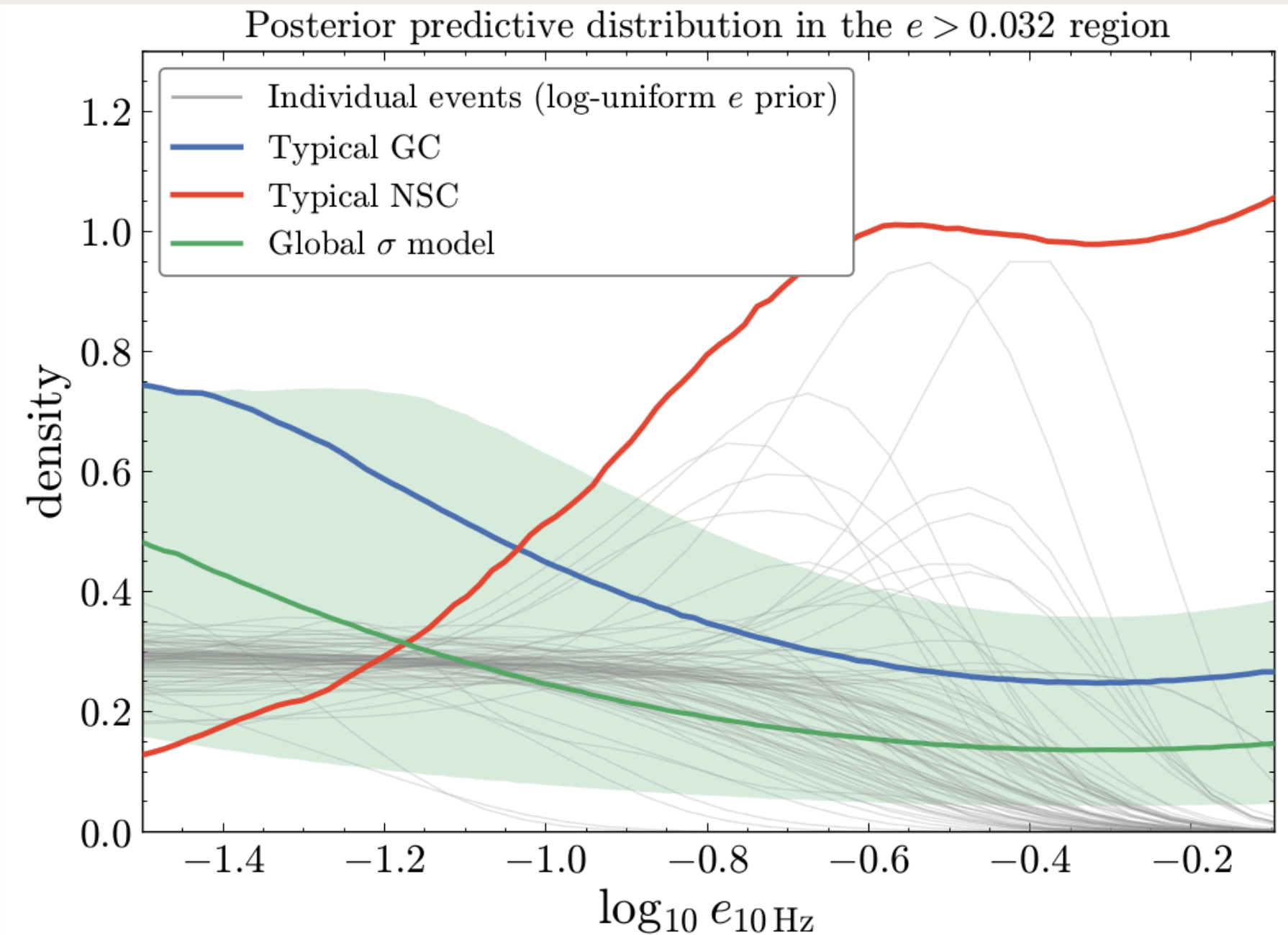
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Gupte+ 2026

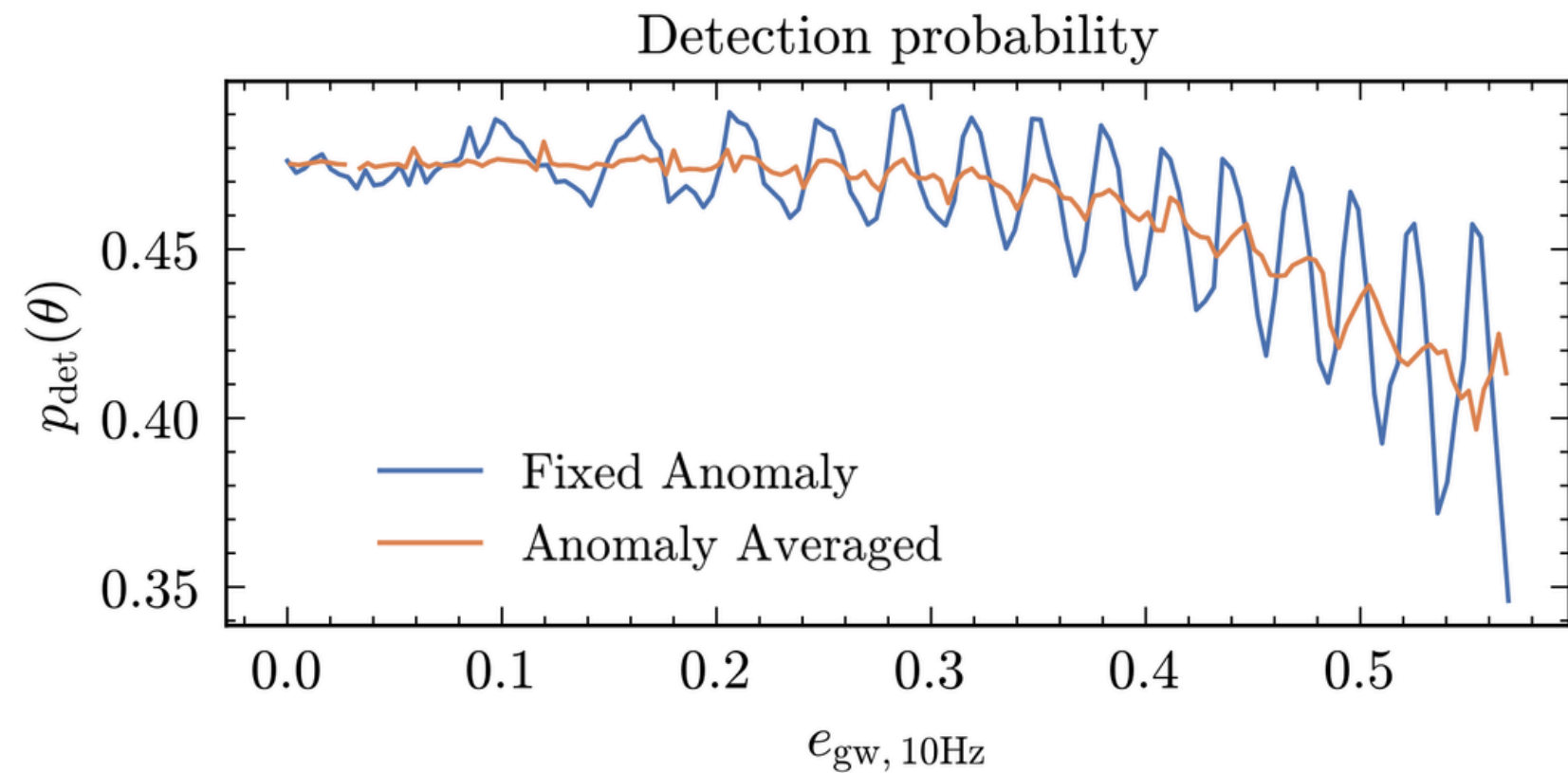
This implies that **less than 47% of binaries come from BH captures in nuclear star clusters**. See also Paul+ 2026!



Gupte+ 2026

SELECTION EFFECTS DUE TO QUASI-CIRCULAR TEMPLATES

> We do not use eccentric templates when searching for BBHs. This can lead to Malmquist bias.

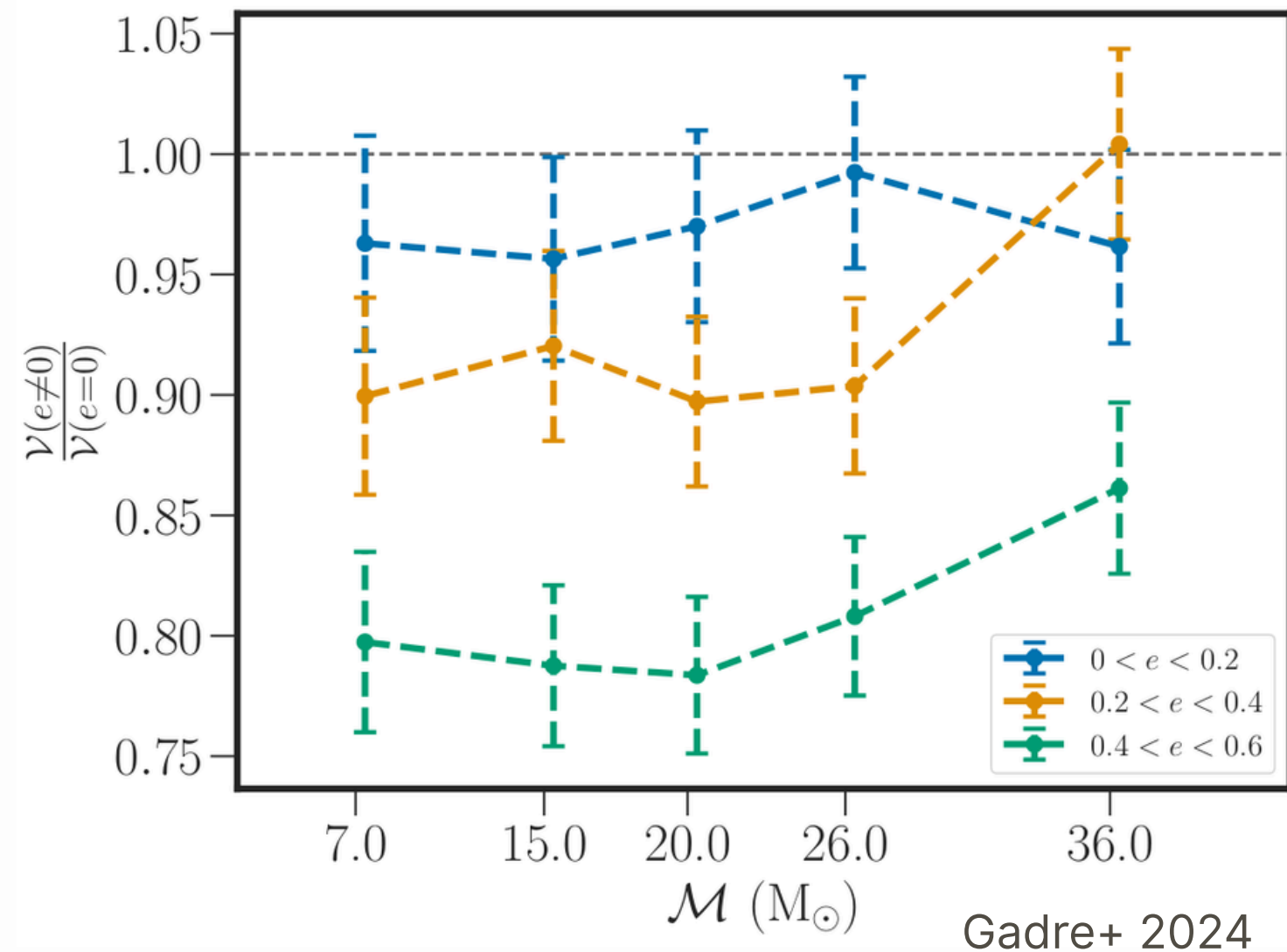


Gupte+ 2026, Vijaykumar+ 2026

> The above plot was produced with a semi-analytical method that estimates sensitivity without using search pipelines.

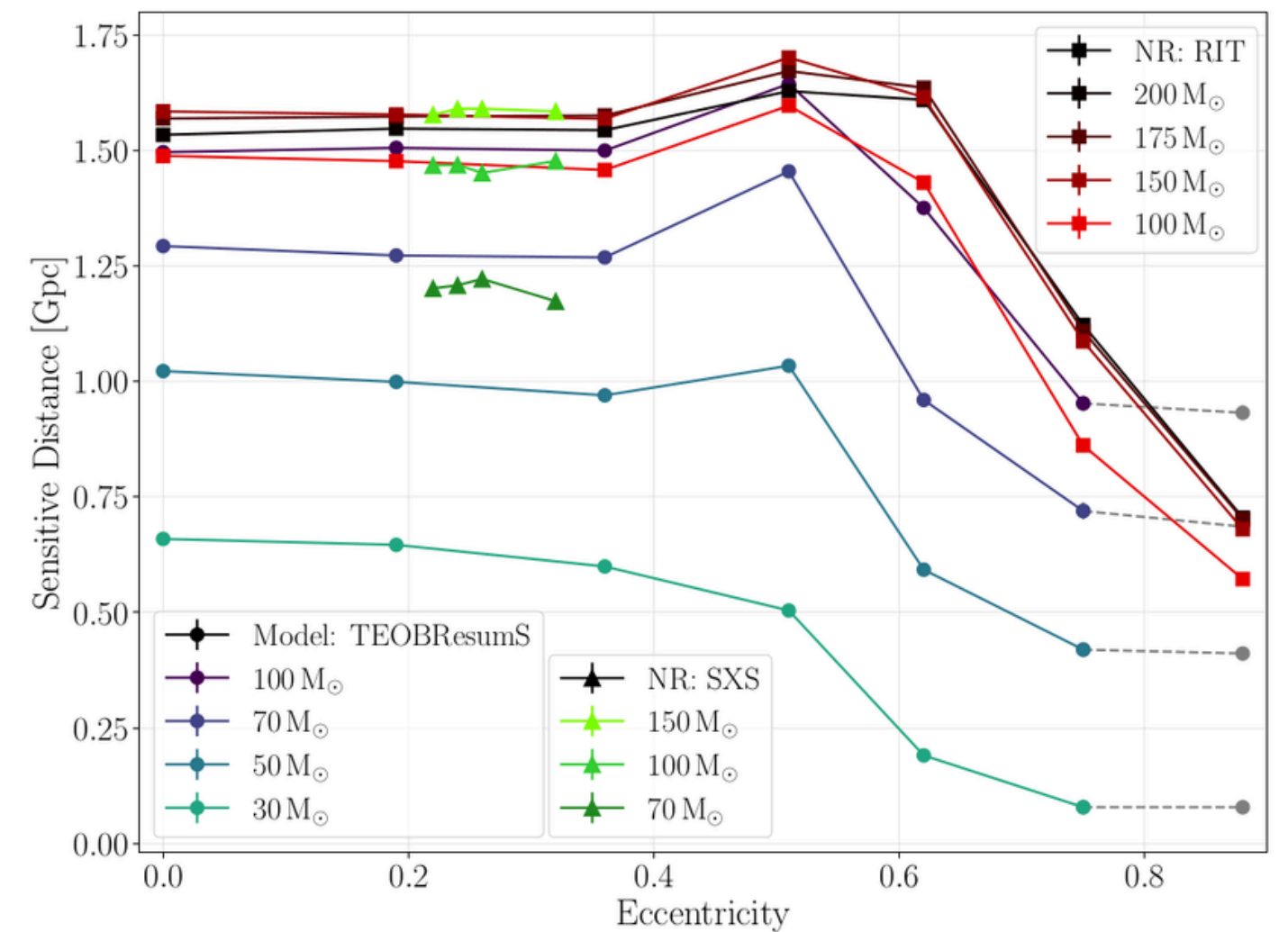
SELECTION EFFECTS DUE TO QUASI-CIRCULAR TEMPLATES

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> There are search papers that quantify our sensitivity to eccentricity

> Note Wang & Nitz 2025 did an eccentric search on O3 data but found no new candidates



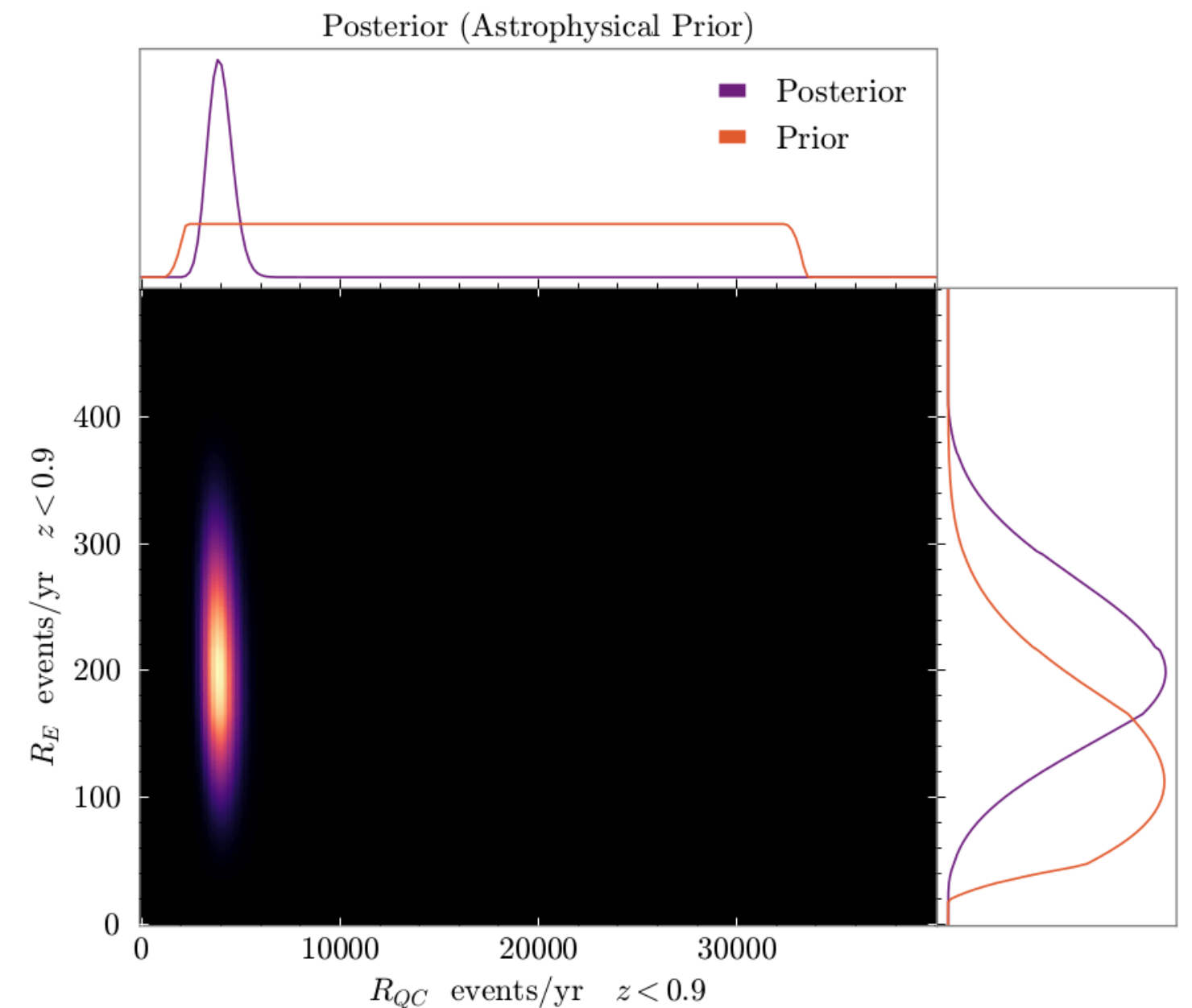
Bhaumik+ 2024

WE COULD ALSO INCORPORATE THE POPULATION INFORMATION VIA PRIOR ODDS

We can “balance” the look elsewhere effect by obtaining a global property of the data. First, we can ask the probability that one event is eccentric.

$$p_{\text{eccentric}} = \frac{\mathcal{O}_{E/QC}}{1 + \mathcal{O}_{E/QC}} = \frac{R_E p(d|M_E)}{R_E p(d|M_E) + R_Q p(d|M_Q)}$$

We can then use all the GW data to obtain a posterior over the rates and marginalize our probability over this.



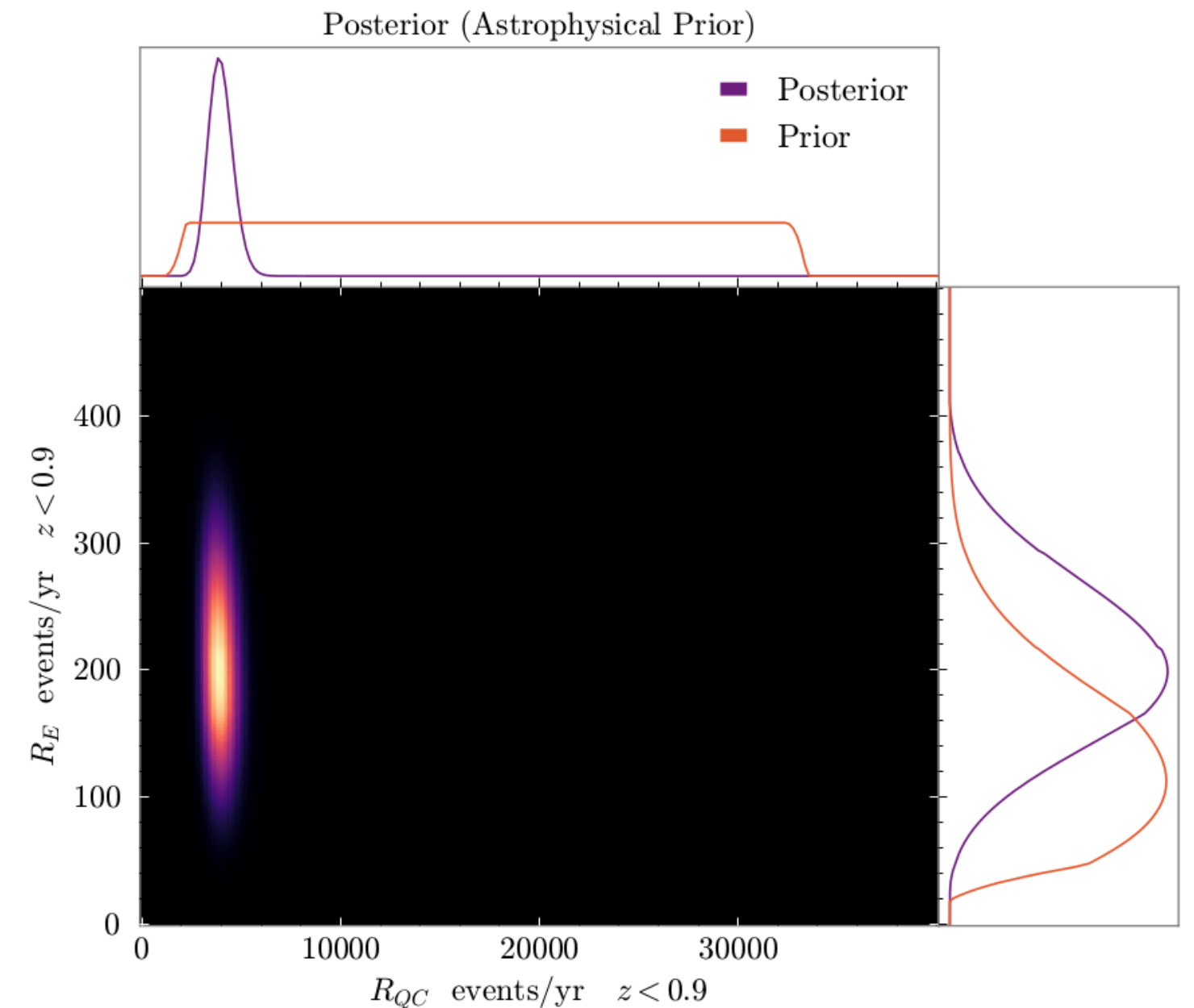
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Finally, we can use the same formalism to ask what the probability is that an event of the 57 is eccentric. This comes out to ~99%.



Gupte+ 2024

CONCLUSIONS

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- > **Modelling eccentricity effects near merger is important**. Just because eccentricity is expected to approach zero near merger does not mean it will for all systems!
- > **Priors are not scary! We just need to choose a good one** (see recent work by Rozner+ 2026)